

기존 HIL 30회 검증 결과

시나리오	시행	목표 접근	경로 이탈	정체	횡오차 RMSE(m)	상위 실패/시도
1	10	9	1	0	0.161	7/787
2	10	0	3	7	0.189	461/1245
3	10	10	0	0	0.182	6/660

세 시나리오 각각 최신 10회 전체를 포함. 성공 회차만 선별하지 않음.
completed는 목표 영역 접근이며 목표 정차 성공이 아님.
횡오차는 현재 하위 추종 경로 기준이며 독립 도로 중심선 오차가 아님.
S1 목표 접근 9회, S3 10회의 실제 최종 정지점은 목표에서 약 32~37m 떨어짐.
S2의 작은 추종오차가 출구 선택 또는 완주 성공을 뜻하지 않음.

실행시간: 계측 범위를 구분한 통계

시나리오	계측 범위	n	중앙값 ms	P95 ms	P99 ms	최대 ms	주기 초과
1	상위 측정 사이클	787	3.202	6.366	8.475	13.362	0
1	하위 솔버만	3928	0.304	0.445	1.127	4.771	전체 미계측
2	상위 측정 사이클	1245	3.550	7.020	9.905	760.812	1
2	하위 솔버만	4052	0.304	0.388	1.029	2.313	전체 미계측
3	상위 측정 사이클	660	3.518	6.990	8.977	13.082	0
3	하위 솔버만	3292	0.307	0.484	1.161	2.418	전체 미계측

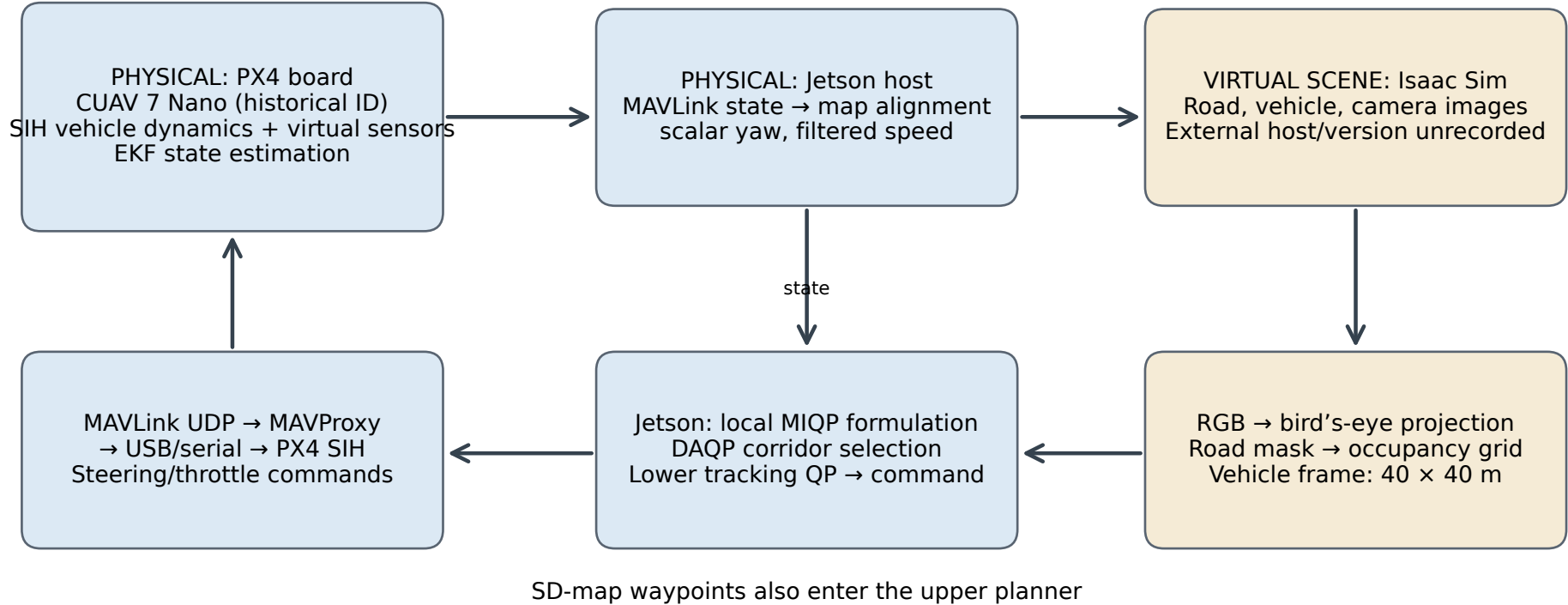
상위 주기 500ms: 자체 보고 비용 포함, callback 진입 전 대기는 제외. S2 1회 초과(0.0803%).
S2 A02 18:42:06.732: 허용 구간 추출 단계 wall 758.635ms / thread CPU 3.409ms.
그때 solver는 0.110ms. 지연의 근본 원인은 미분리.
하위 주기 100ms 전체 callback은 미계측. 솔버 통계로 deadline miss 0회를 주장하지 않음.
성공만/실패 포함/배치 전체의 분리 통계는 runtime_statistics.csv. 최대값은 WCET가 아님.

실제 설정과 확인 범위

구성	값	증거
상위	2Hz, dt .75s, 최대 N6	실행 설정 + 로그
하위	10Hz, dt .1s, N30	실행 설정 + trace
목표속도 / 휠베이스	6m/s / 2.8m	실행 설정 + 로그
길이 / 폭	4.7m / 미확보	parameter event / 미확보
격자	128×128, .3125m, x0~40/y±20m	73,250 원 프레임
하위 조향 / 변화율	.3rad / 1.0472rad/s 설정	실행 설정, 실제 명령 별도 검사
B/soft	relative factor1.5, ratio .7, margin2m	실행 설정
물리 / 가상	Jetson·PX4 보드 / SIH 차량·센서·Isaac 도로	구성·과거 보드 기록
편향 기능	8개 배치 모두 disabled	보존된 effective_config

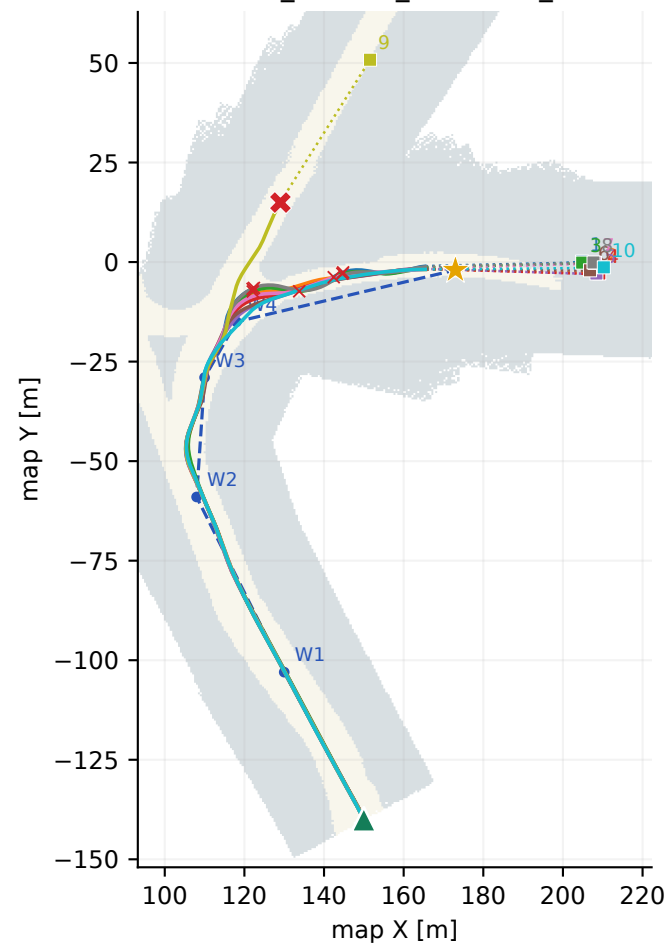
fullscale: 제어 단위 m. Isaac stage의 실제 축척·버전·외부 BEV YAML은 미확보.
 DAQP vendor 소스 버전0.7.2. 이번 상위 후보 조합은 모두256 이하: 조합별 연속 QP 비교.
 odom.orientation.z는 scalar yaw. 위치는 EKF 유래 가상 상태이며 모션캡처 실측이 아님.
 실험 화면·하드웨어 사진·실물 바퀴각·독립 도로 경계·차량 폭·편향 HIL 시험은 미확보.
 현재 호스트/과거 보드 확인값과 해당 주행 적용값을 CONFIGURATION_KO.txt에서 구분.

HIL functional configuration — physical processors, simulated vehicle/environment

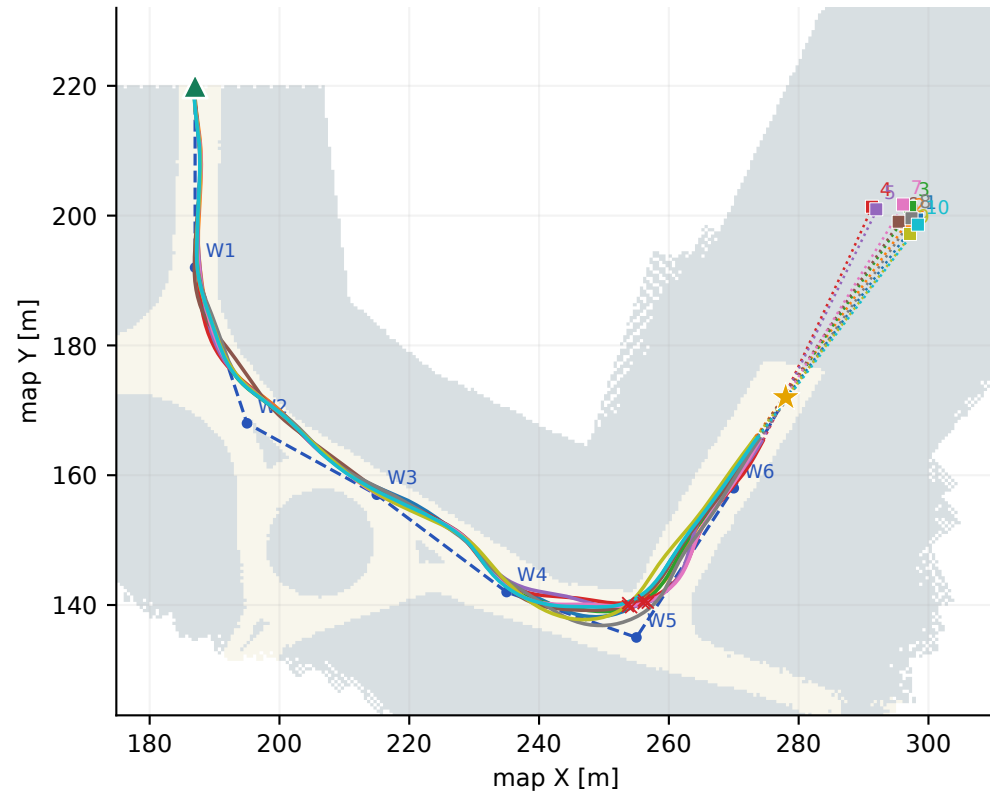


All 30 HIL attempts — reconstructed road context

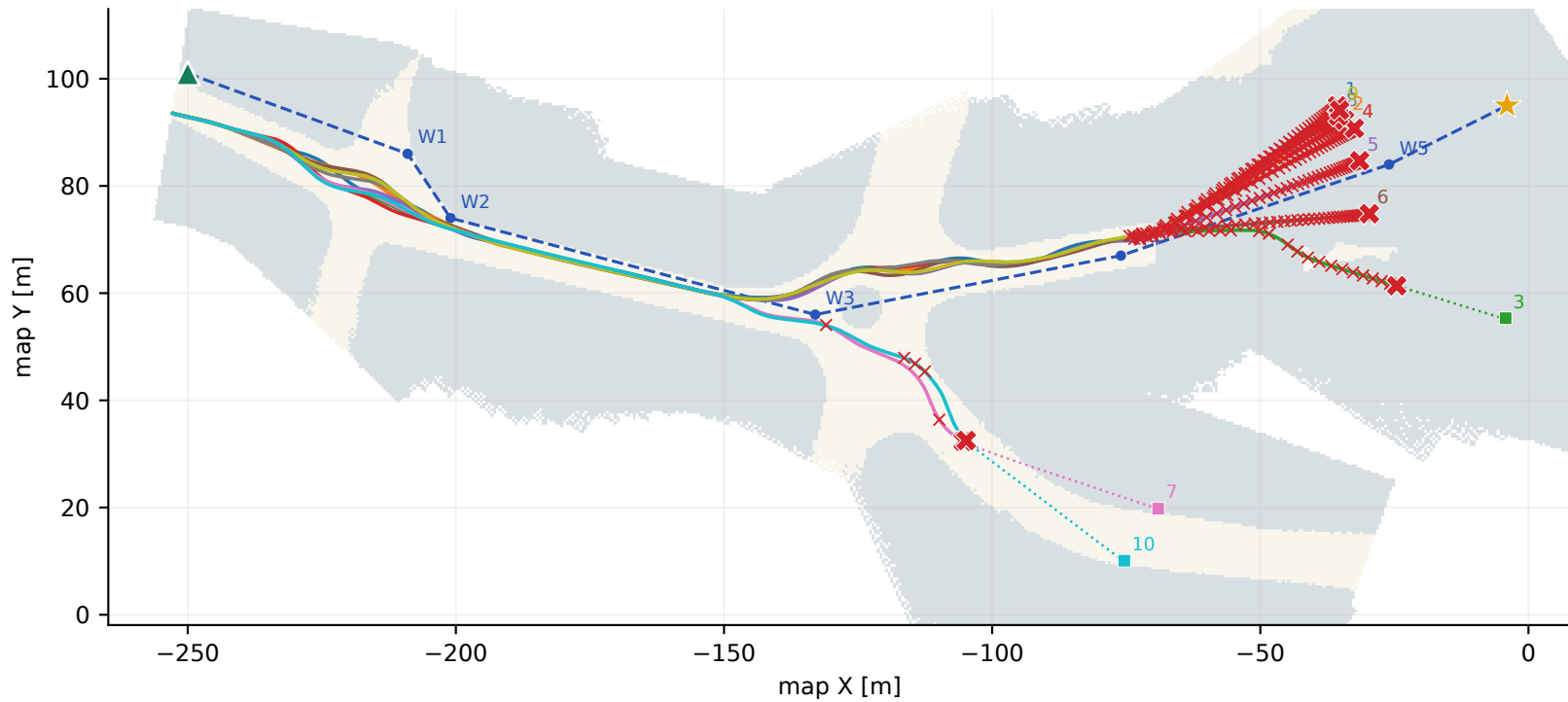
Scenario 1: 9/10 goal approaches
20261008_182344_scenario1_0cefb5



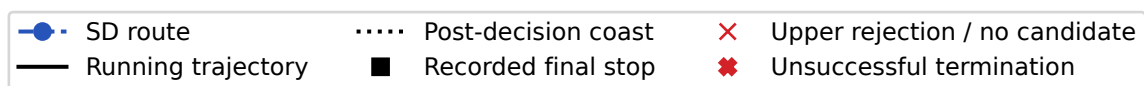
Scenario 3: 10/10 goal approaches
20261008_185426_scenario3_72c4df



Scenario 2: 0/10 goal approaches
20261008_183952_scenario2_5524df

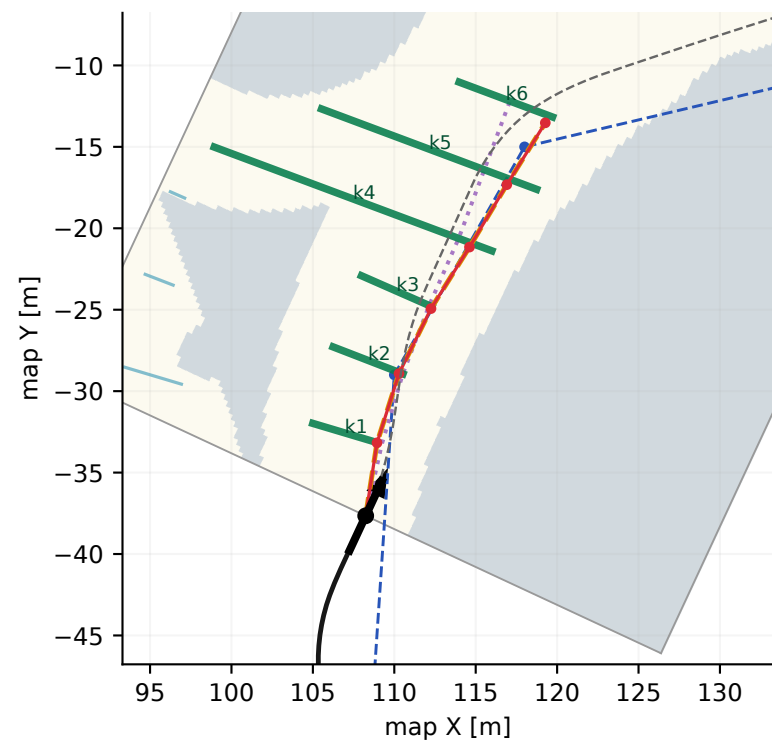


Background: grid-vote mosaic, aligned after recording; not an independent road boundary. W labels are route vertices.

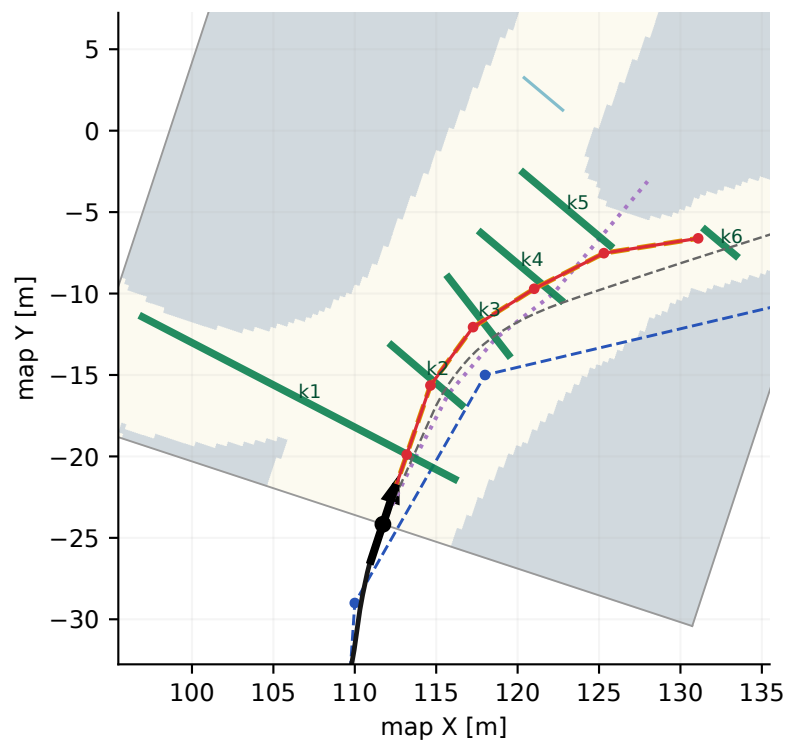


Scenario 1 / A06 / branch — zoom 20261008_182344_scenario1_0cefb5

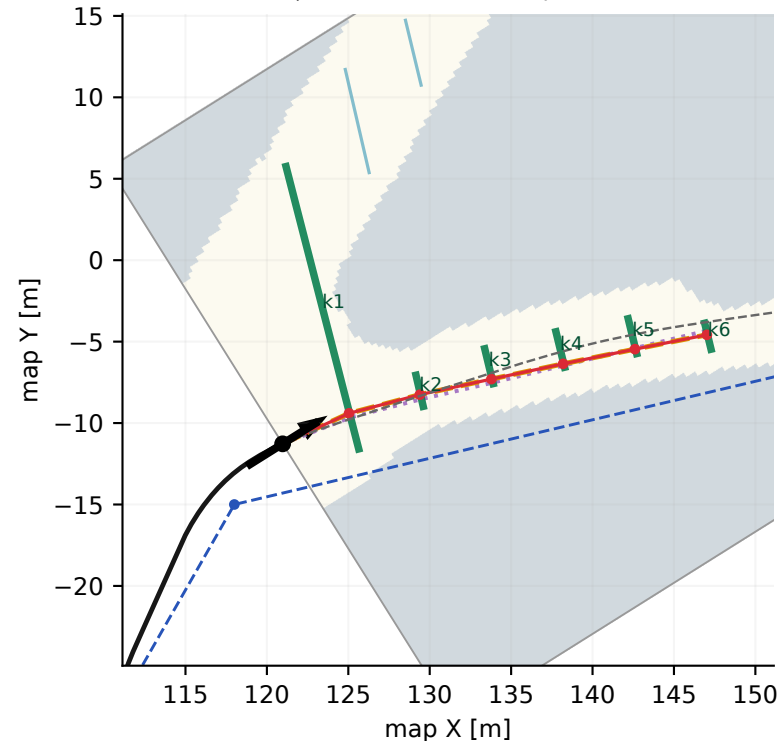
1. 18:29:40.128 KST / plan 696
N=6, max candidates=2 / success



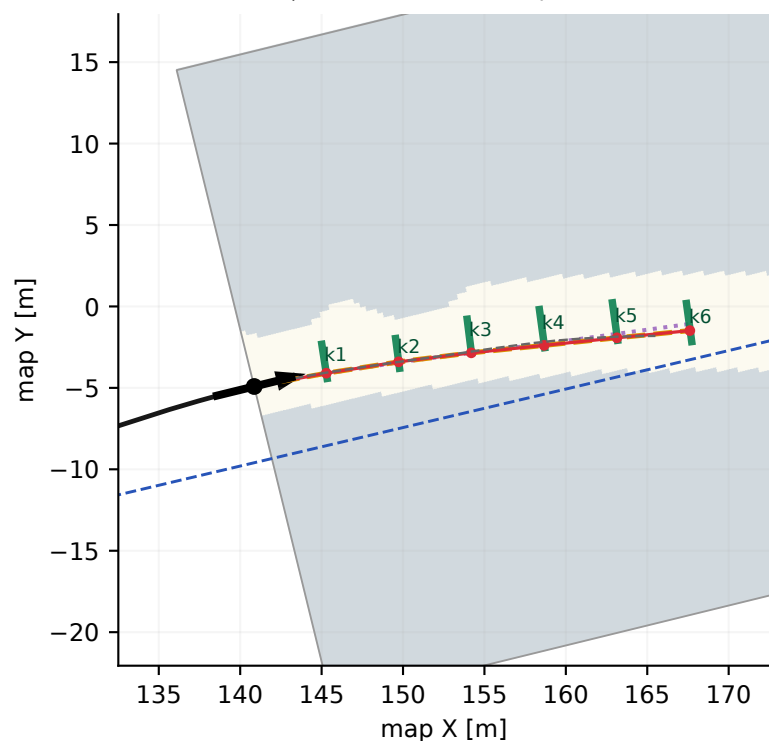
2. 18:29:43.128 KST / plan 702
N=6, max candidates=2 / success



3. 18:29:46.628 KST / plan 709
N=6, max candidates=2 / success



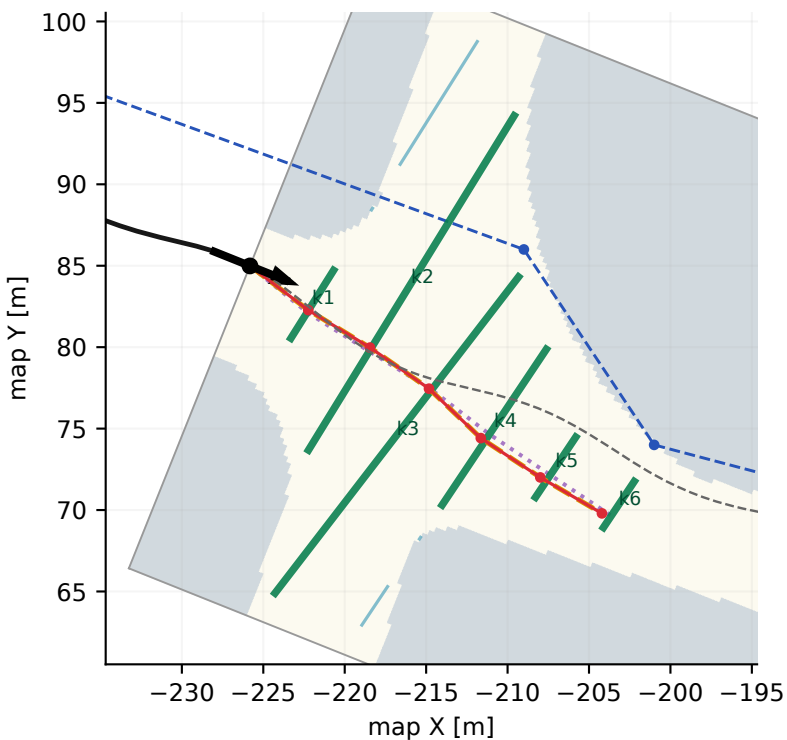
4. 18:29:51.128 KST / plan 718
N=6, max candidates=1 / success



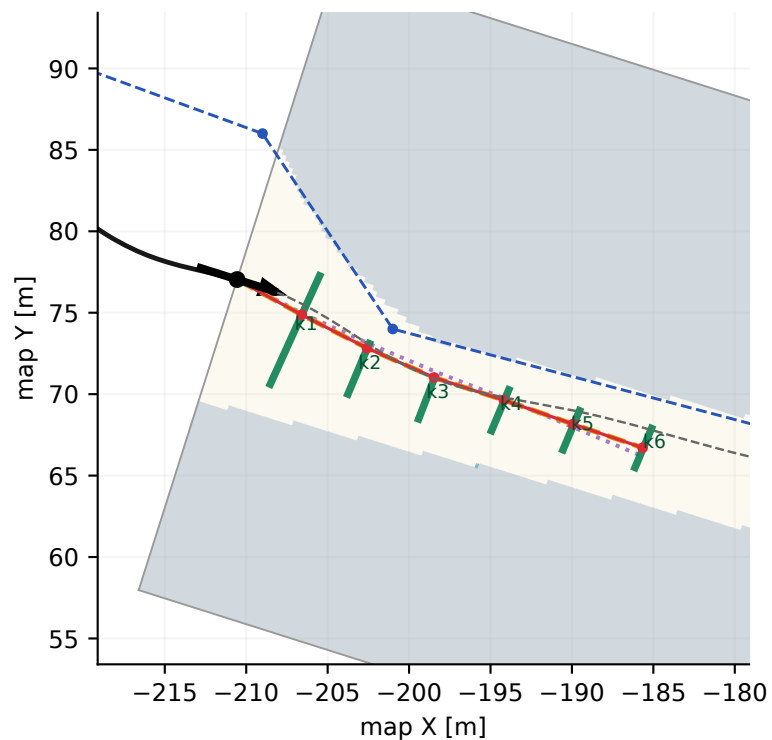
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 2 / A01 / branch — zoom 20261008_183952_scenario2_5524df

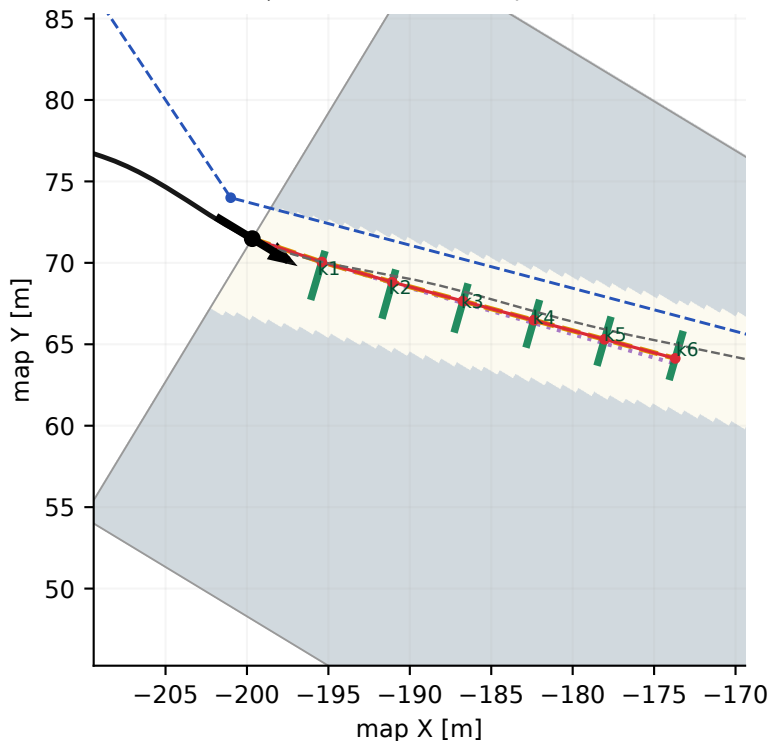
1. 18:40:09.729 KST / plan 20
N=6, max candidates=3 / success



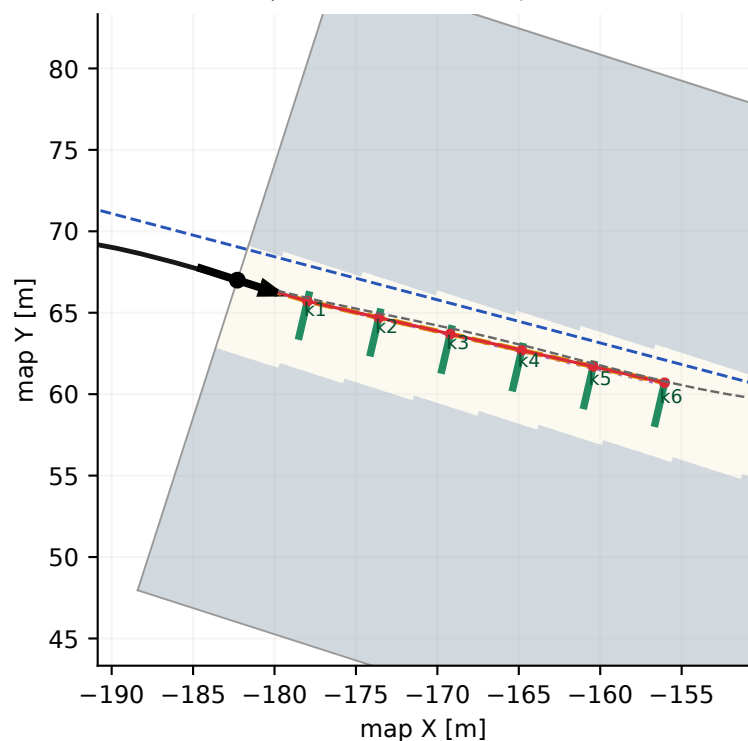
2. 18:40:13.229 KST / plan 27
N=6, max candidates=2 / success



3. 18:40:15.729 KST / plan 32
N=6, max candidates=1 / success



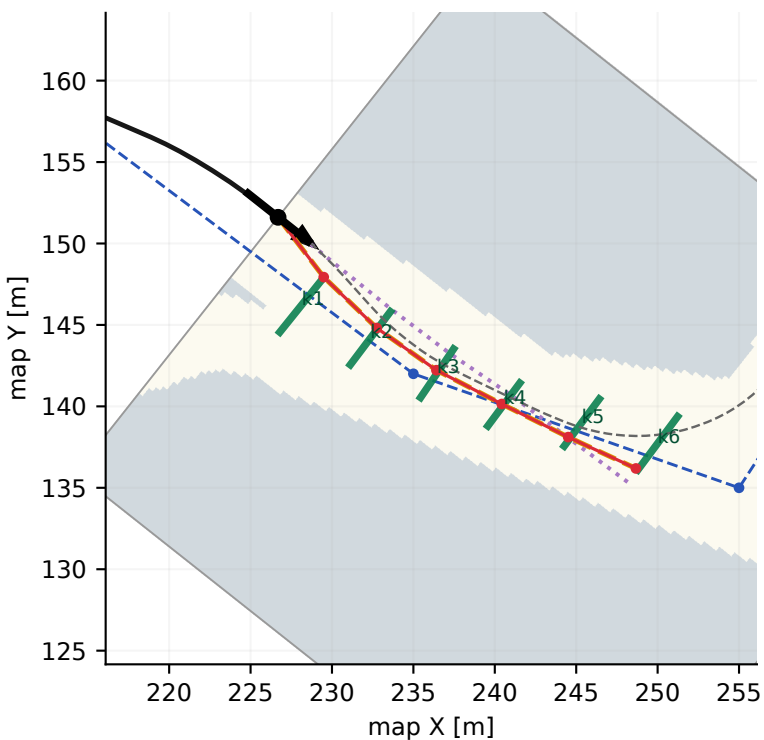
4. 18:40:19.229 KST / plan 39
N=6, max candidates=1 / success



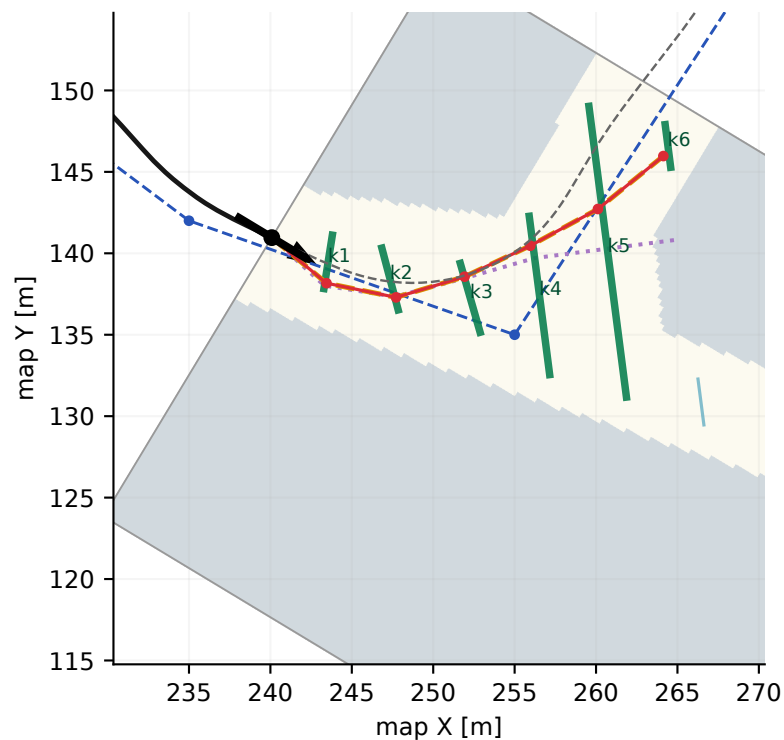
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 3 / A01 / branch — zoom 20261008_185426_scenario3_72c4df

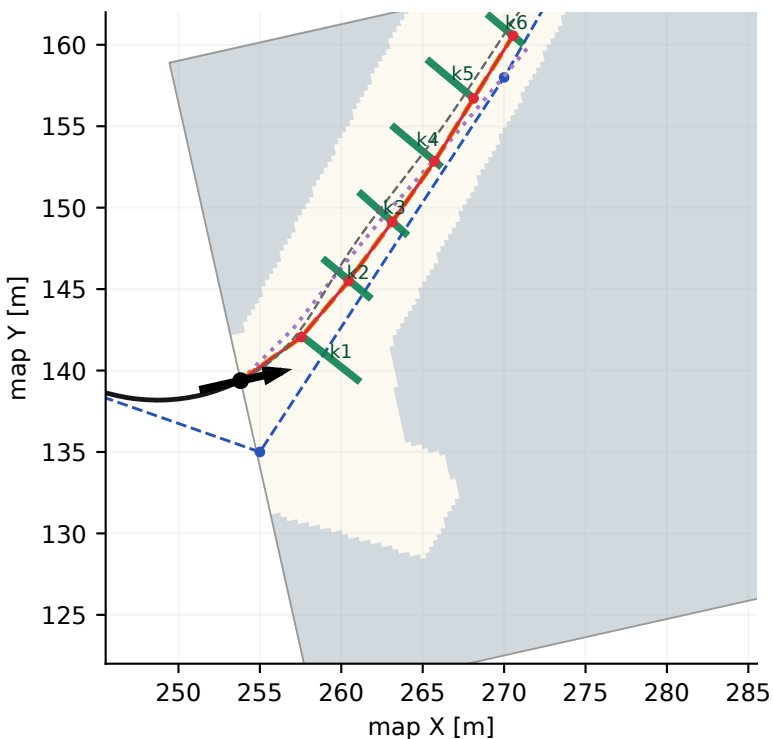
1. 18:54:55.521 KST / plan 43
N=6, max candidates=1 / success



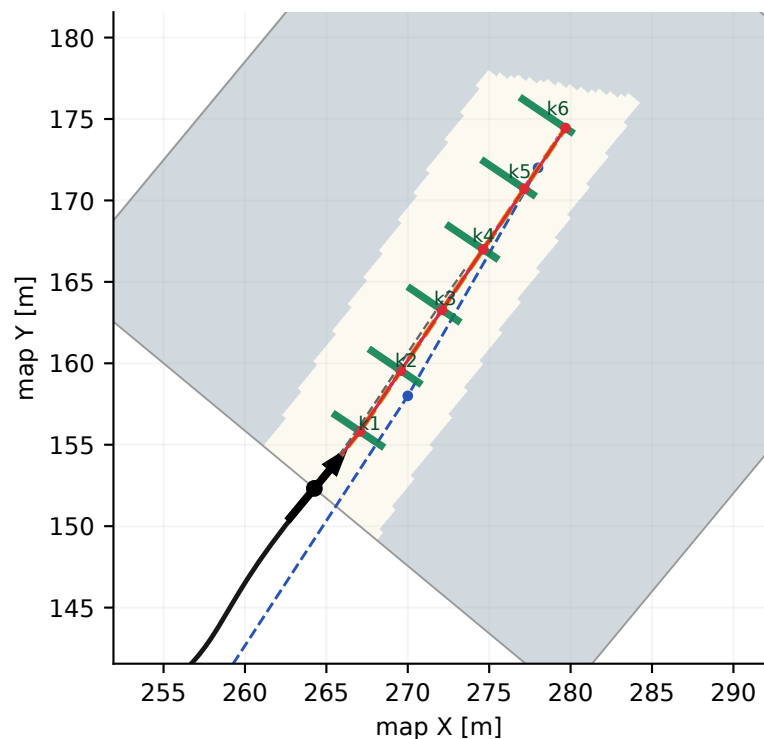
2. 18:54:59.021 KST / plan 50
N=6, max candidates=2 / success



3. 18:55:02.021 KST / plan 56
N=6, max candidates=1 / success



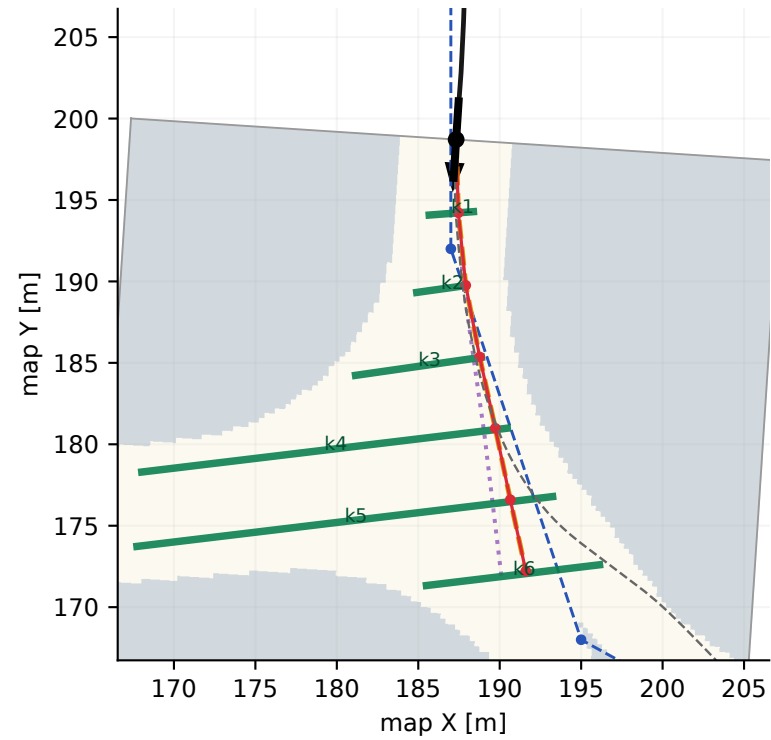
4. 18:55:05.521 KST / plan 63
N=6, max candidates=1 / success



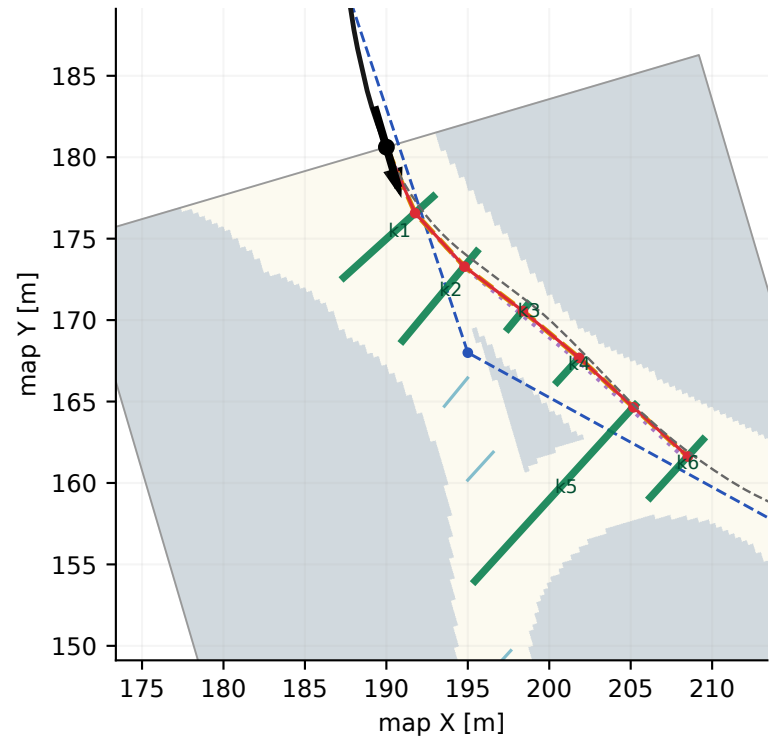
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 3 / A01 / curve — zoom 20261008_185426_scenario3_72c4df

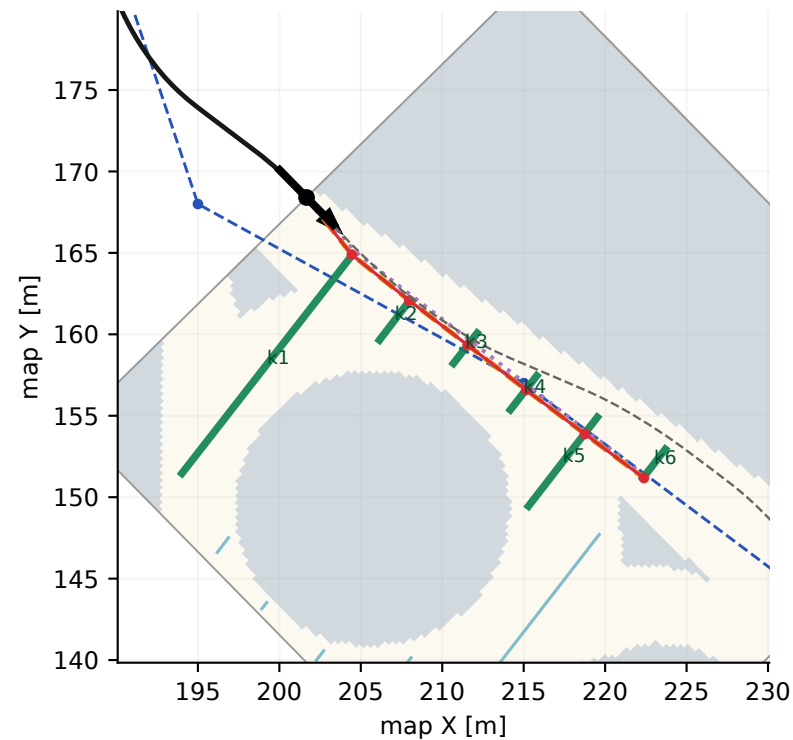
1. 18:54:42.521 KST / plan 17
N=6, max candidates=1 / success



2. 18:54:46.021 KST / plan 24
N=6, max candidates=2 / success

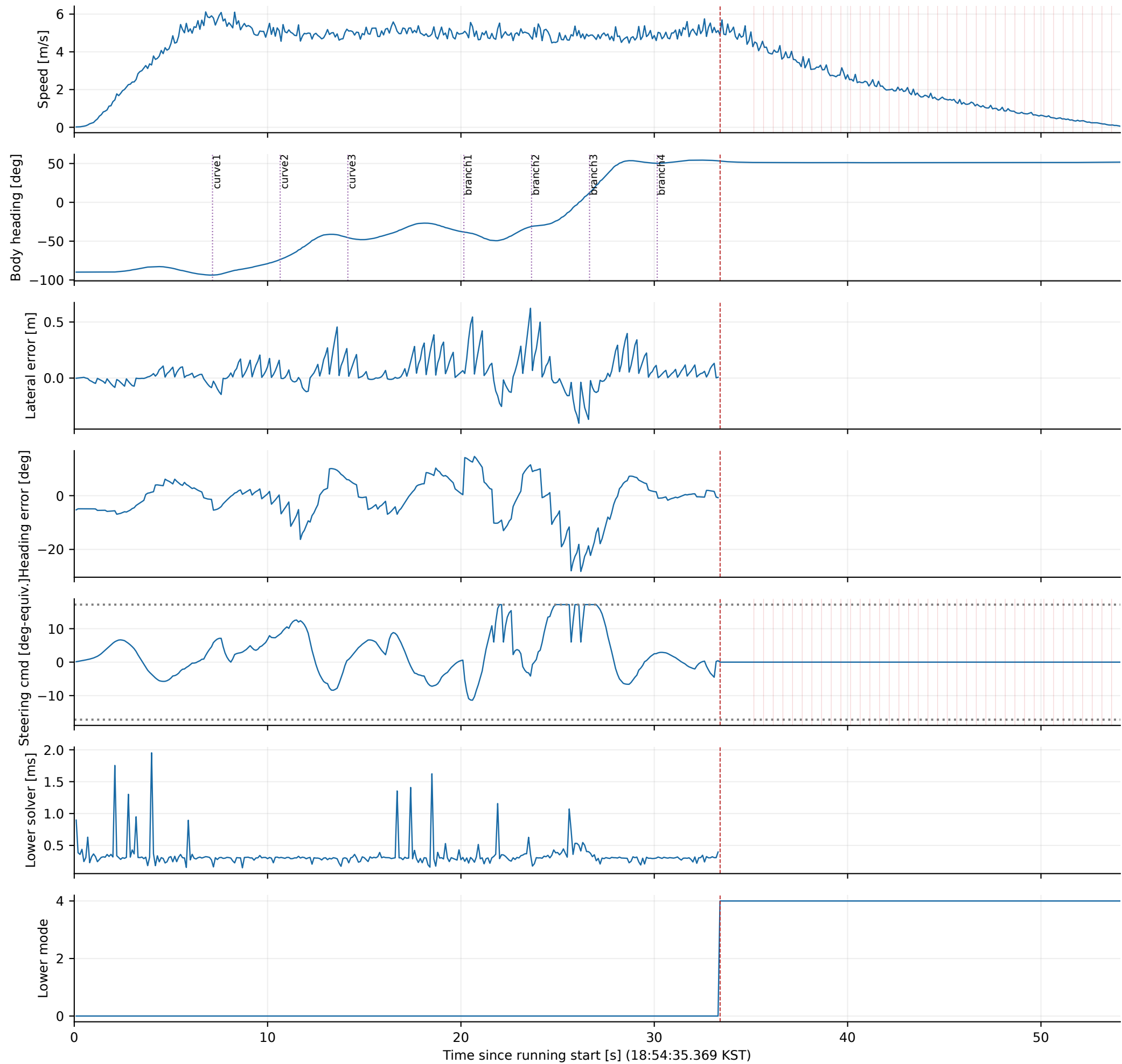


3. 18:54:49.521 KST / plan 31
N=6, max candidates=2 / success



Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

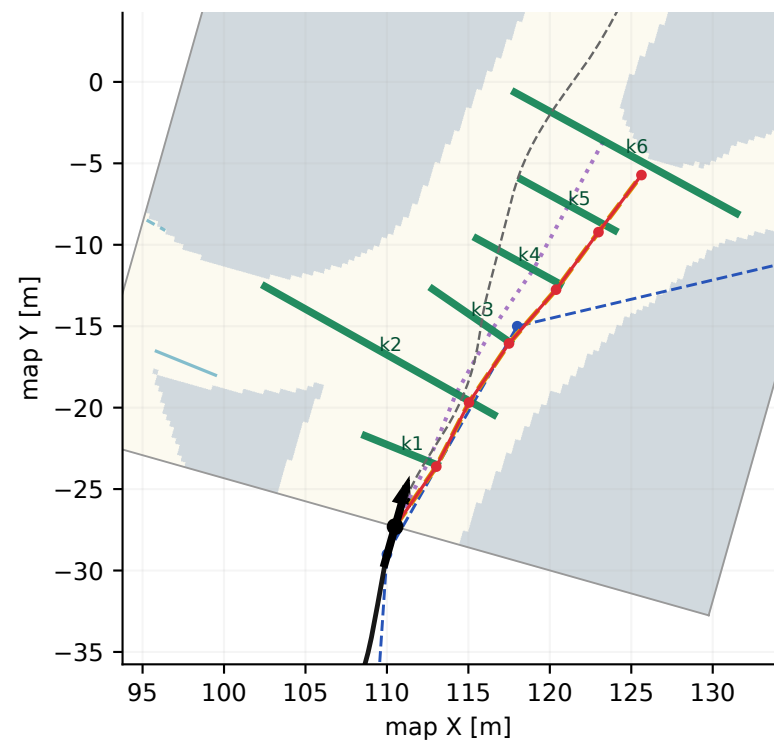
Scenario 3 / A01: completed
20261008_185426_scenario3_72c4df



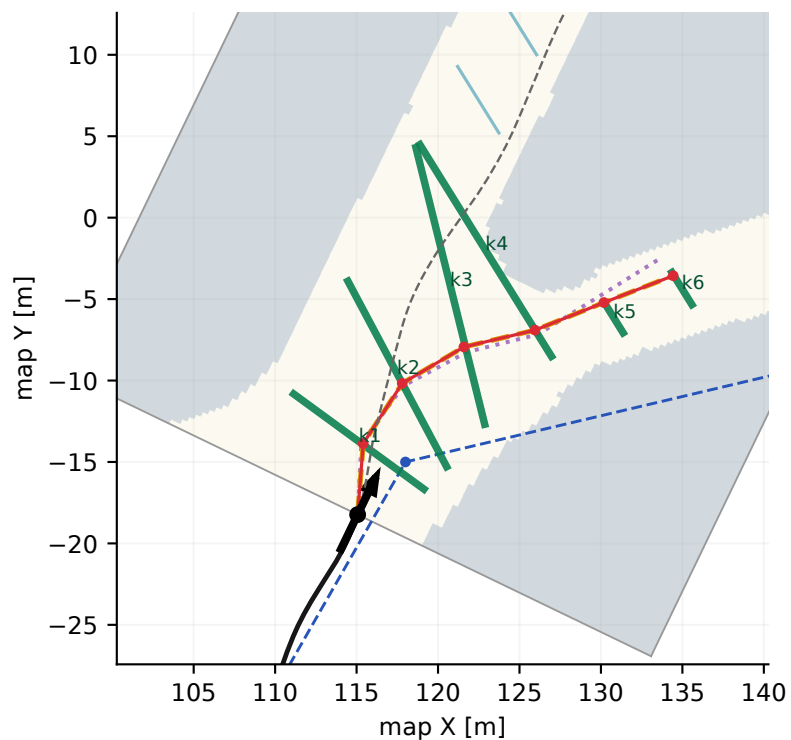
Errors are relative to the current lower path; neutral placeholder errors excluded. Red lines: upper failure. Dashed boundary: runner decision. No measured steering.

Scenario 1 / A09 / wrong exit — zoom 20261008_182344_scenario1_0cefb5

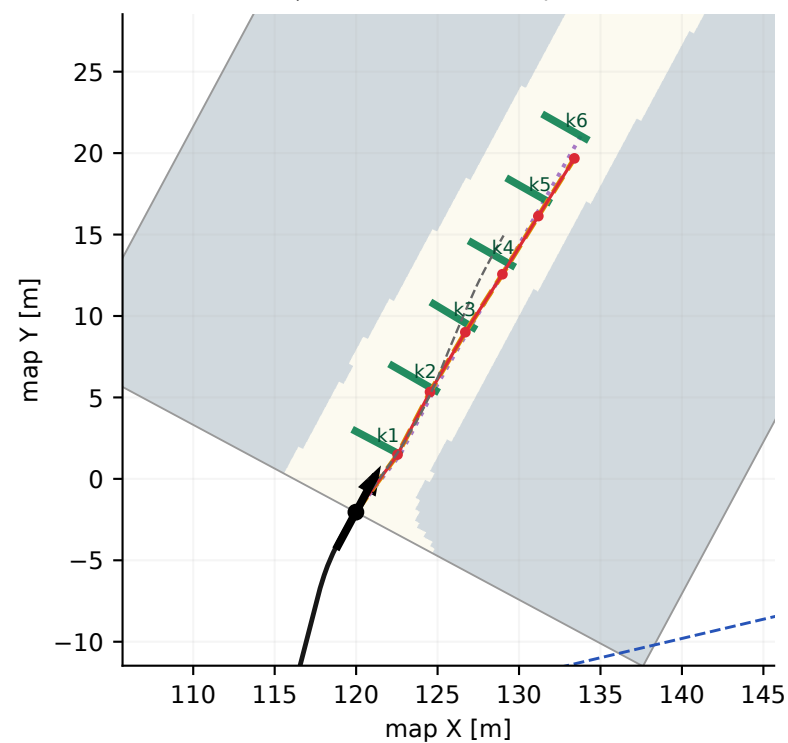
1. 18:32:56.133 KST / plan 1087
N=6, max candidates=3 / success



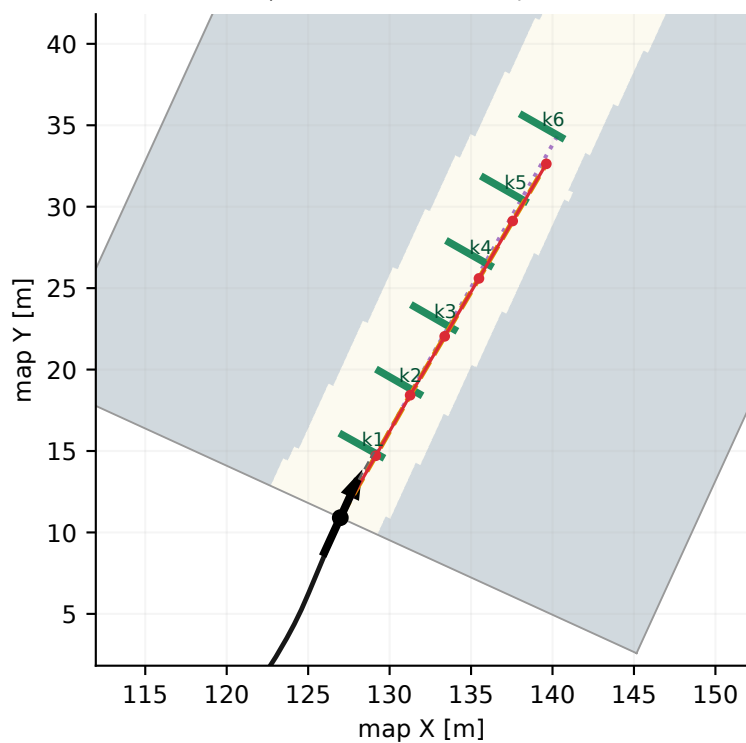
2. 18:32:58.133 KST / plan 1091
N=6, max candidates=2 / success



3. 18:33:01.633 KST / plan 1098
N=6, max candidates=1 / success



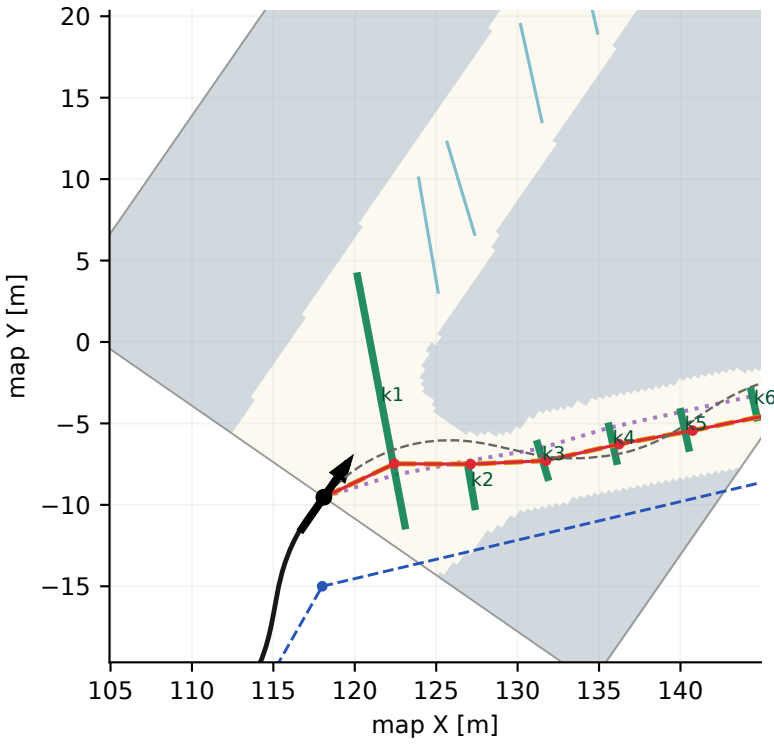
4. 18:33:04.633 KST / plan 1104
N=6, max candidates=1 / success



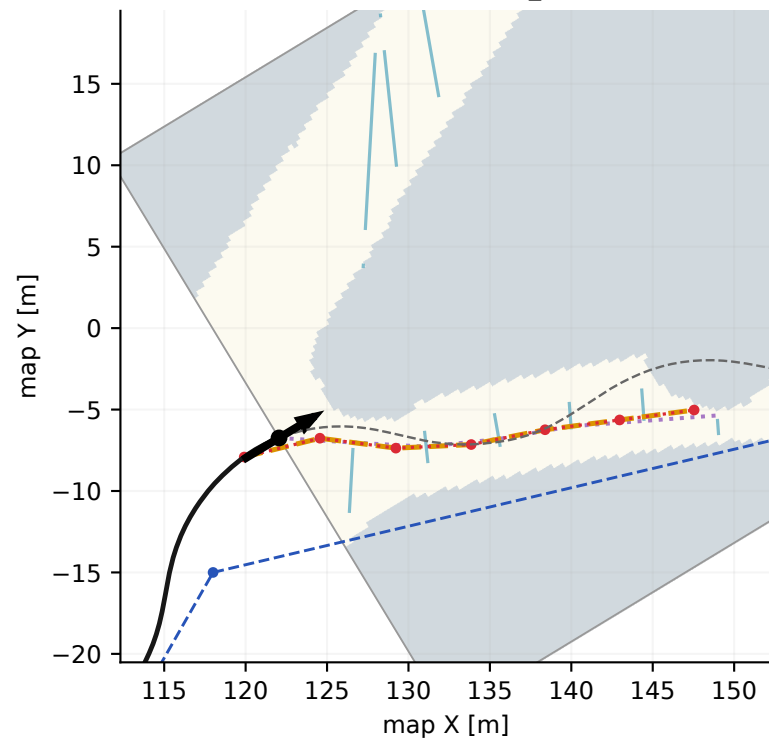
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 1 / A01 / failure — zoom 20261008_182344_scenario1_0cefb5

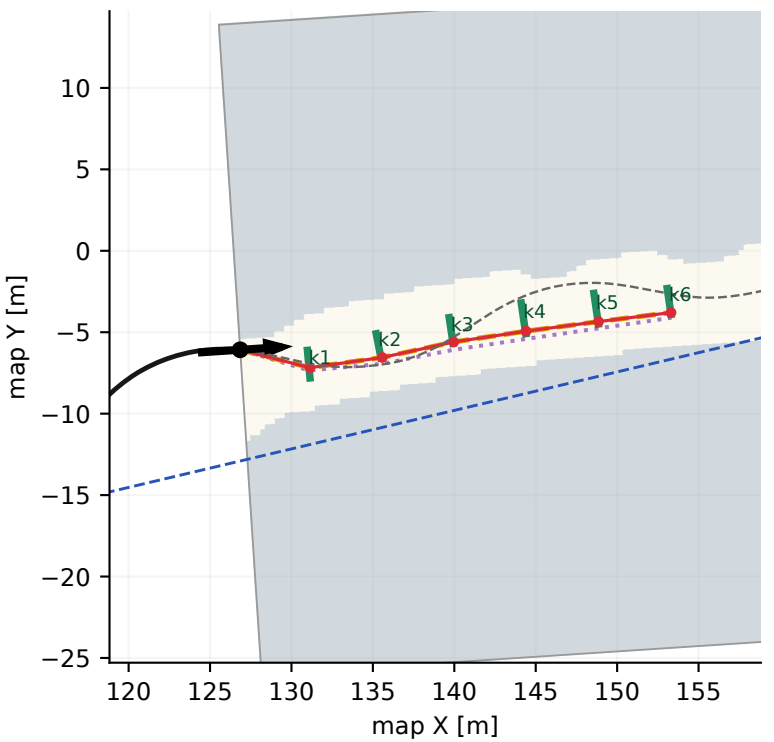
1. 18:24:23.119 KST / plan 63
N=6, max candidates=2 / success



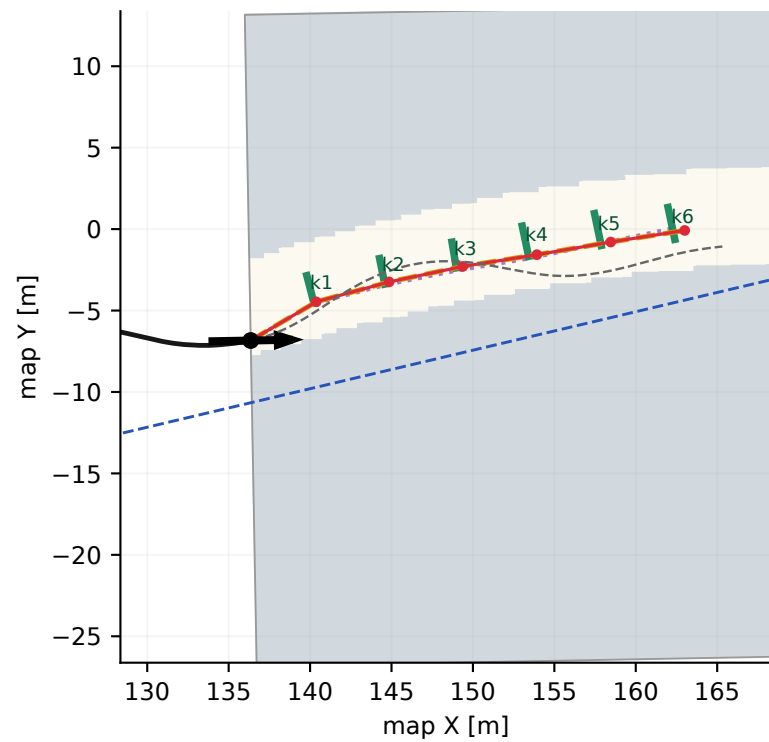
2. 18:24:24.119 KST / plan 65
N=6, max candidates=3 / solver_rejected; retained



3. 18:24:25.119 KST / plan 67
N=6, max candidates=1 / success



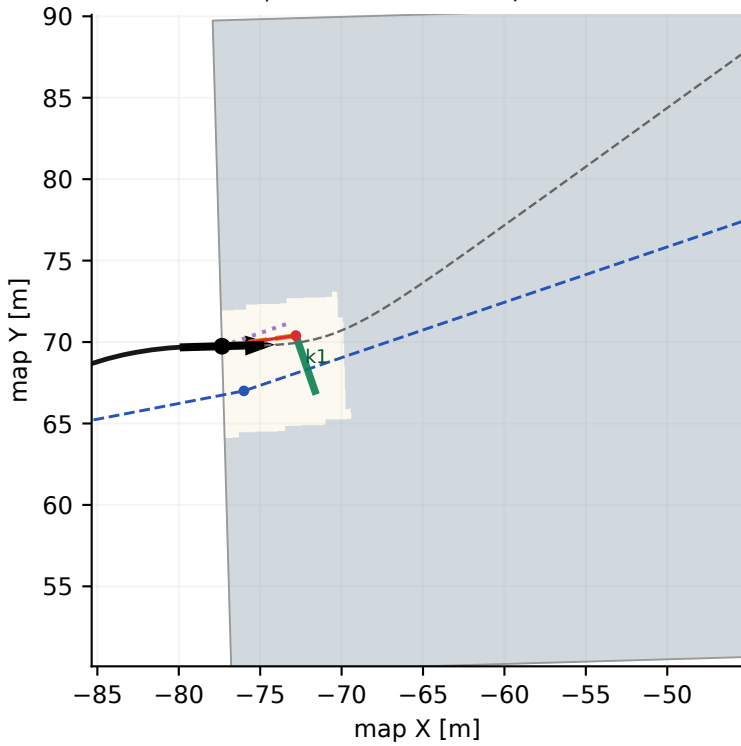
4. 18:24:27.119 KST / plan 71
N=6, max candidates=1 / success



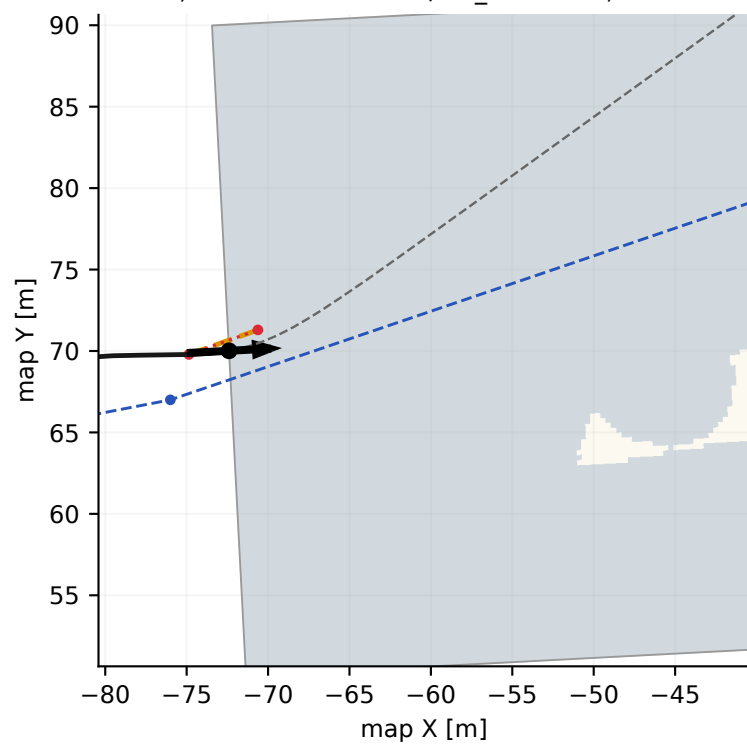
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 2 / A01 / failure — zoom 20261008_183952_scenario2_5524df

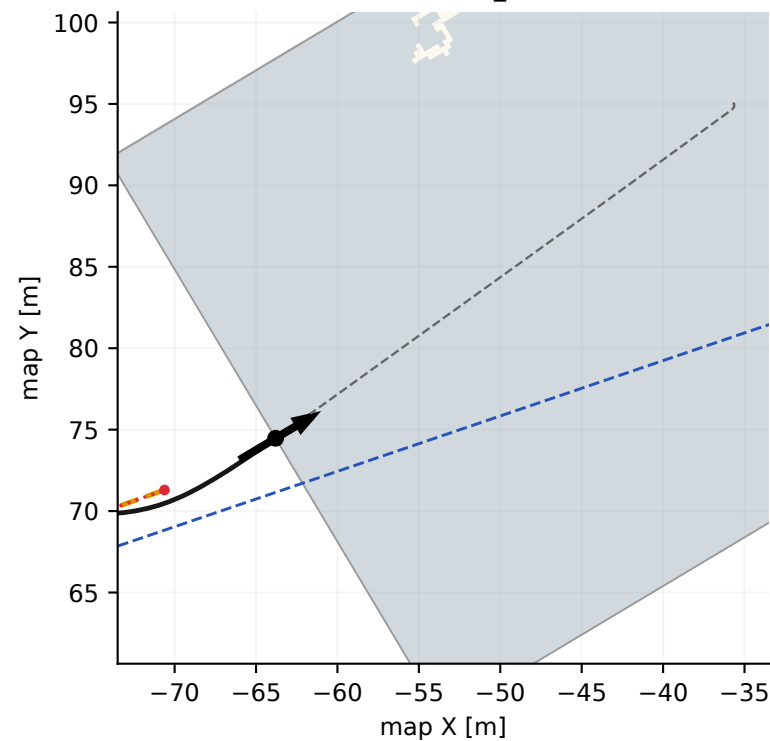
1. 18:40:40.230 KST / plan 81
N=1, max candidates=1 / success



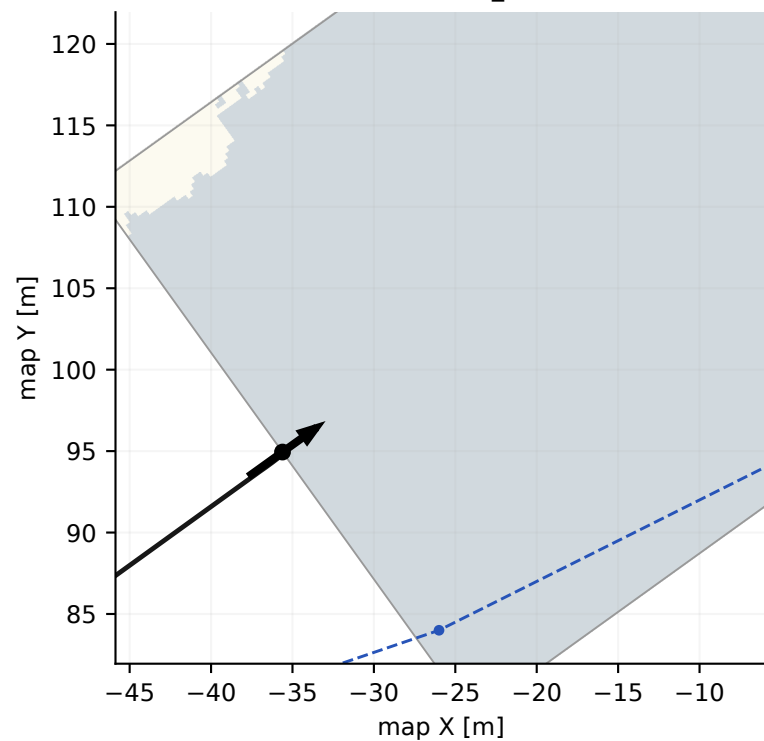
2. 18:40:41.230 KST / plan 83
N=0, max candidates=0 / no_candidate; retained



3. 18:40:43.230 KST / plan 87
N=0, max candidates=0 / no_candidate; retained



4. 18:41:14.231 KST / plan 149
N=0, max candidates=0 / no_candidate; retained

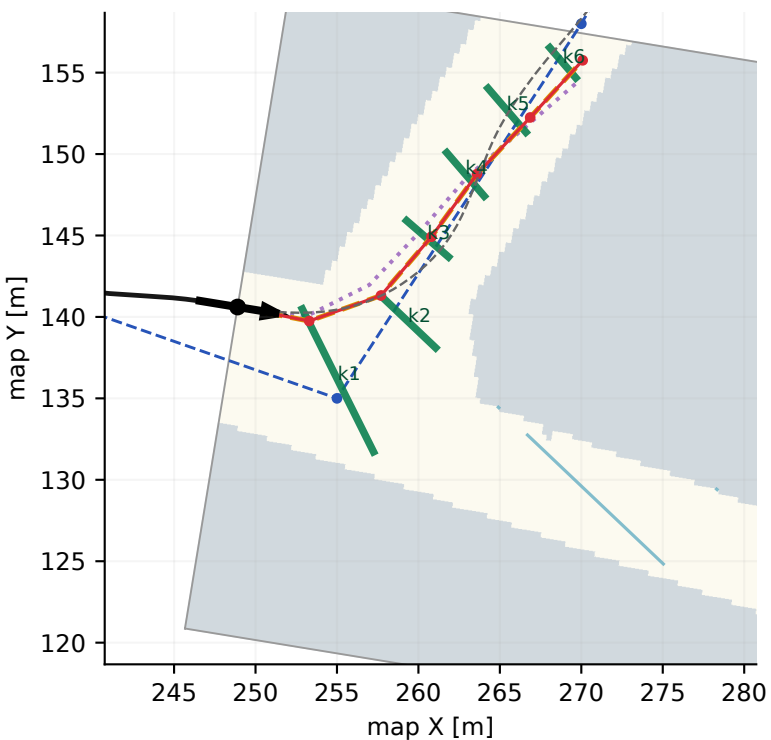


--- SD waypoints --- Allowed interval ● Published plan — Executed past trajectory
... Carried local reference --- Selected interval --- Recorded lower path --- Later execution (hindsight)

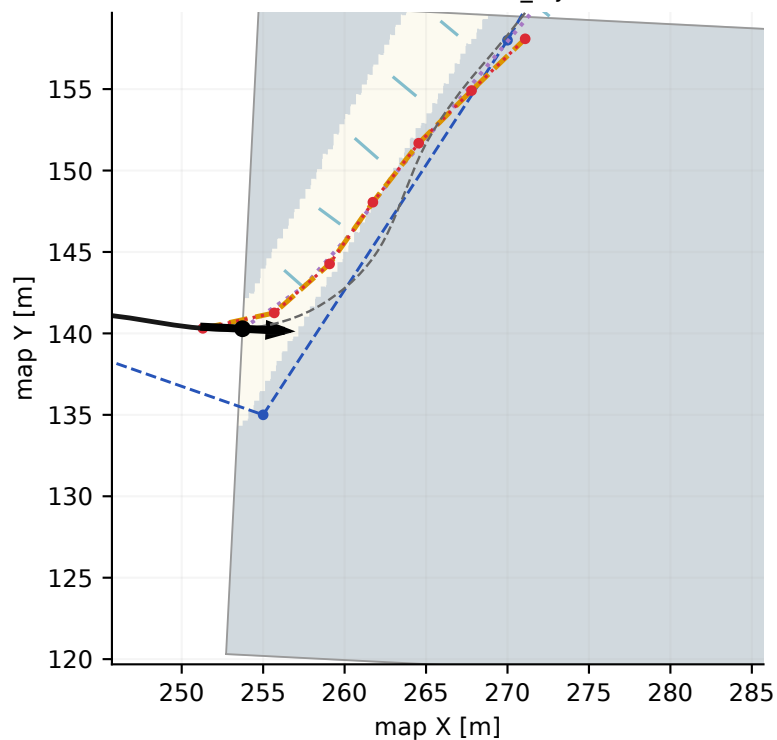
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 3 / A04 / failure — zoom 20261008_185426_scenario3_72c4df

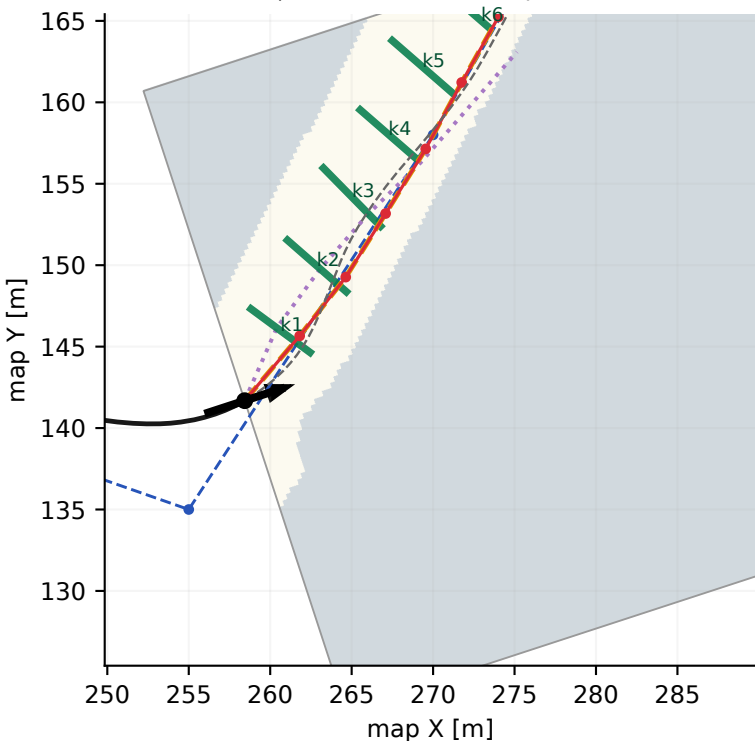
1. 18:57:53.526 KST / plan 398
N=6, max candidates=3 / success



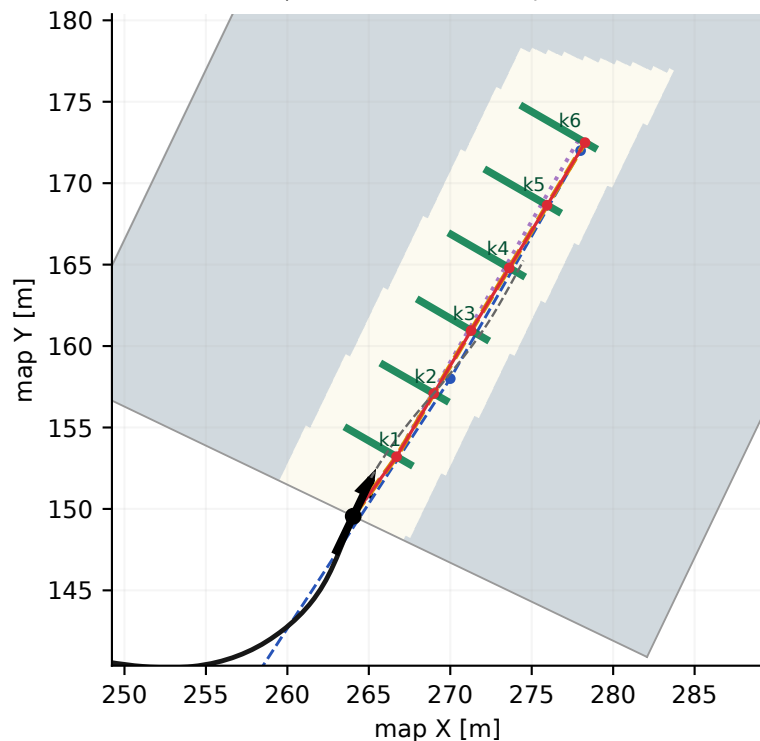
2. 18:57:54.526 KST / plan 400
N=6, max candidates=1 / solver_rejected; retained



3. 18:57:55.526 KST / plan 402
N=6, max candidates=1 / success

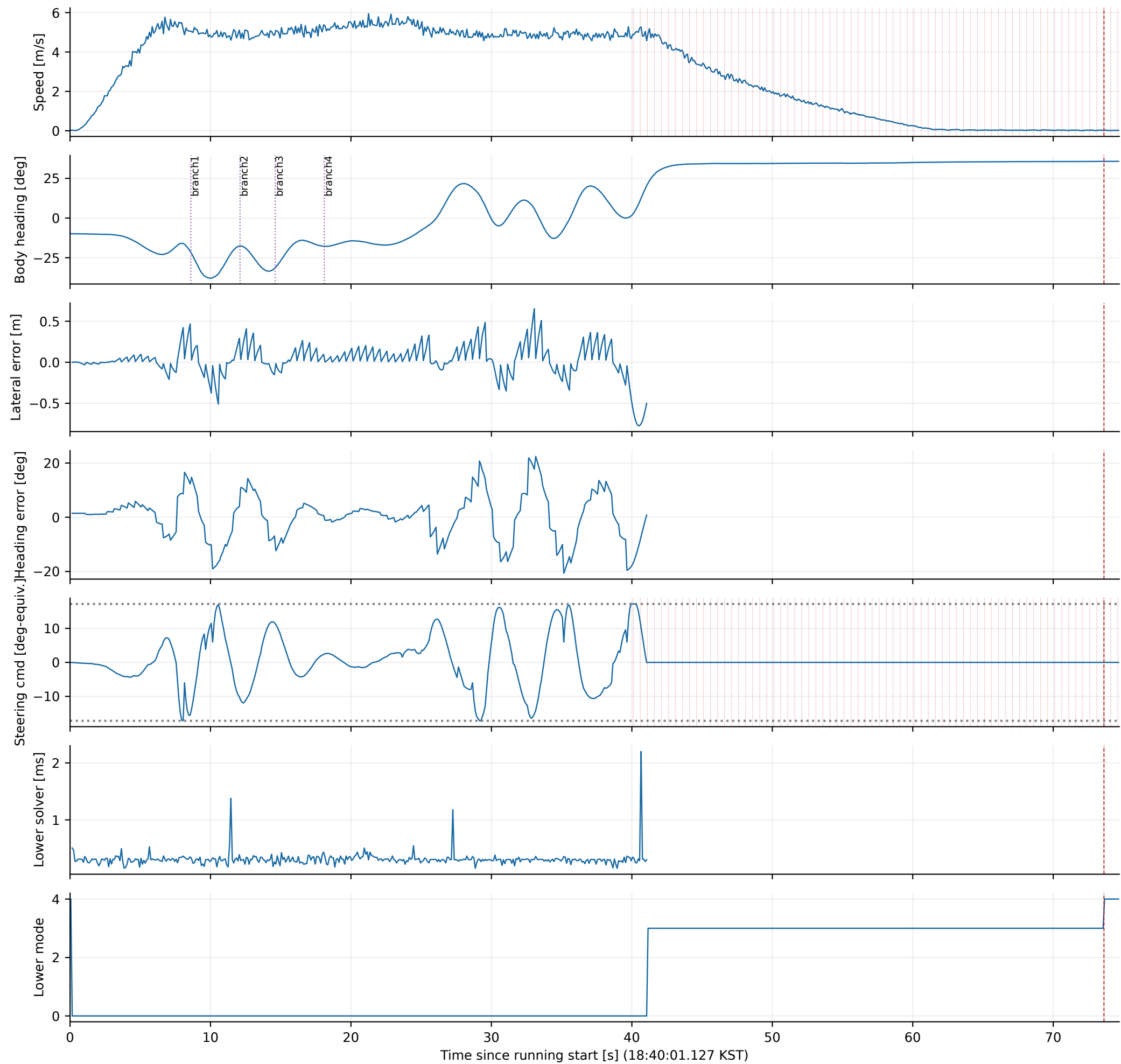


4. 18:57:57.526 KST / plan 406
N=6, max candidates=1 / success



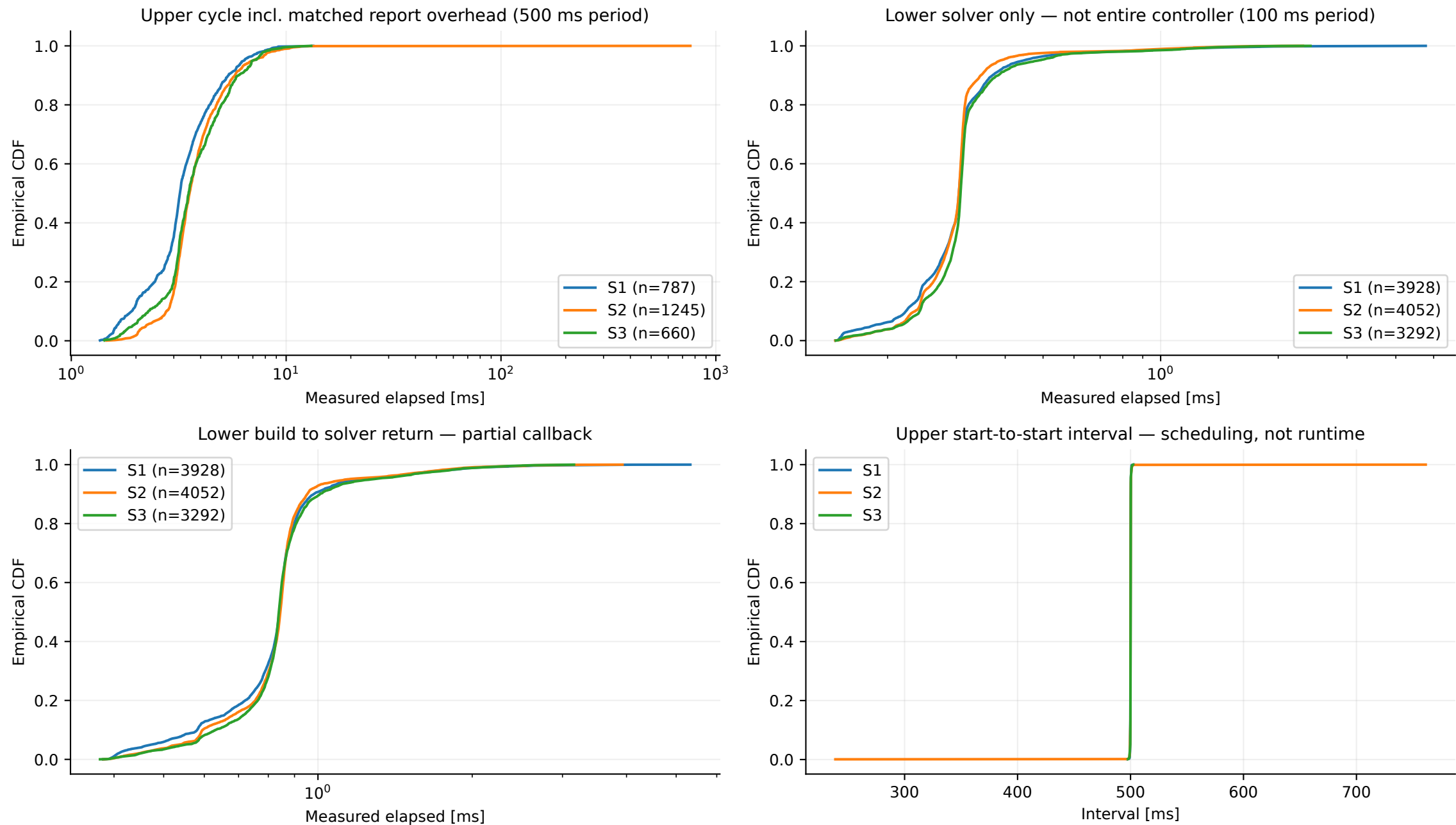
Recorded plans/paths; grid matched retrospectively by receipt time. Black vehicle axis: length 4.7 m; width unavailable (not a footprint).

Scenario 2 / A01: stopped_10s
20261008_183952_scenario2_5524df



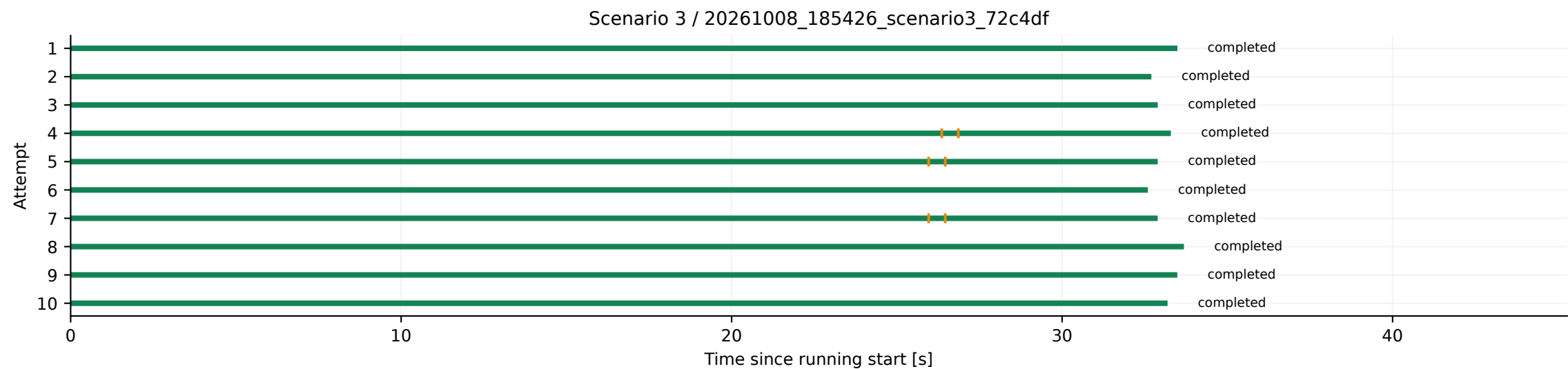
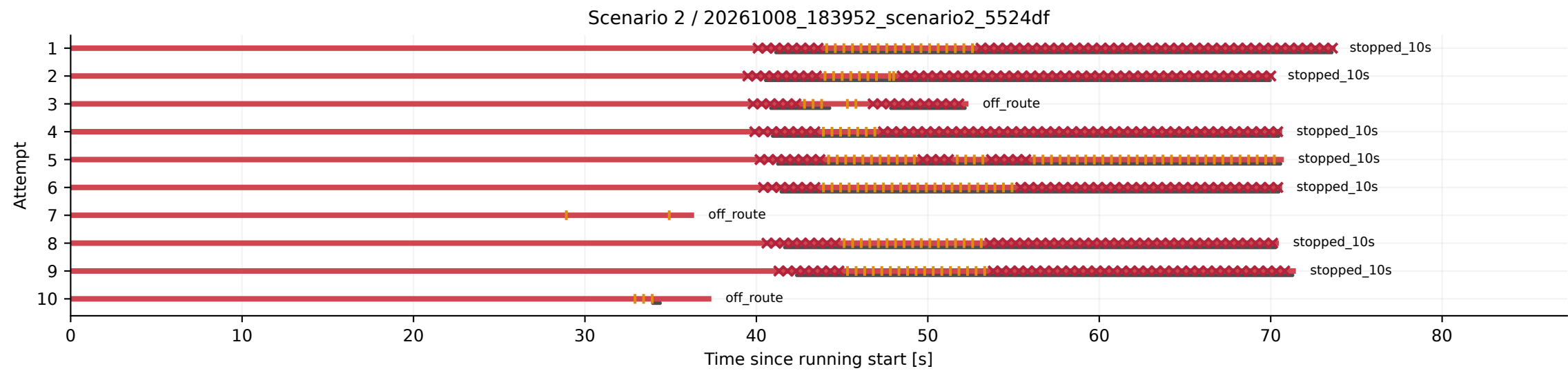
Errors are relative to the current lower path; neutral placeholder errors excluded. Red lines: upper failure. Dashed boundary: runner decision. No measured steering.

Runtime evidence during running phases — failures included when solver was called



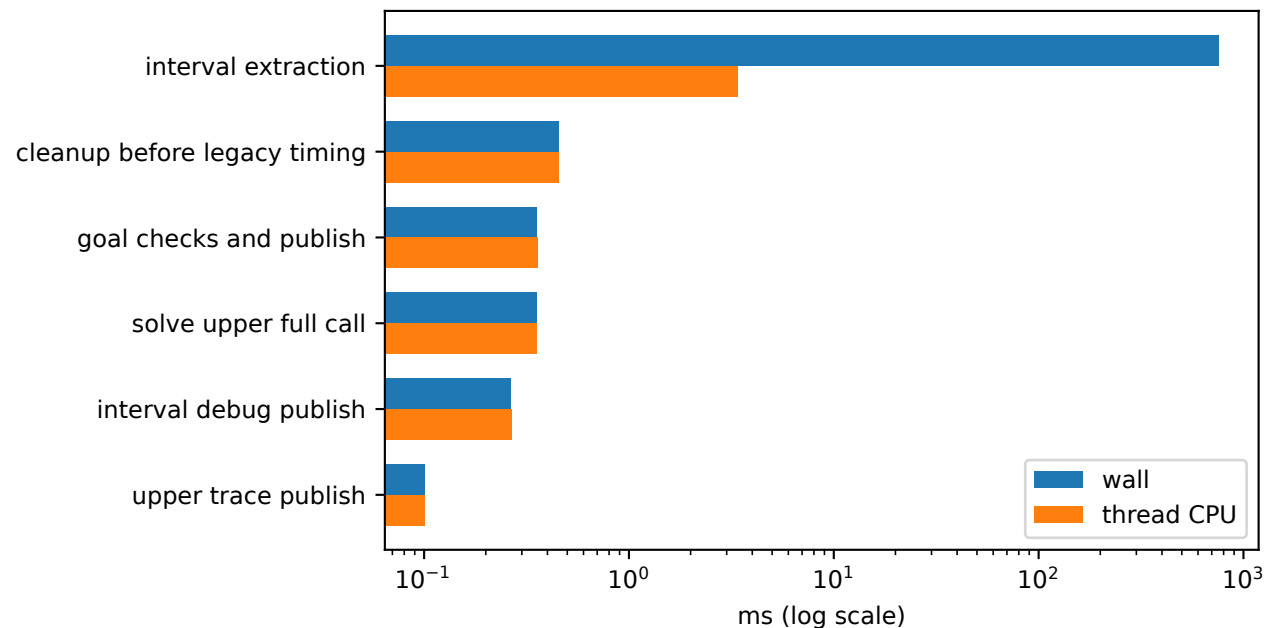
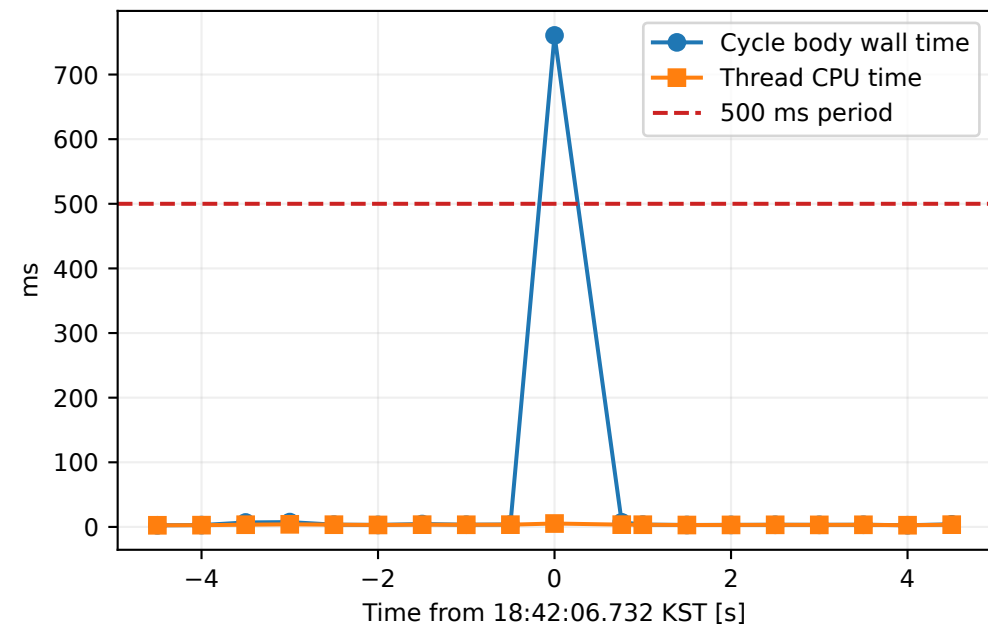
Upper measured cycle excludes callback dispatch delay. Lower complete callback was not instrumented. Observed maxima are not WCET bounds.

All failures and interruptions during the 30 primary trials



Green/red bars: goal approach / unsuccessful termination. Orange |: rejected solve. Red x: no candidate. Gray dots: lower neutral mode 3.

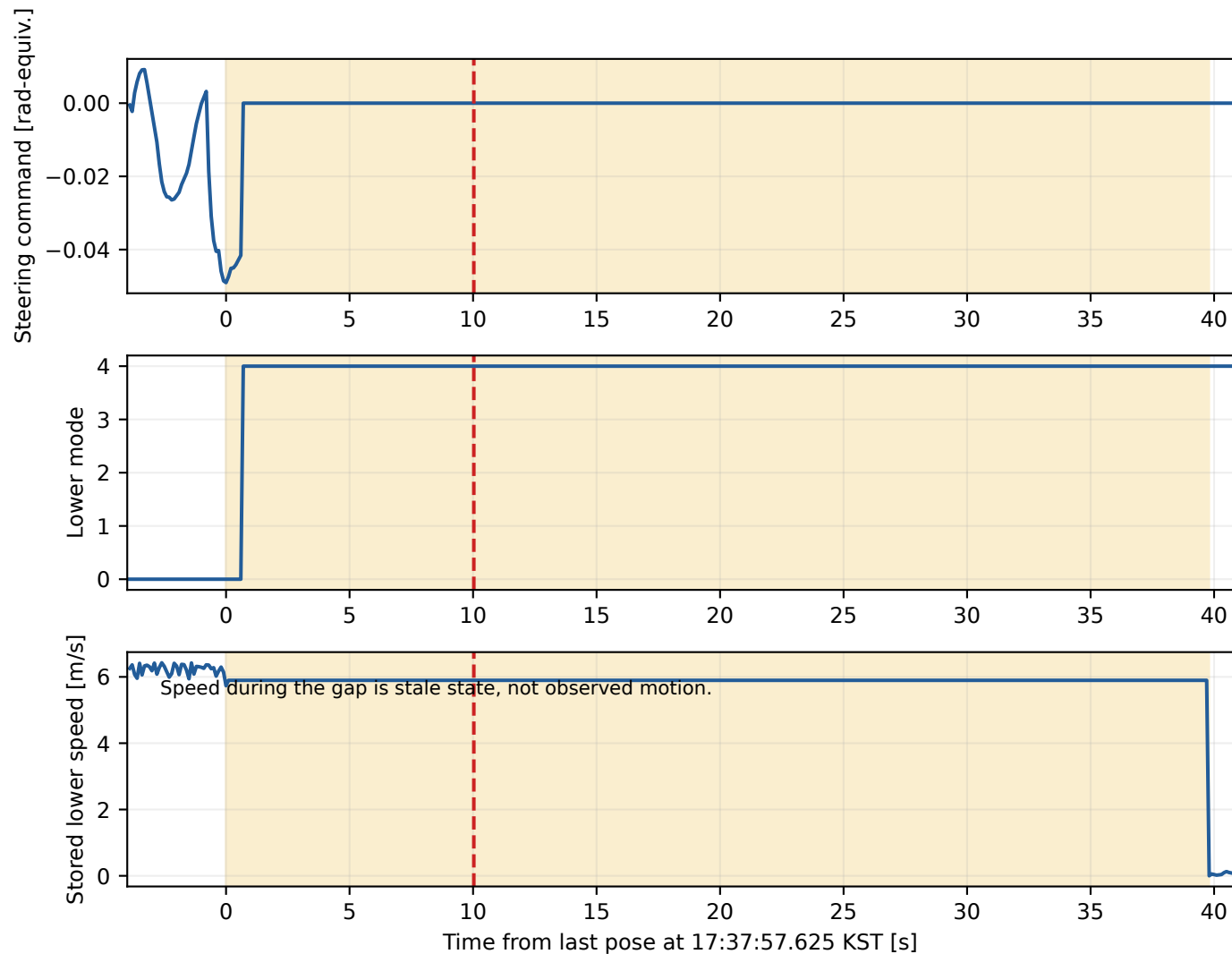
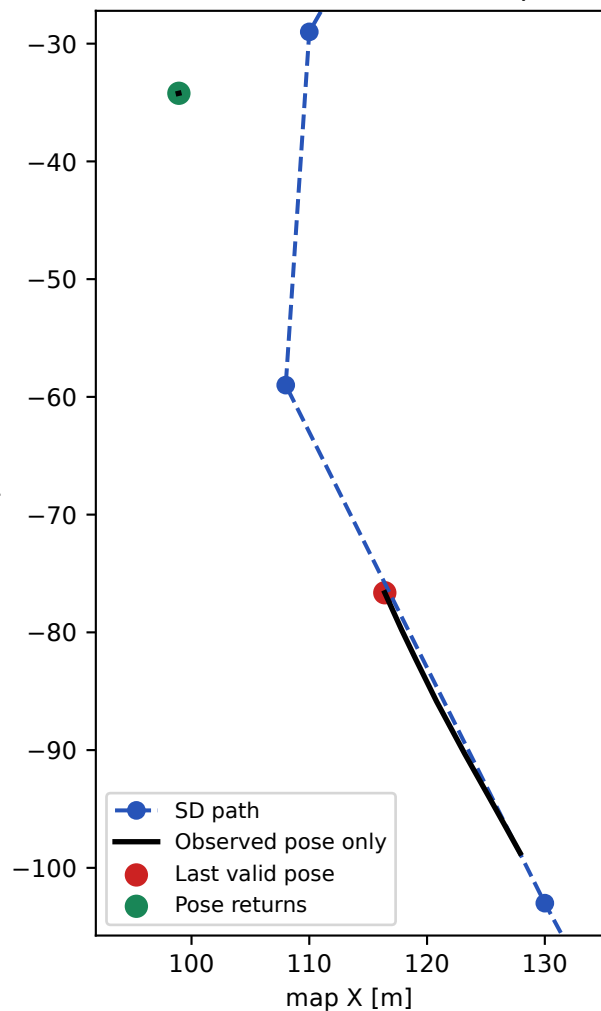
S2 A02: observed 500 ms cycle overrun / plan 254
20261008_183952_scenario2_5524df — 18:42:06.732 KST



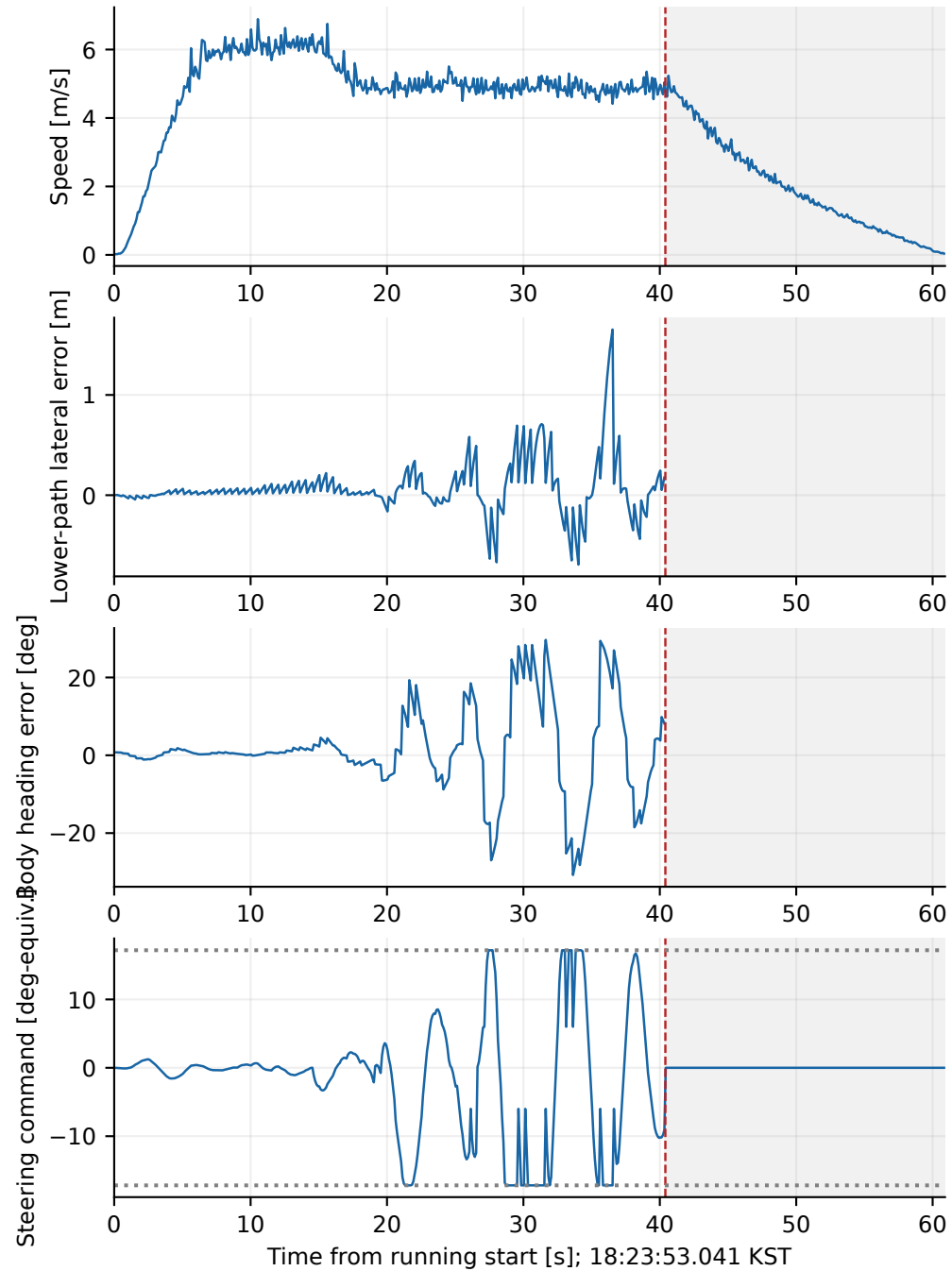
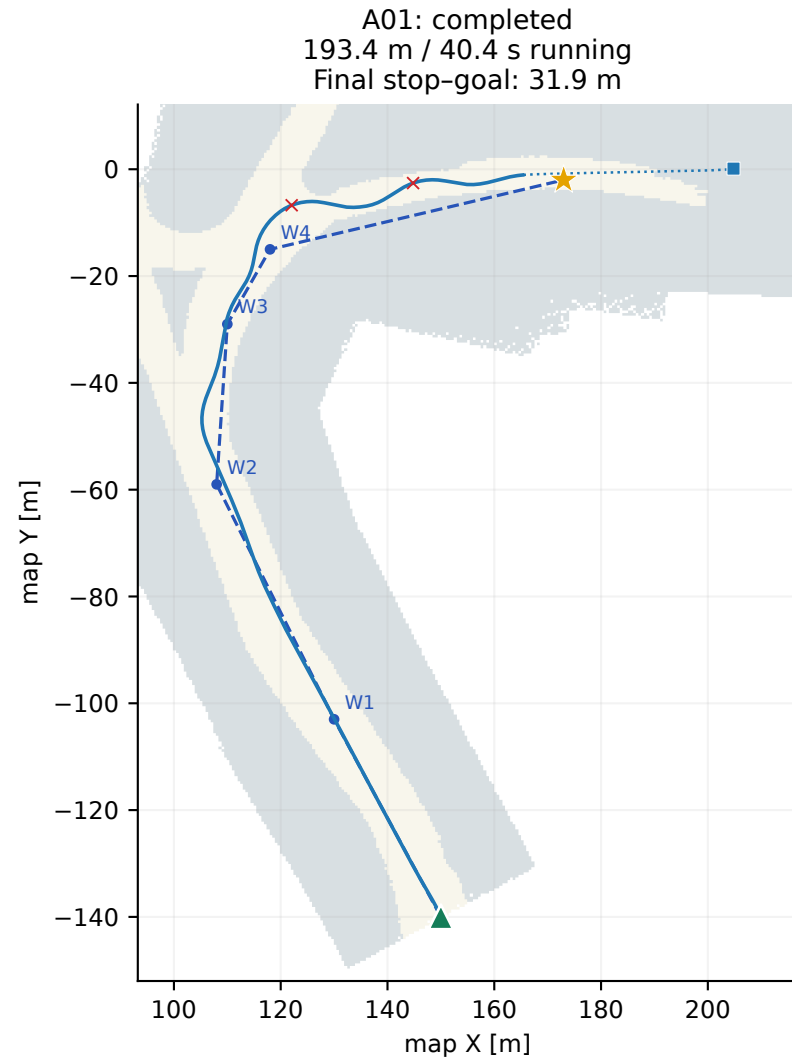
Body 760.540 ms; matched own-report overhead 0.272 ms. Wall/CPU difference does not establish the waiting or scheduling cause.

Historical input interruption: S1 A05 — 20261008_173306_scenario1_756dad
Pose gap 39.789 s; pose_timeout at 17:38:07.656 KST

Unobserved motion is not interpolated

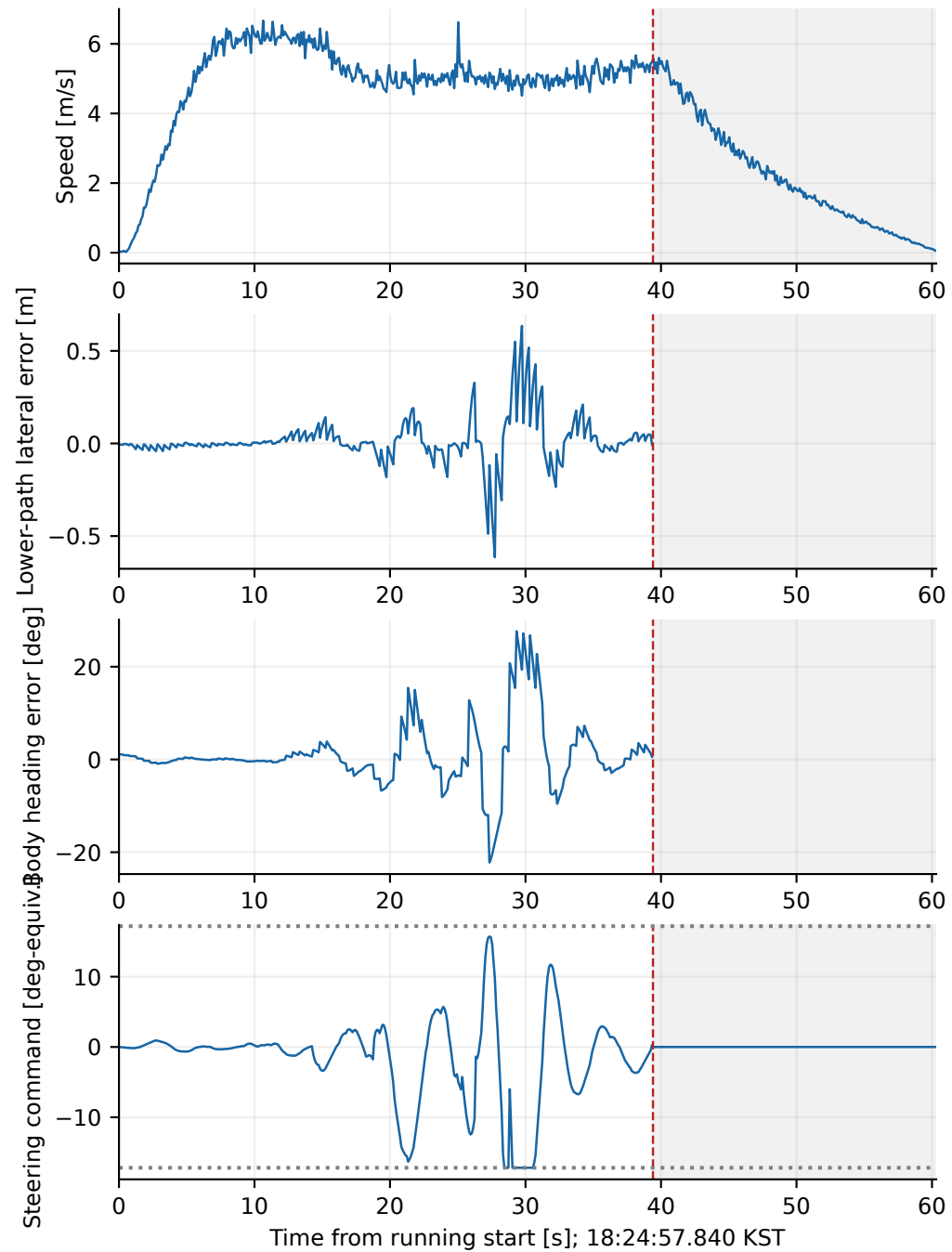
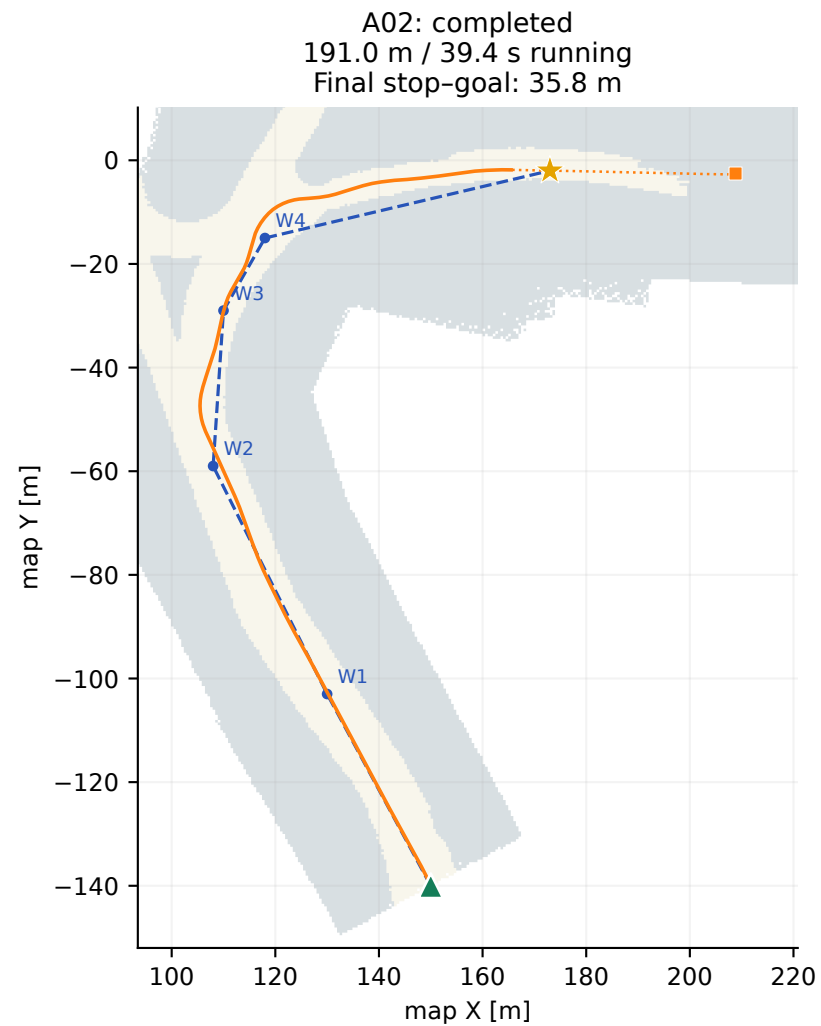


Scenario 1 / Attempt 01 — 20261008_182344_scenario1_0cefb5



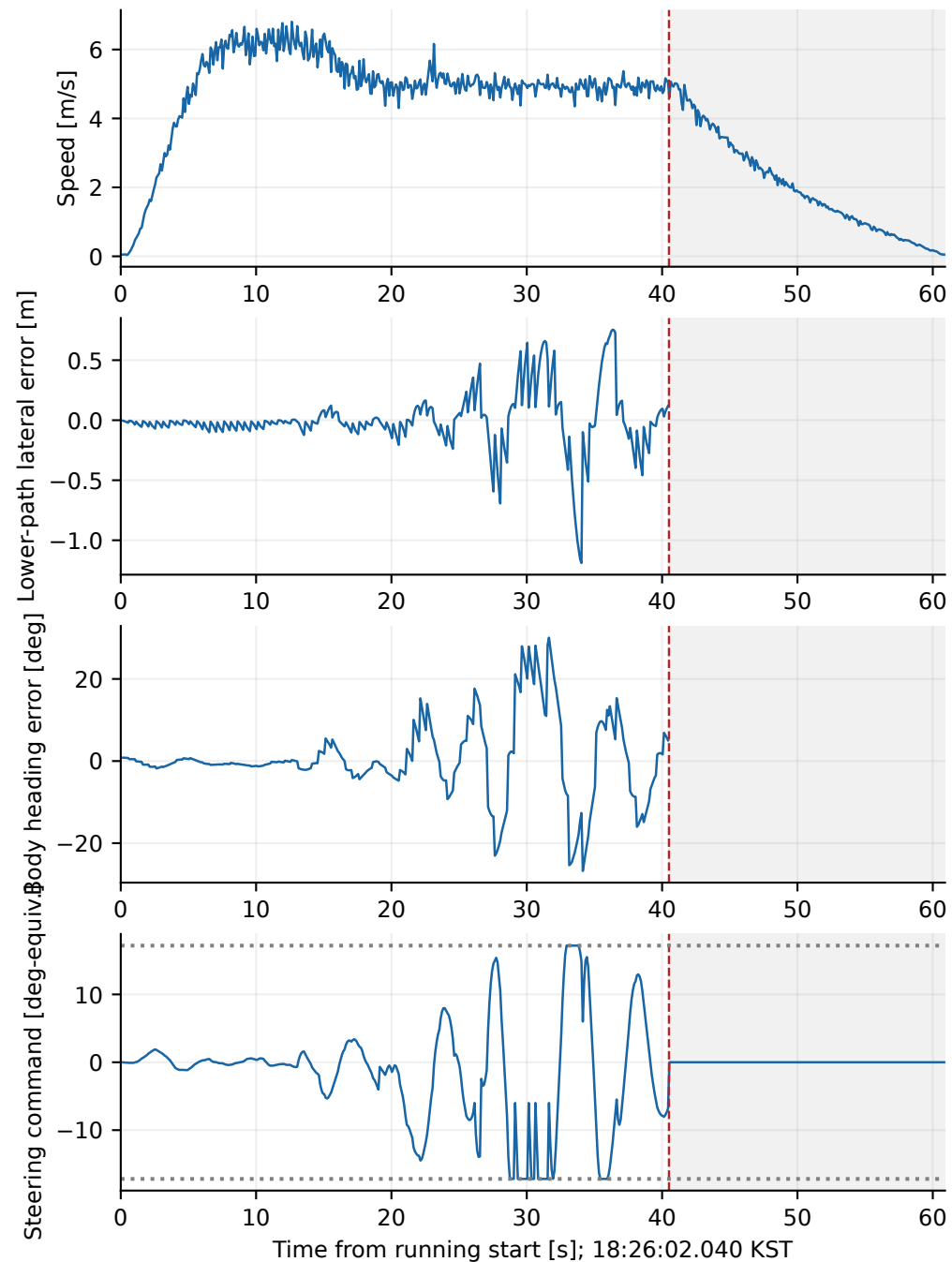
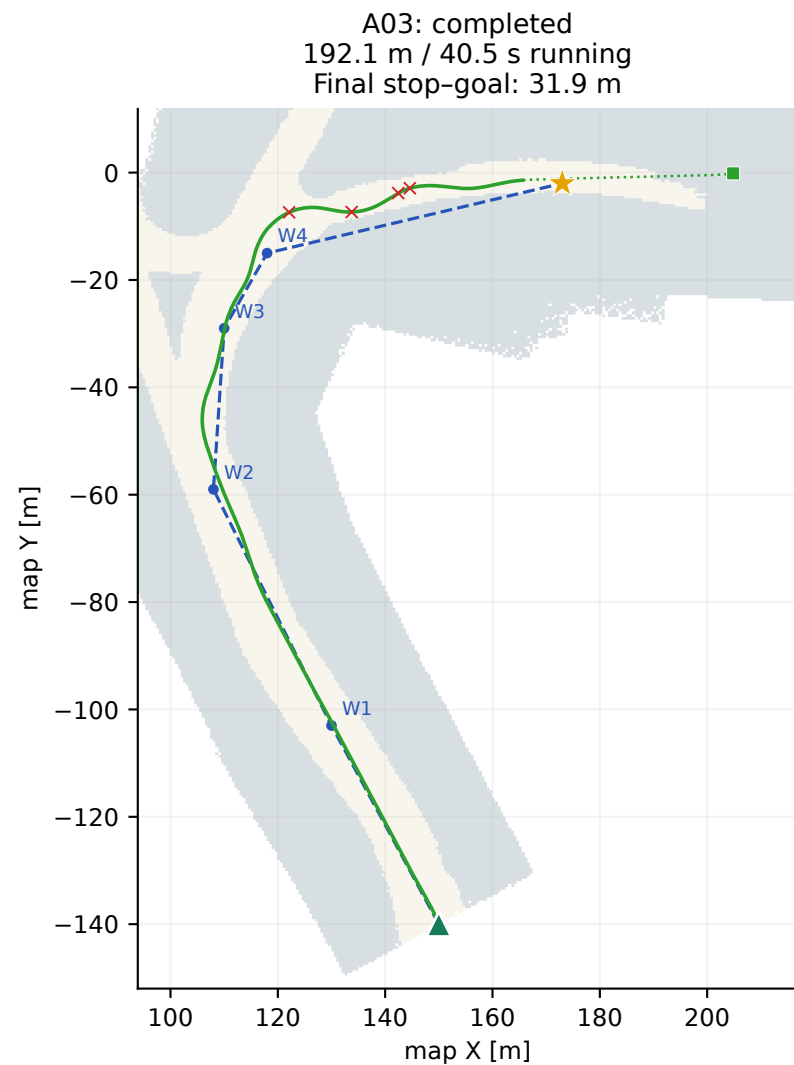
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 02 — 20261008_182344_scenario1_0cefb5



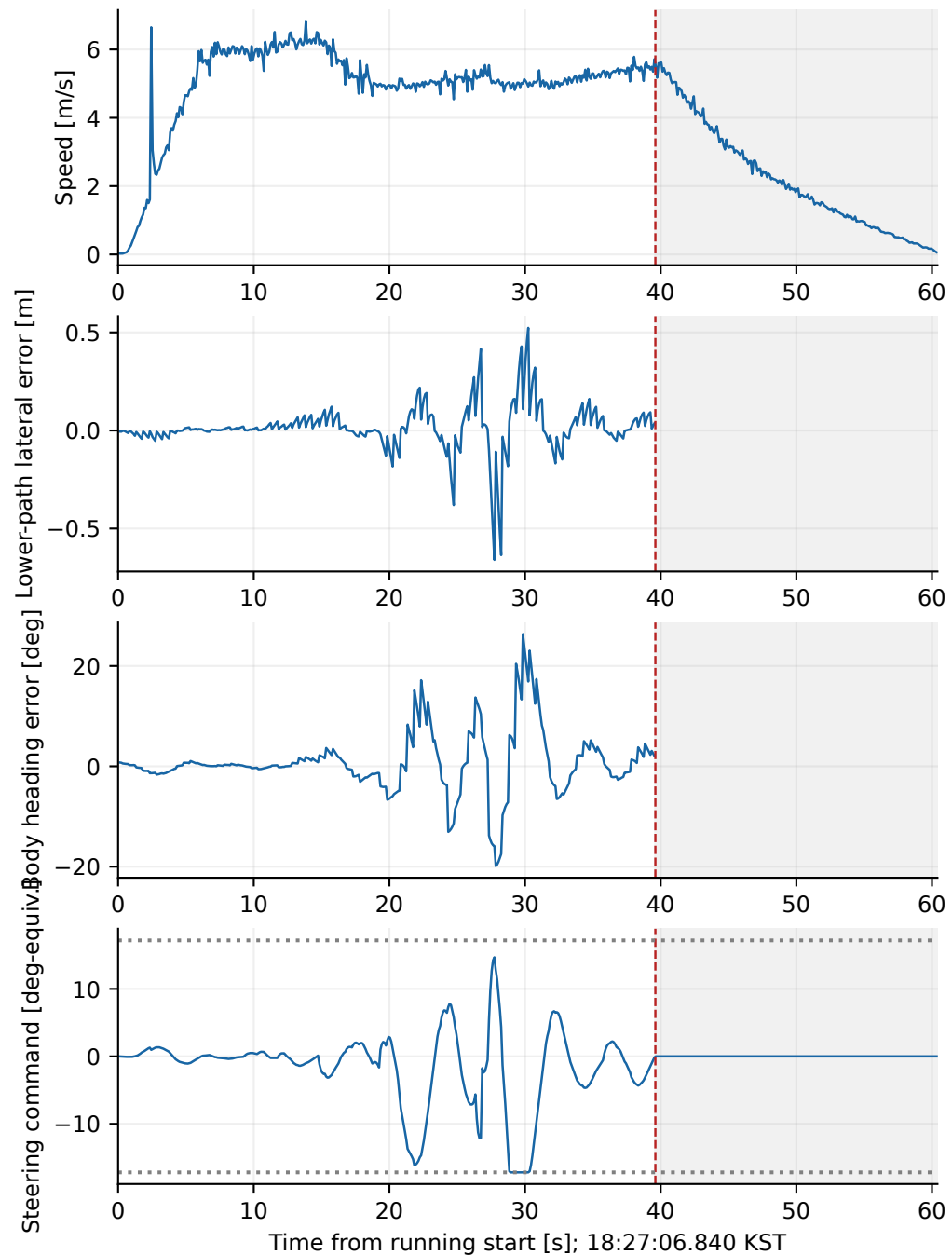
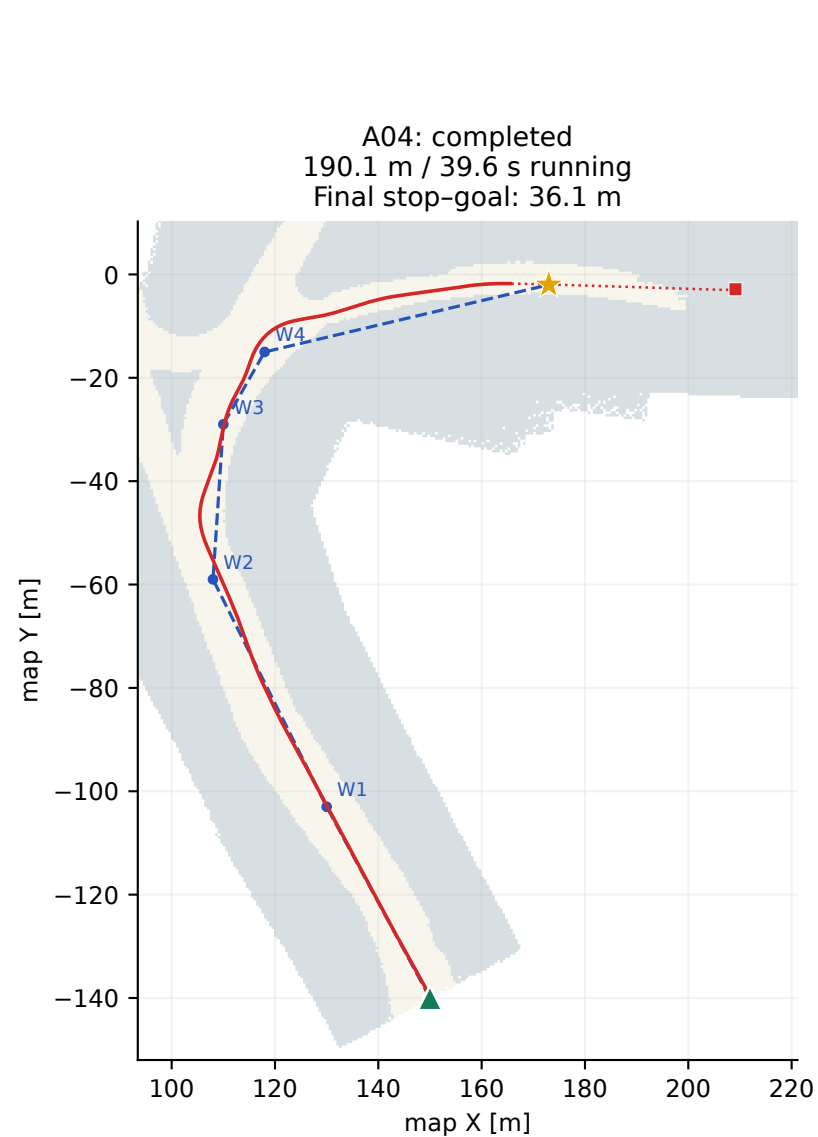
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 03 — 20261008_182344_scenario1_0cefb5



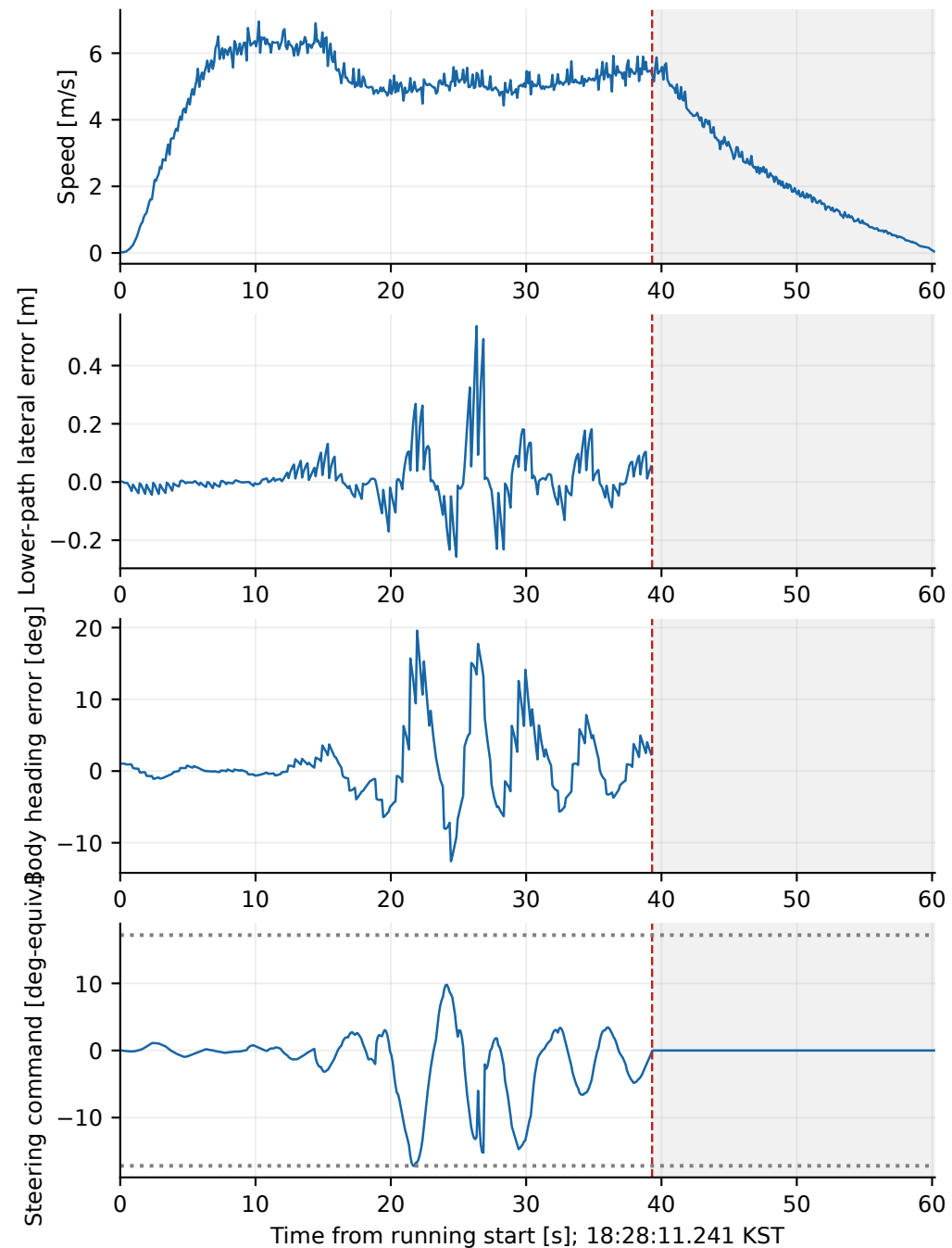
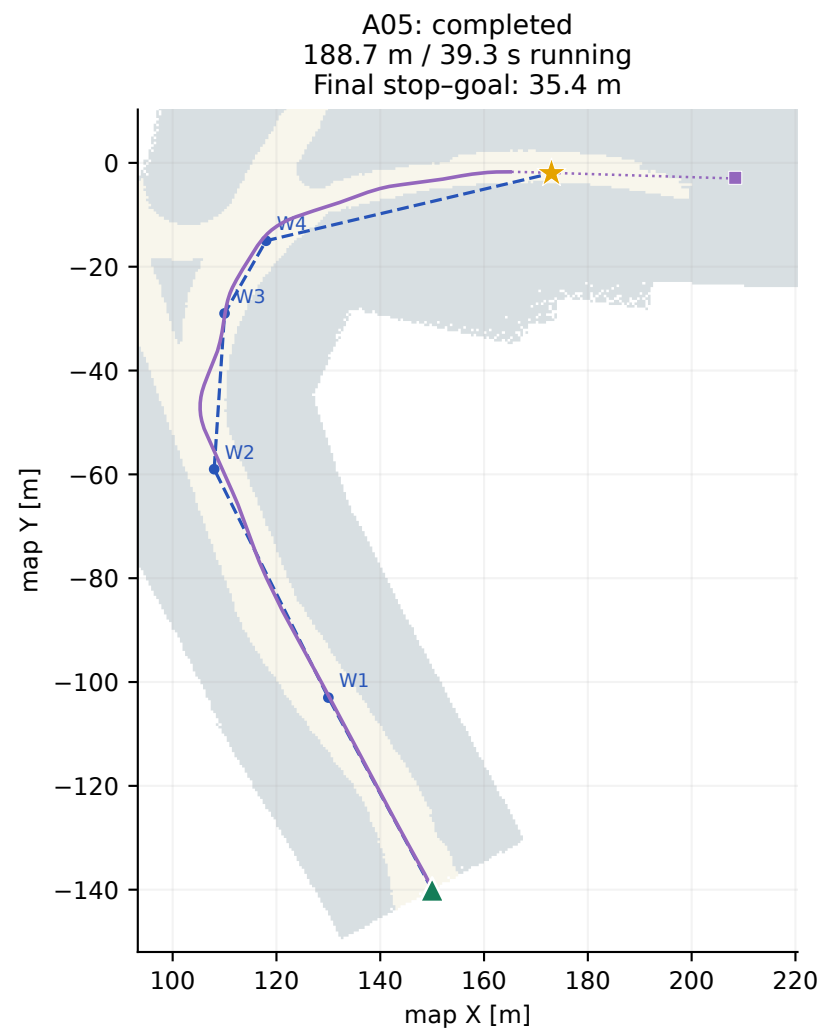
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 04 — 20261008_182344_scenario1_0cefb5



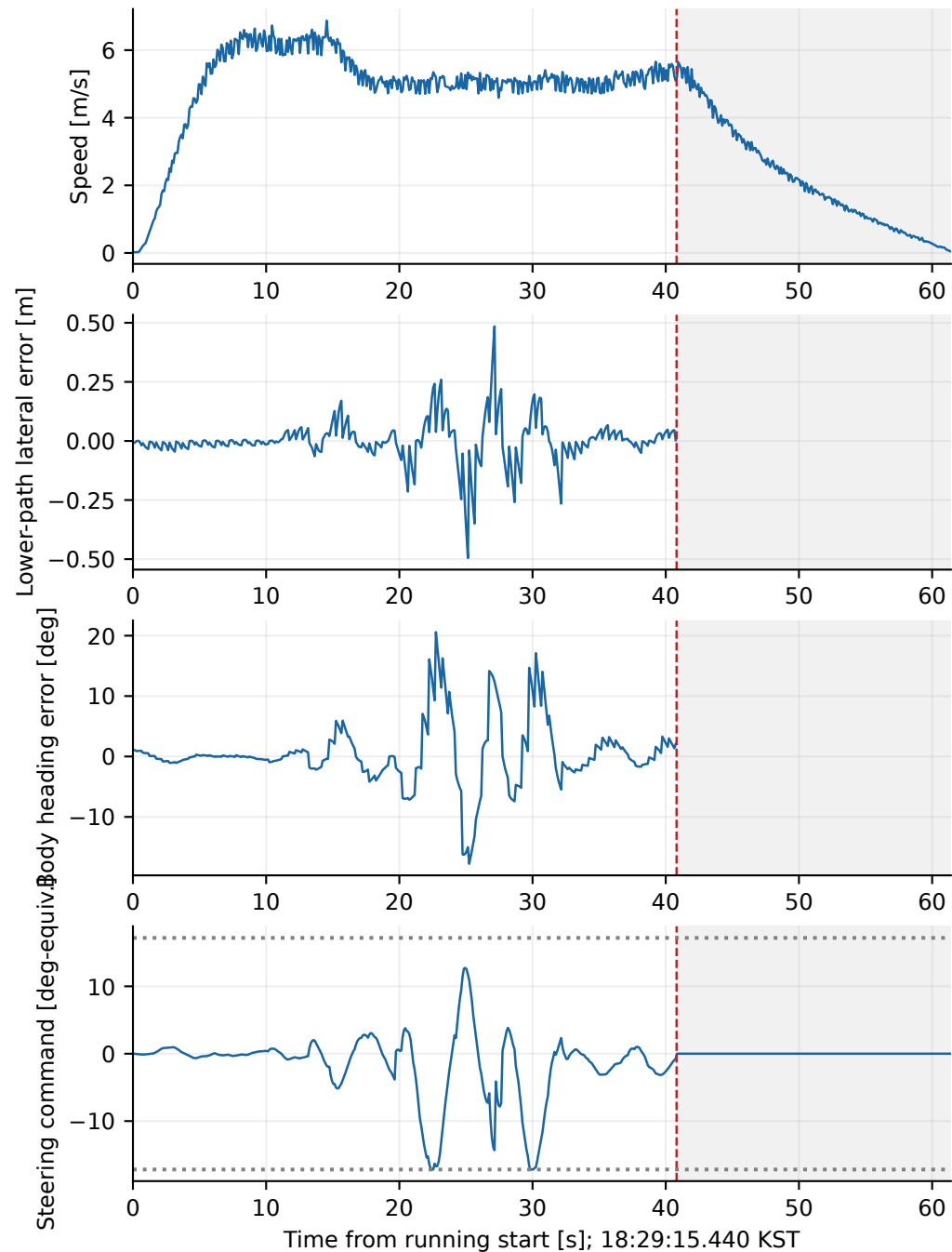
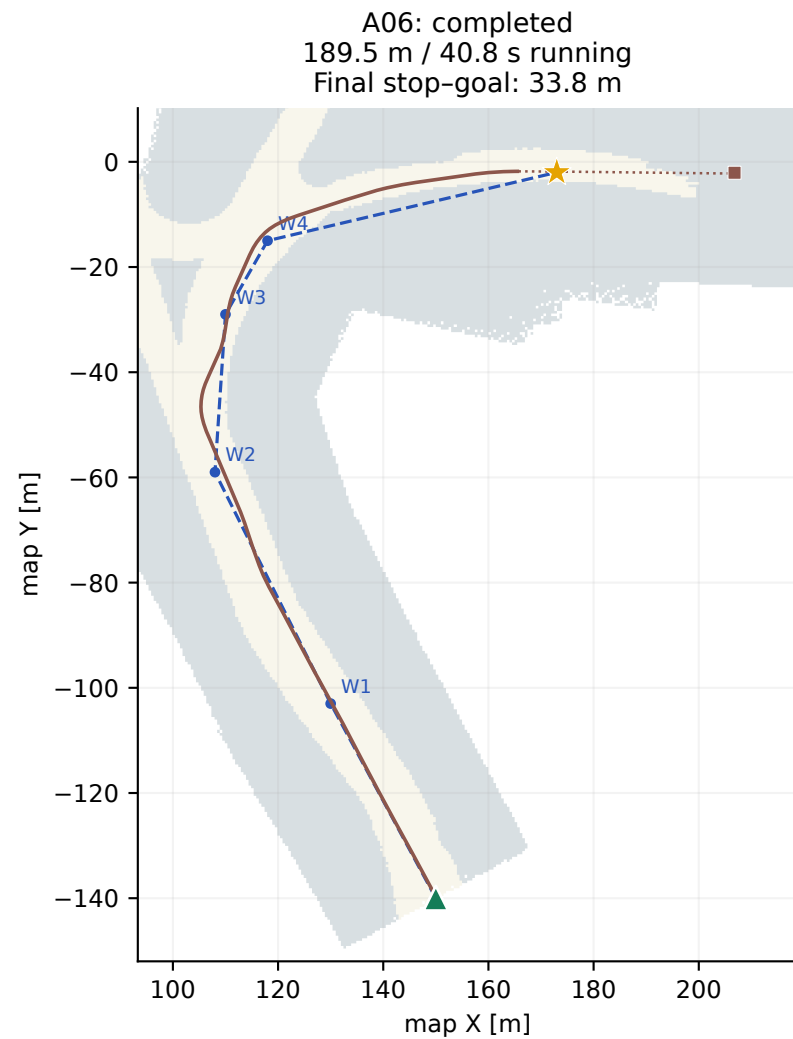
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 05 — 20261008_182344_scenario1_0cefb5



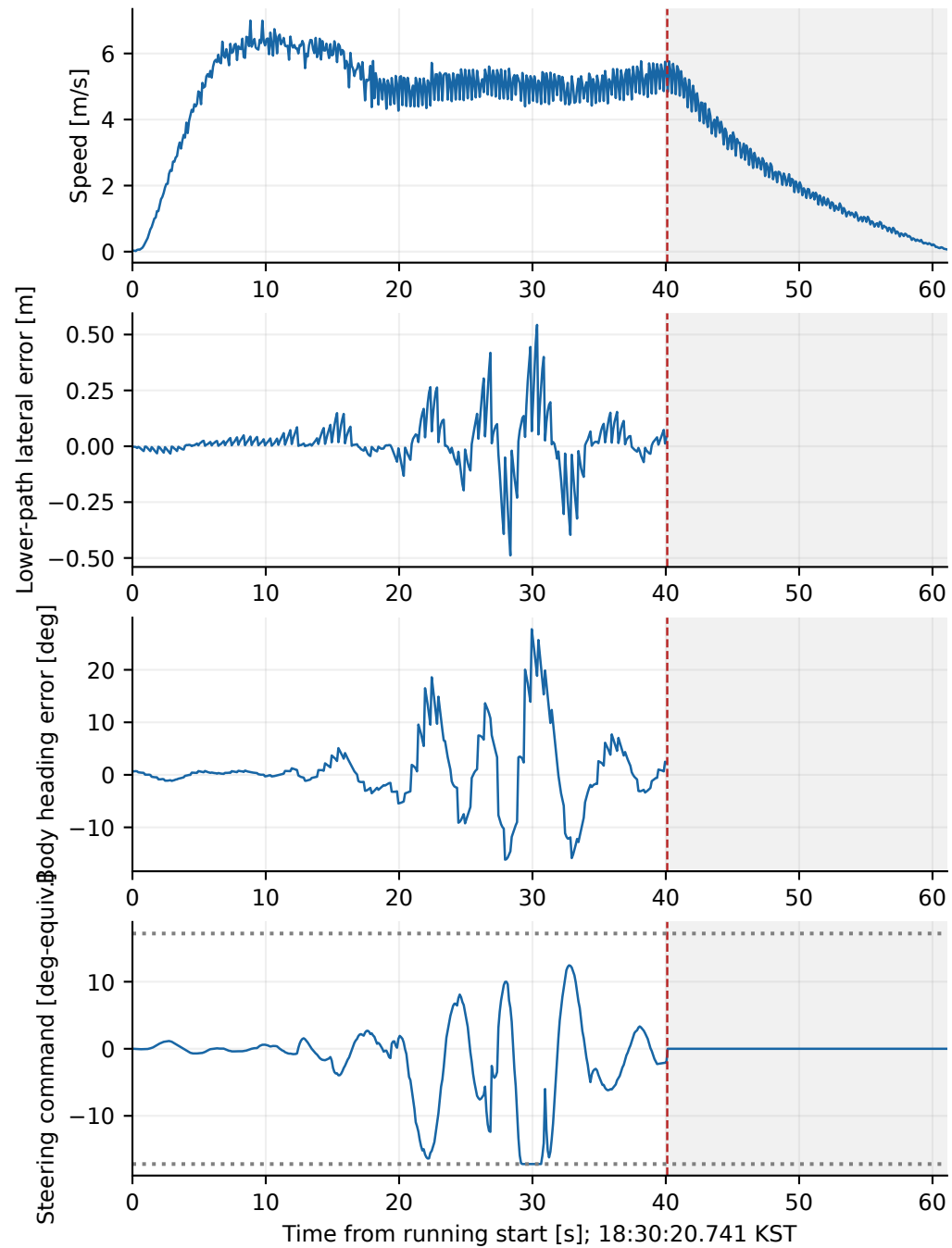
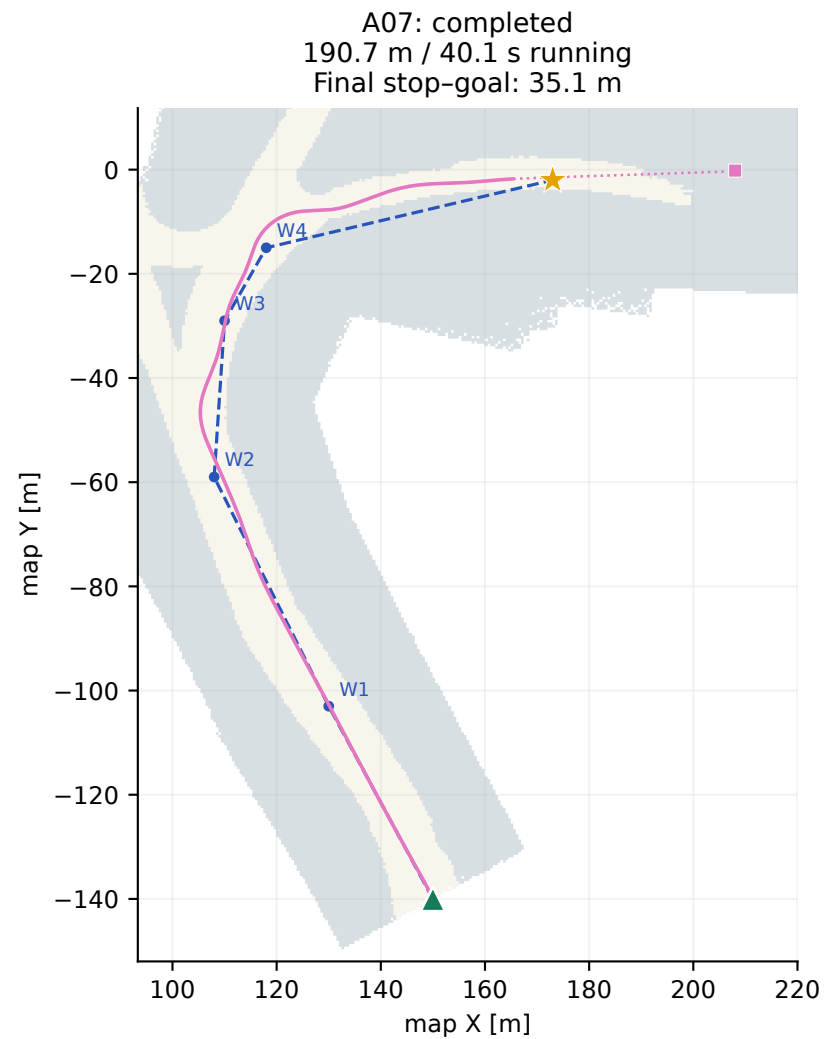
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 06 — 20261008_182344_scenario1_0cefb5



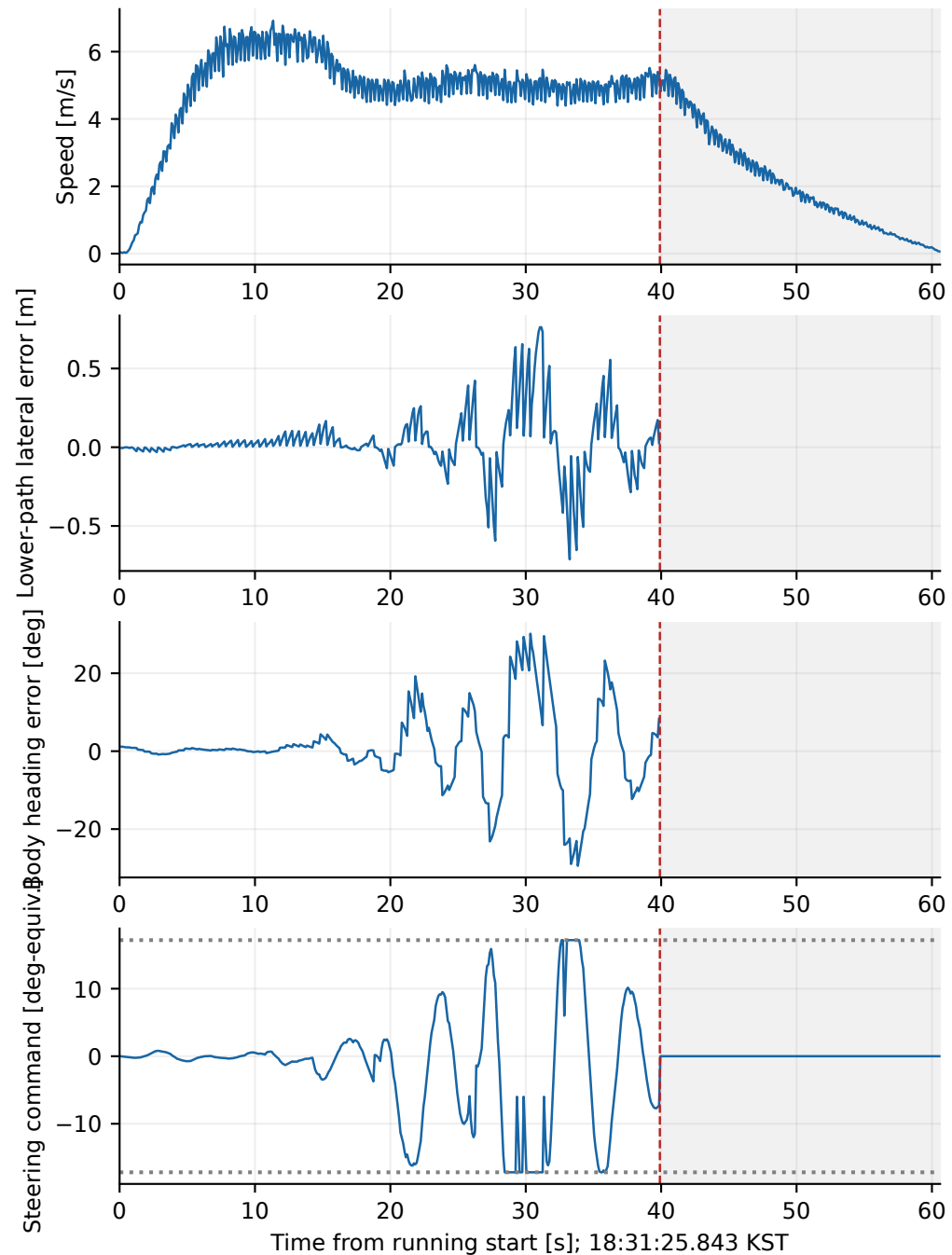
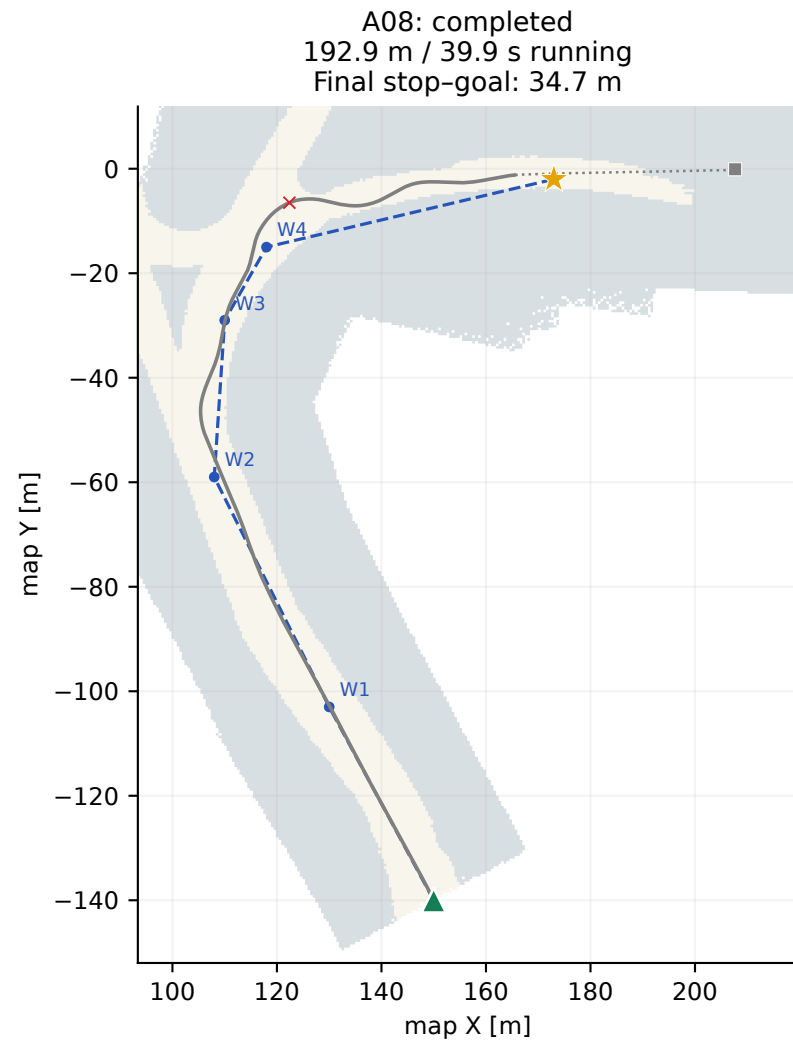
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 07 — 20261008_182344_scenario1_0cefb5



Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 08 — 20261008_182344_scenario1_0cefb5

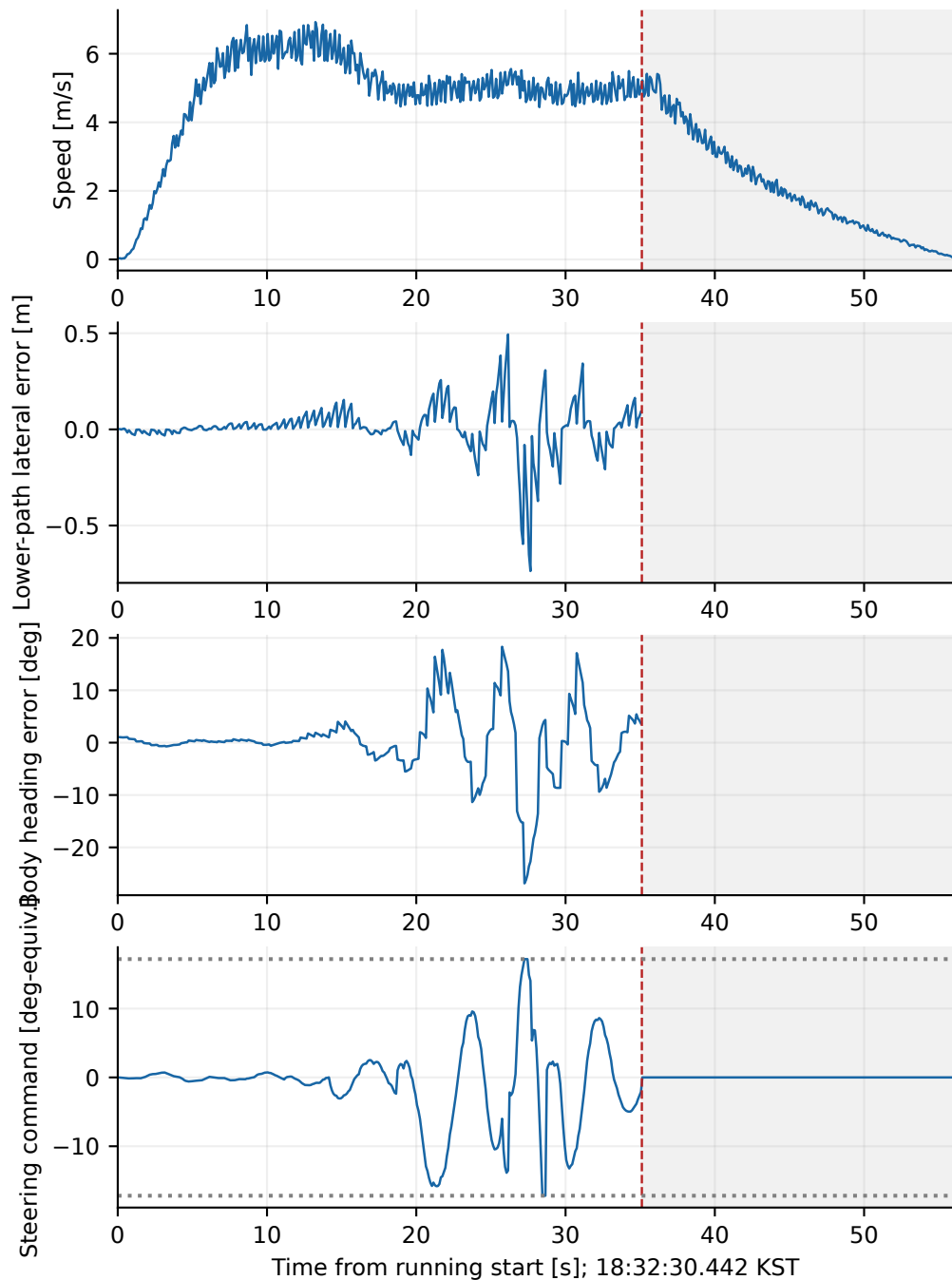
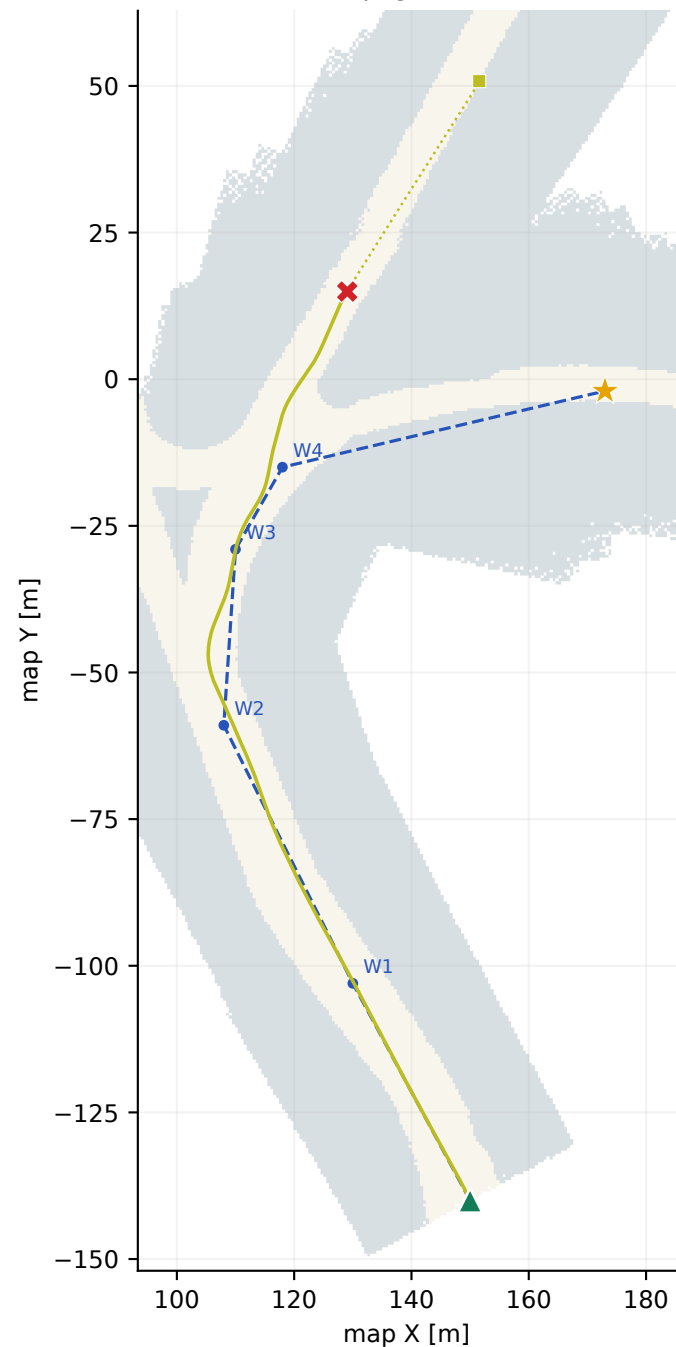


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 09 — 20261008_182344_scenario1_0cefb5

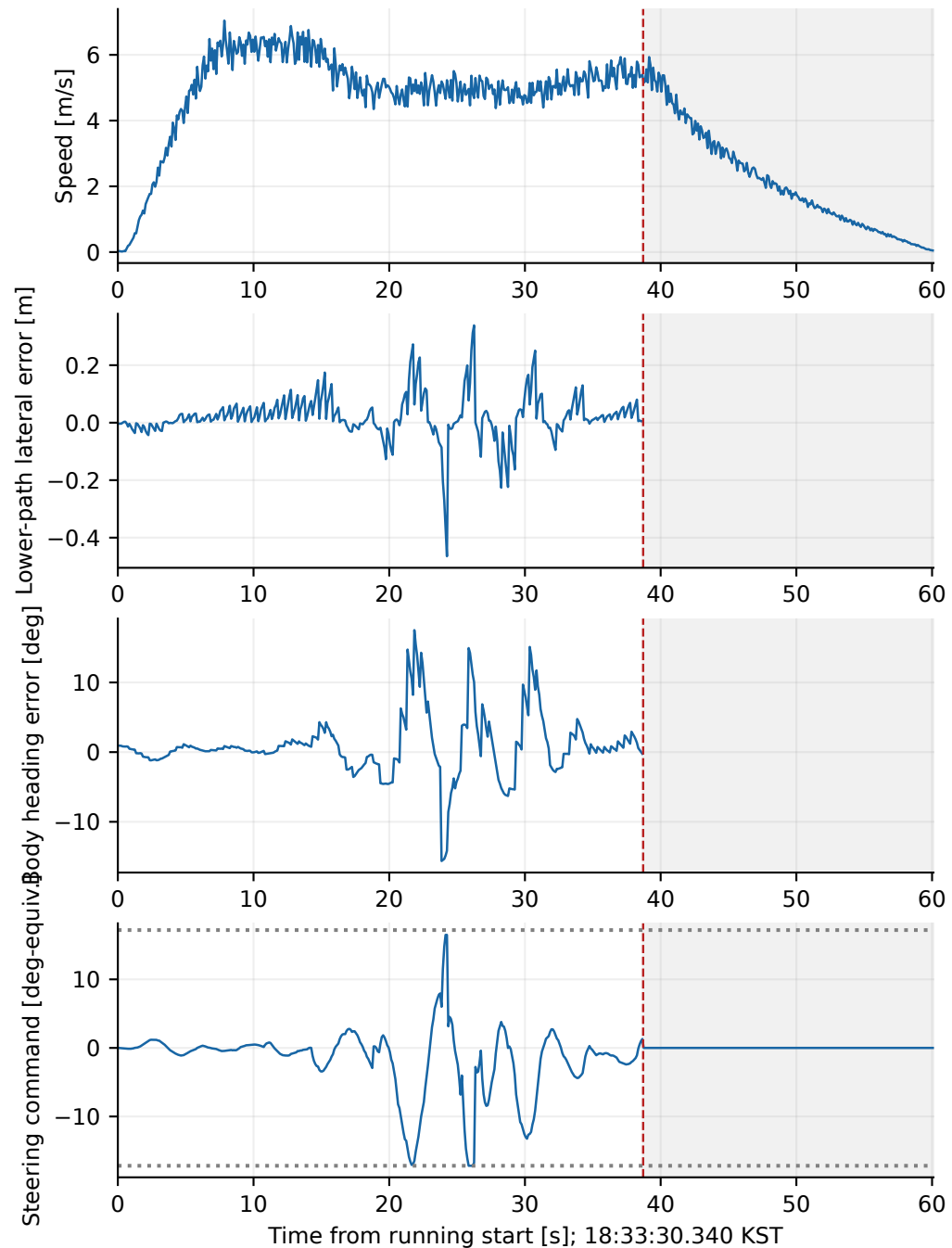
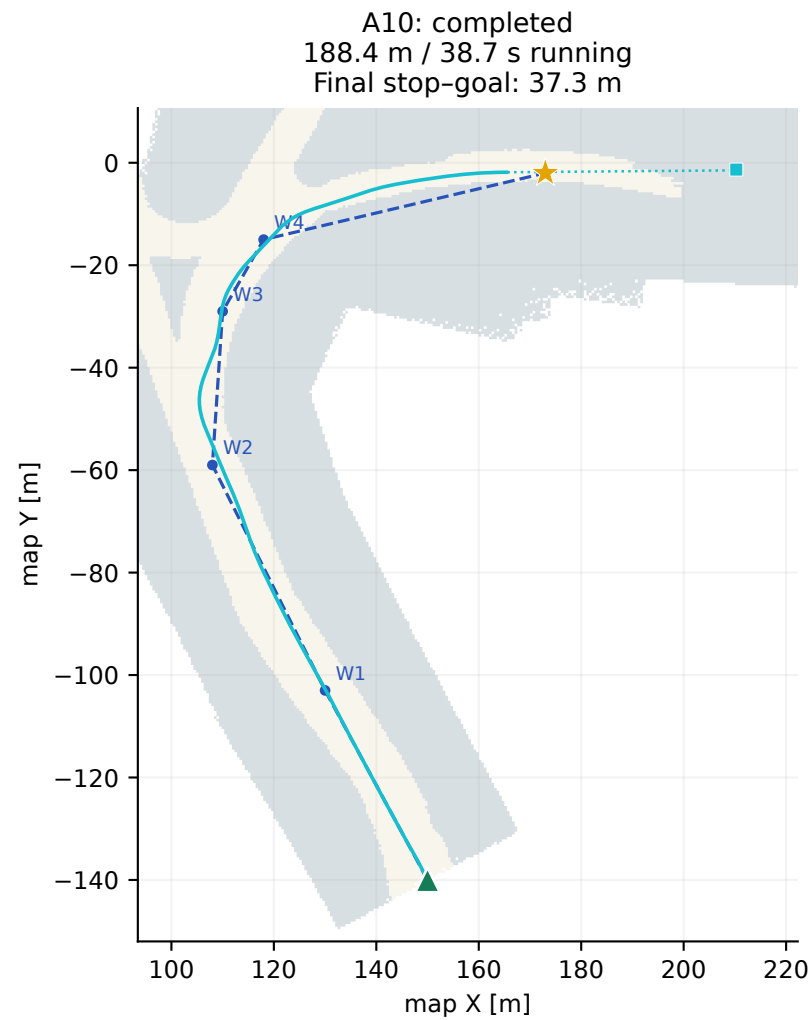
170.5 m / 35.1 s running

Final stop-goal: 57.0 m



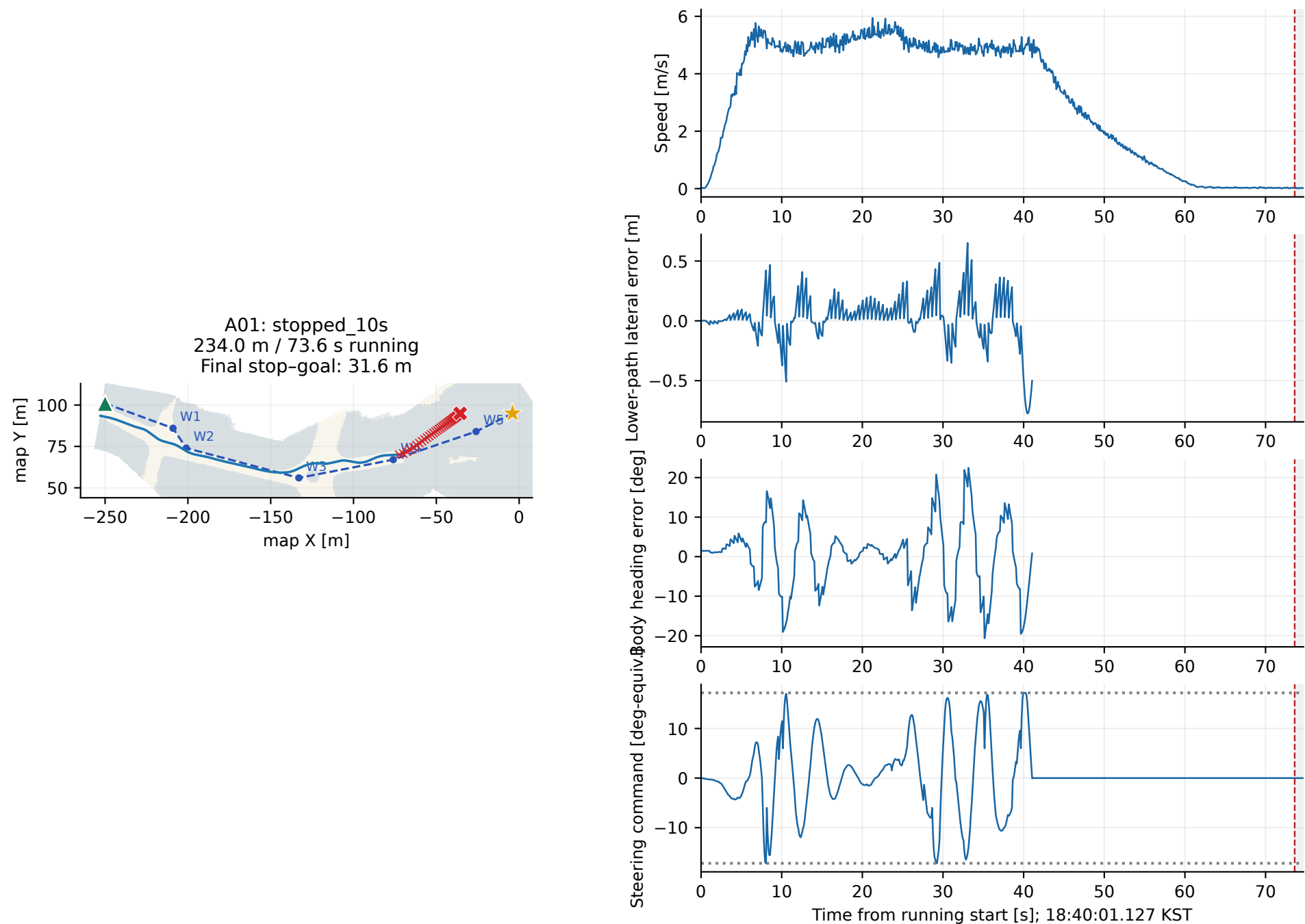
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 1 / Attempt 10 — 20261008_182344_scenario1_0cefb5

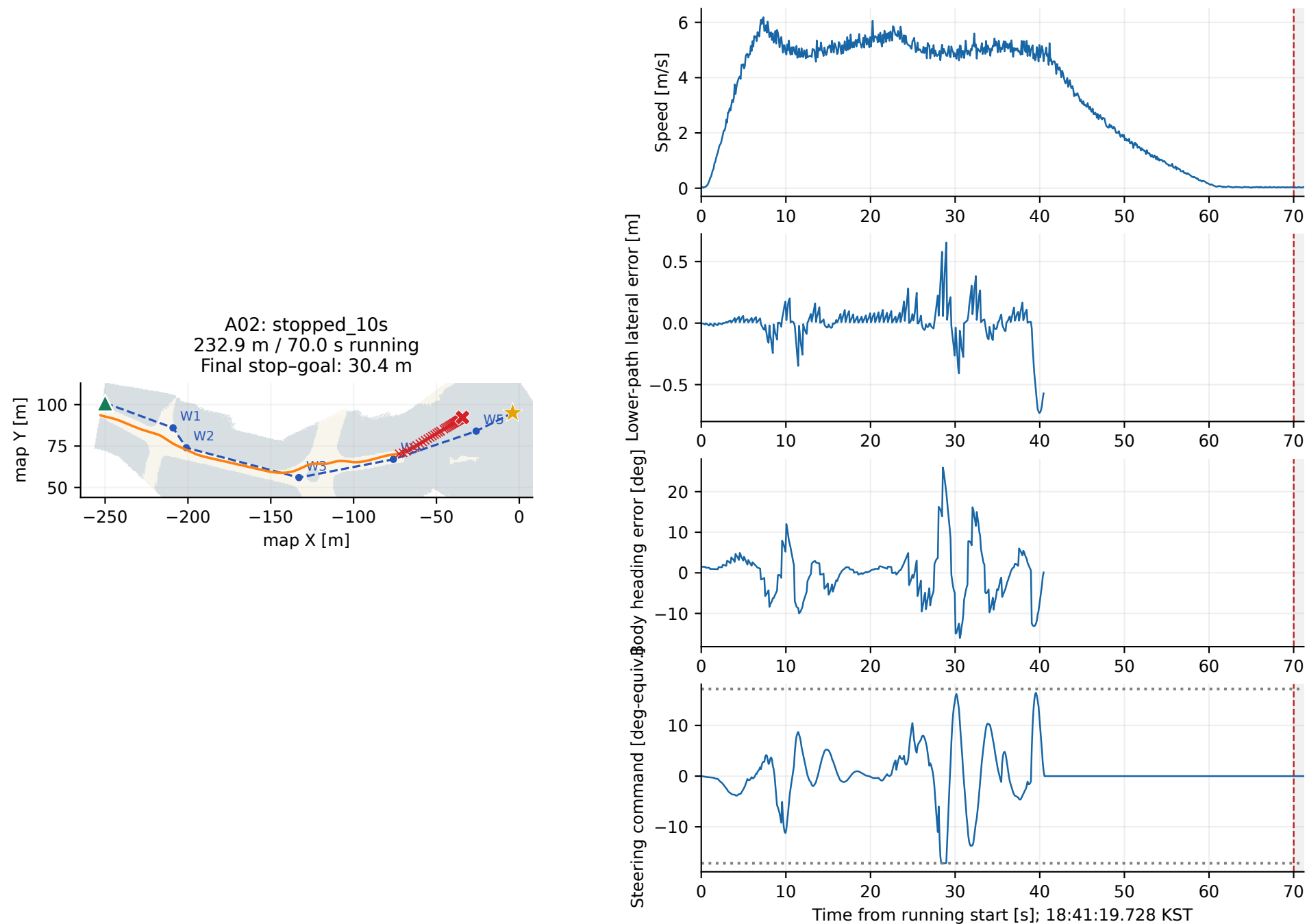


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

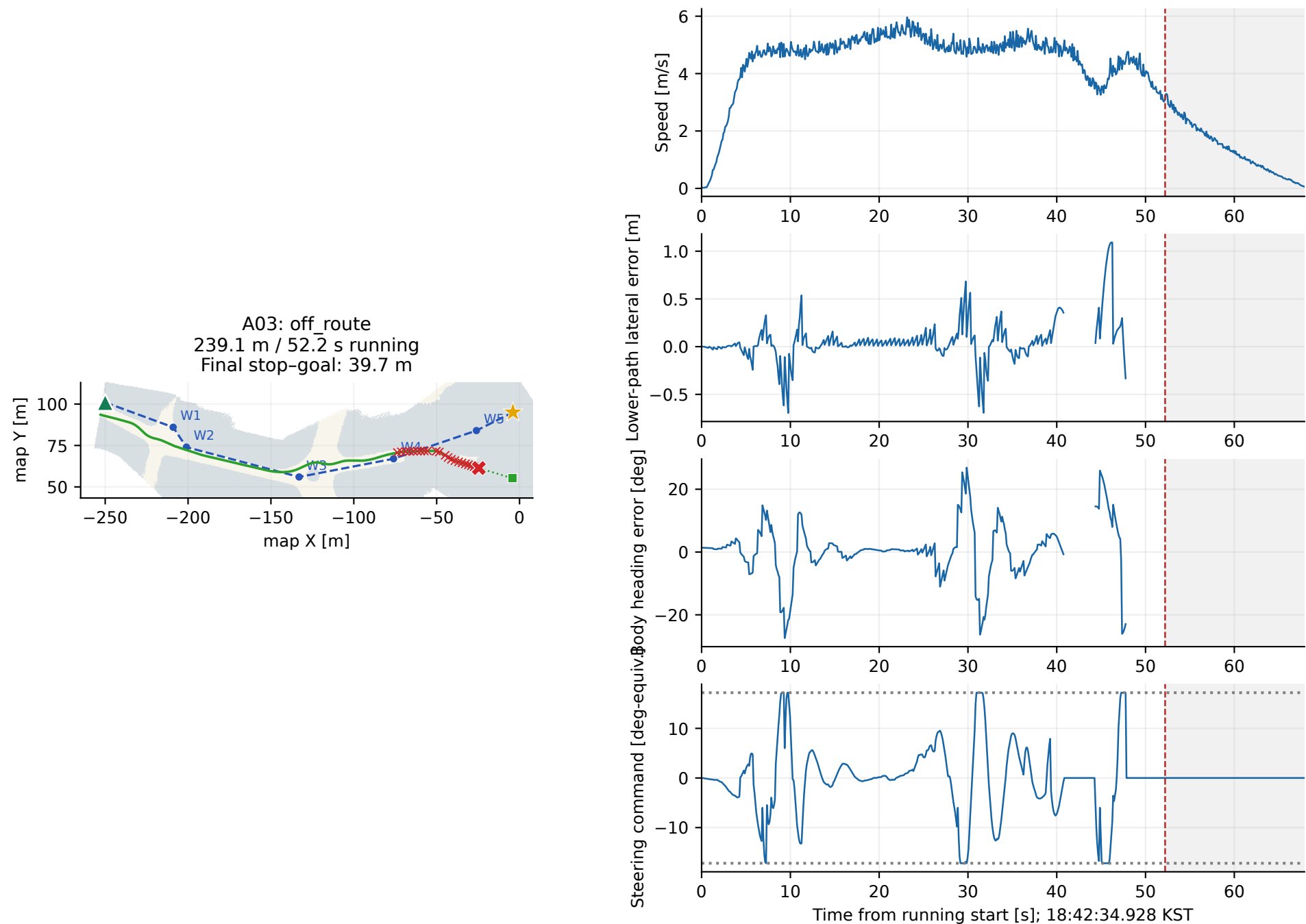
Scenario 2 / Attempt 01 — 20261008_183952_scenario2_5524df



Scenario 2 / Attempt 02 — 20261008_183952_scenario2_5524df

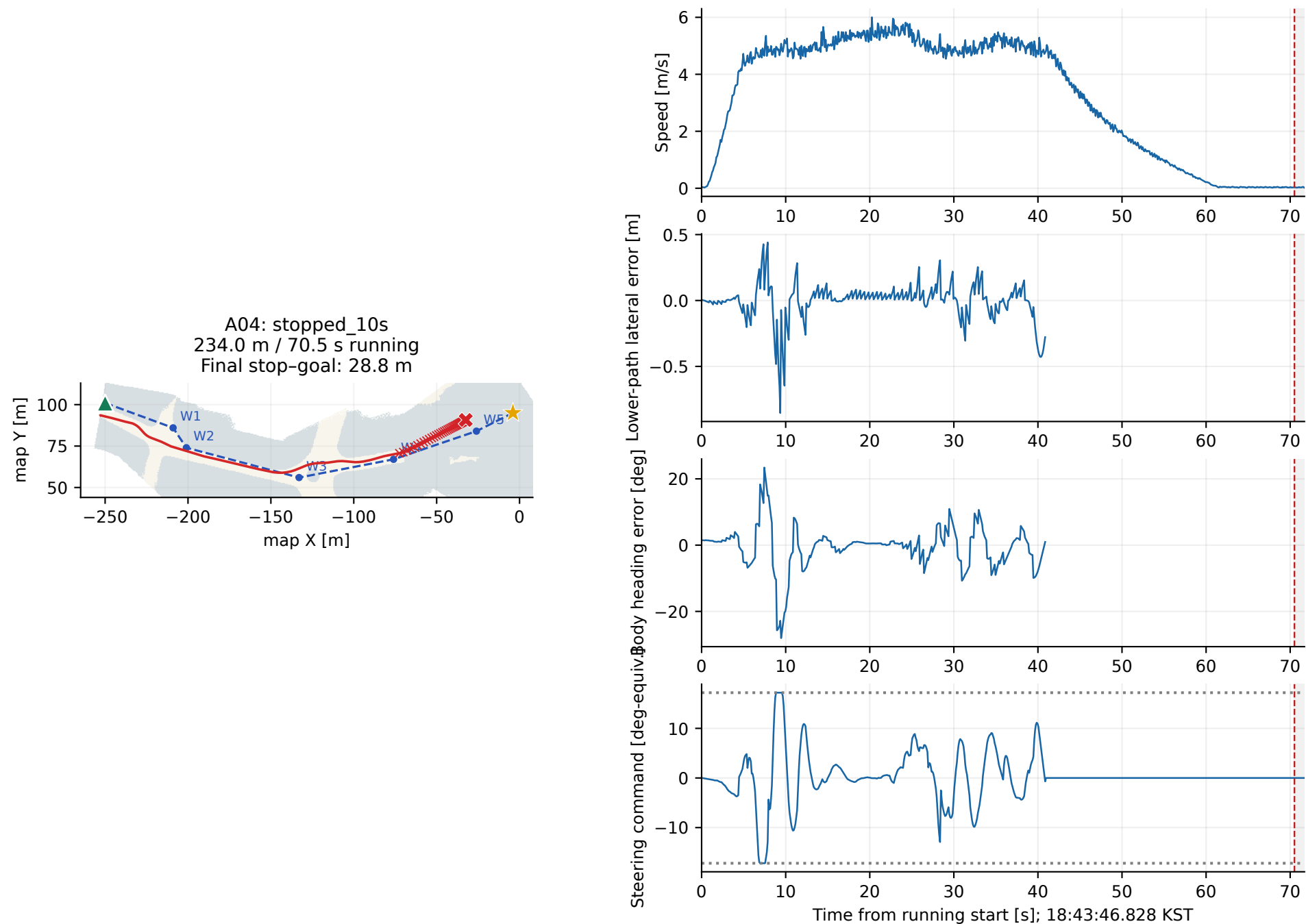


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.



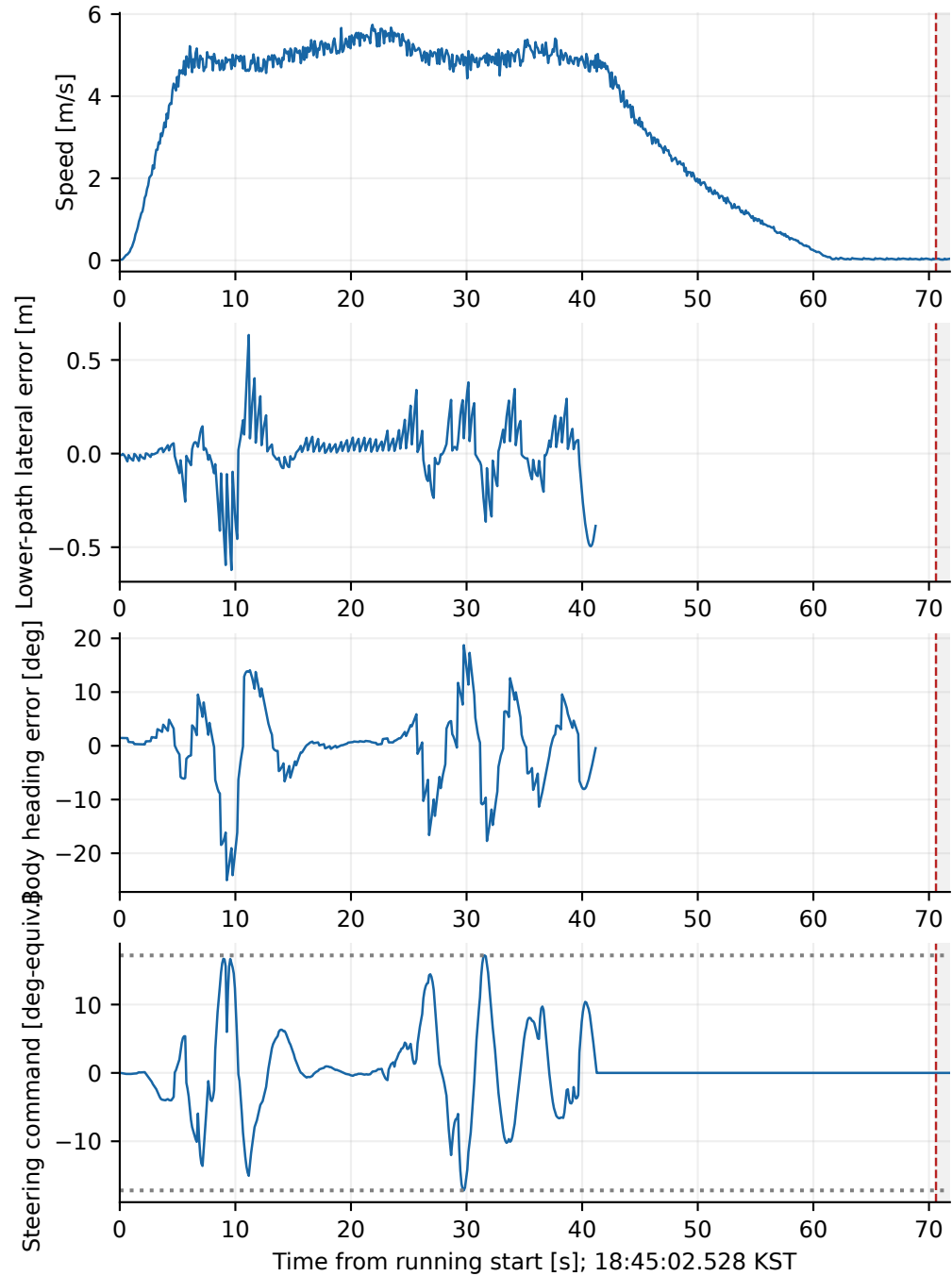
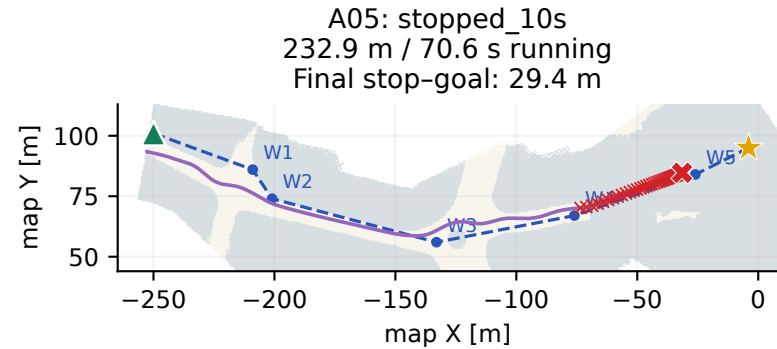
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 04 — 20261008_183952_scenario2_5524df



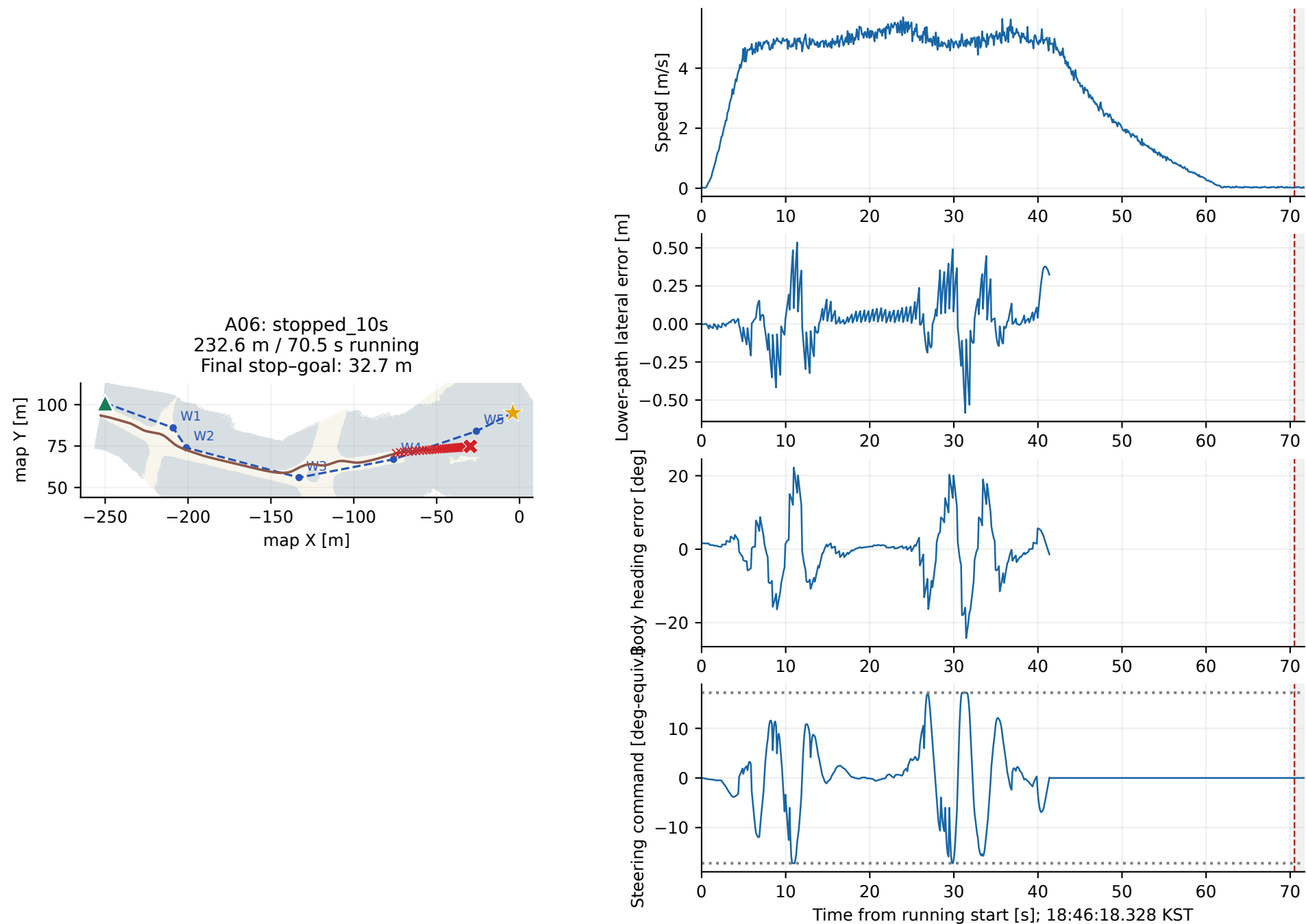
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 05 — 20261008_183952_scenario2_5524df



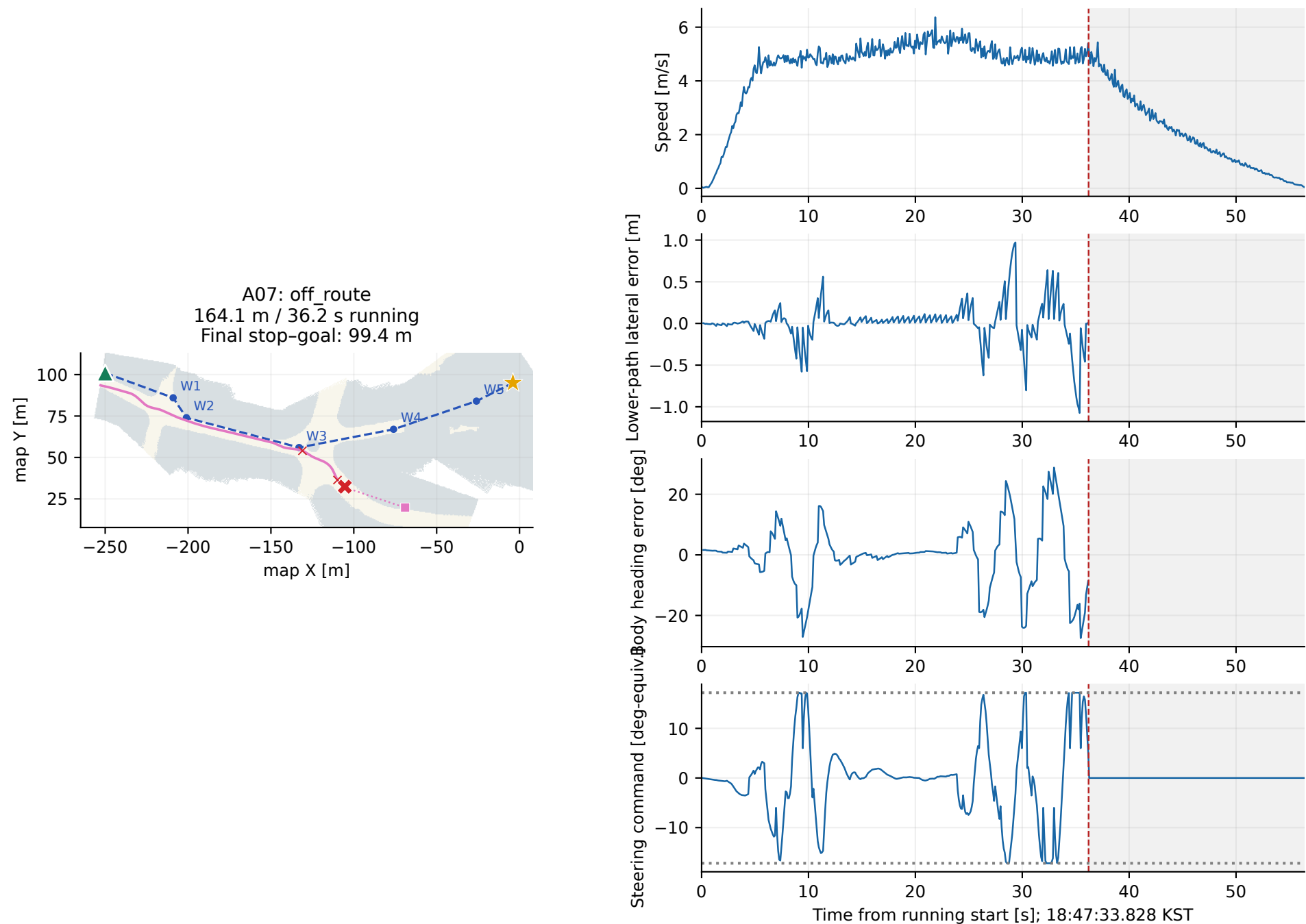
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 06 — 20261008_183952_scenario2_5524df



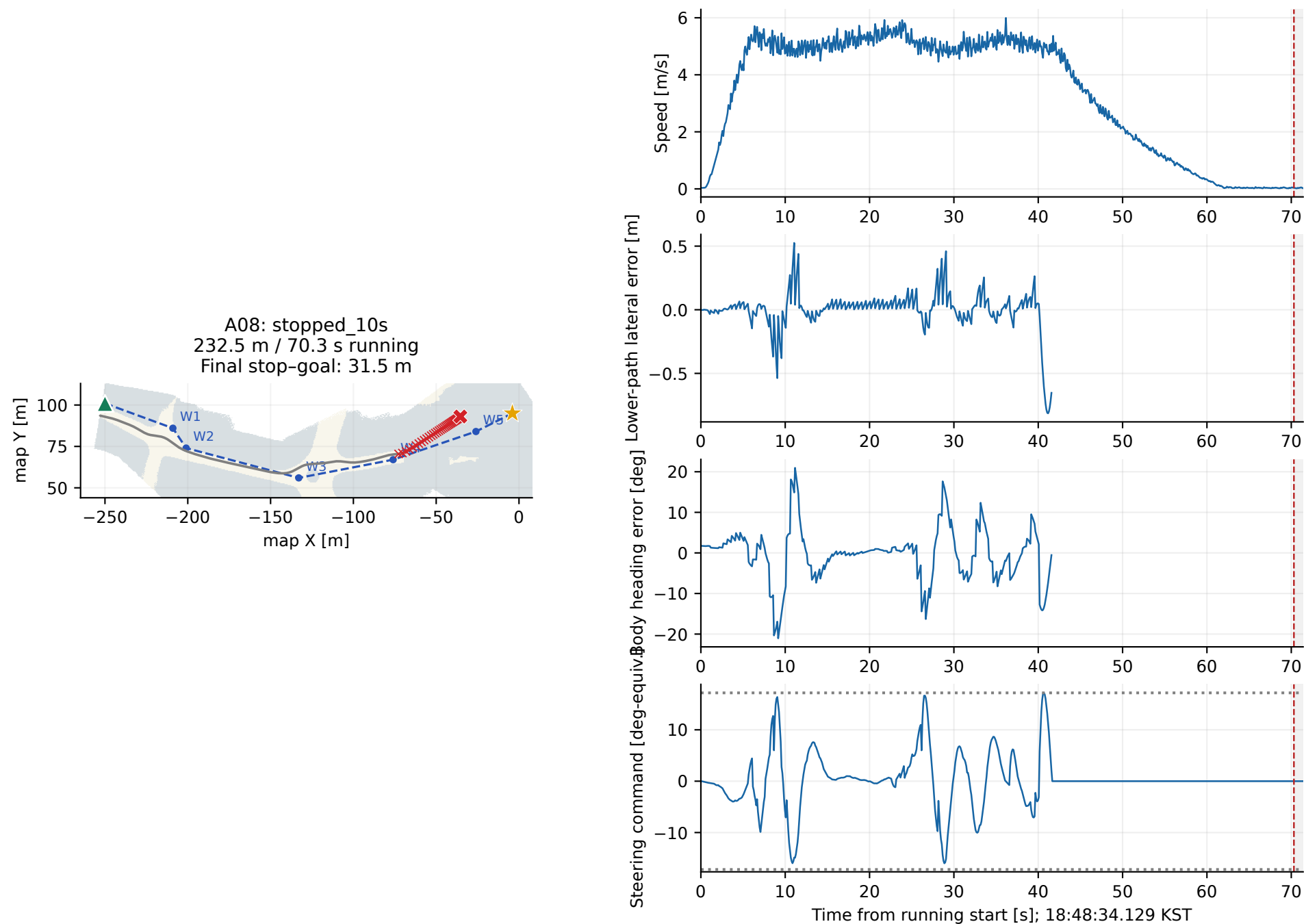
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 07 — 20261008_183952_scenario2_5524df



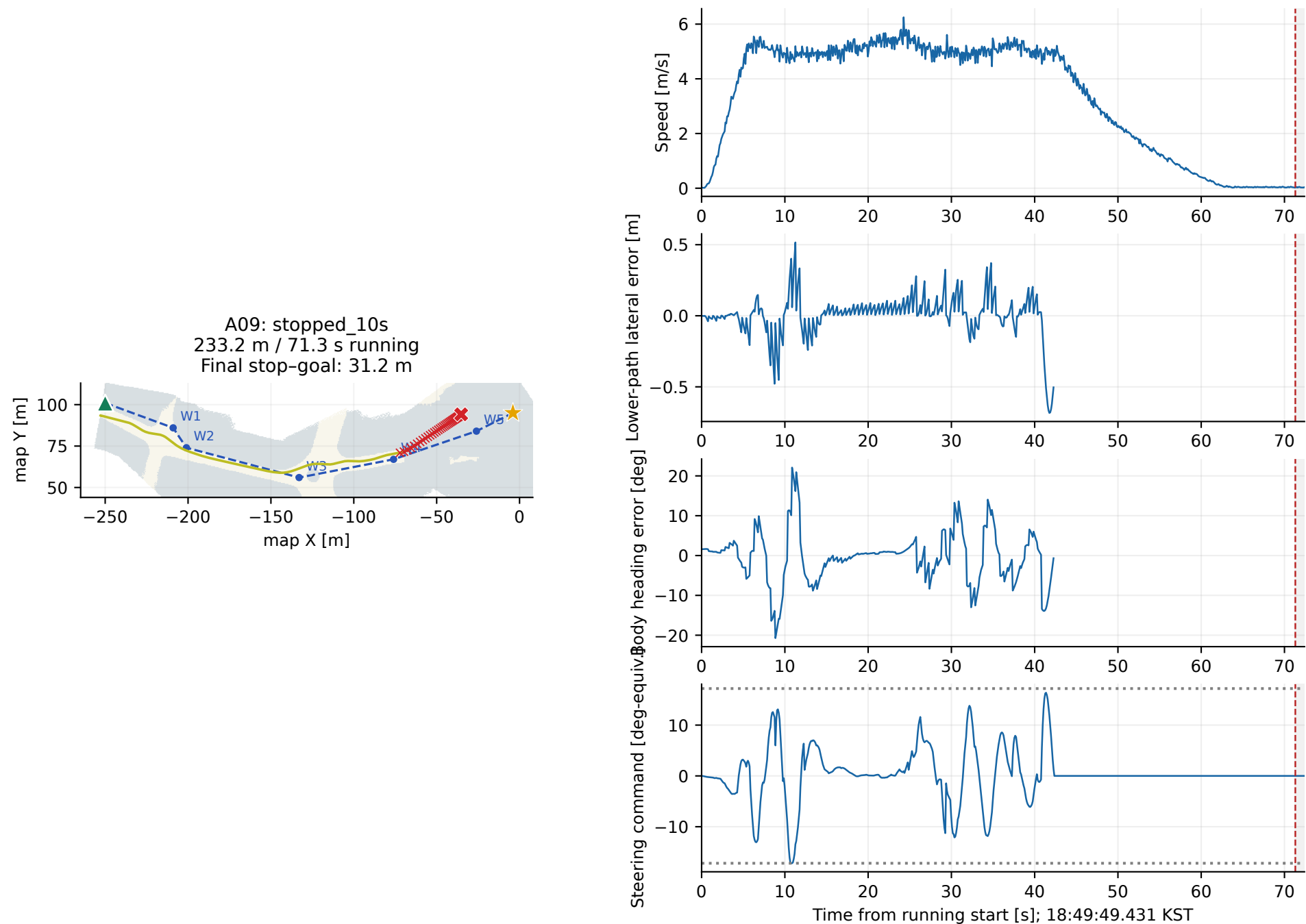
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 08 — 20261008_183952_scenario2_5524df



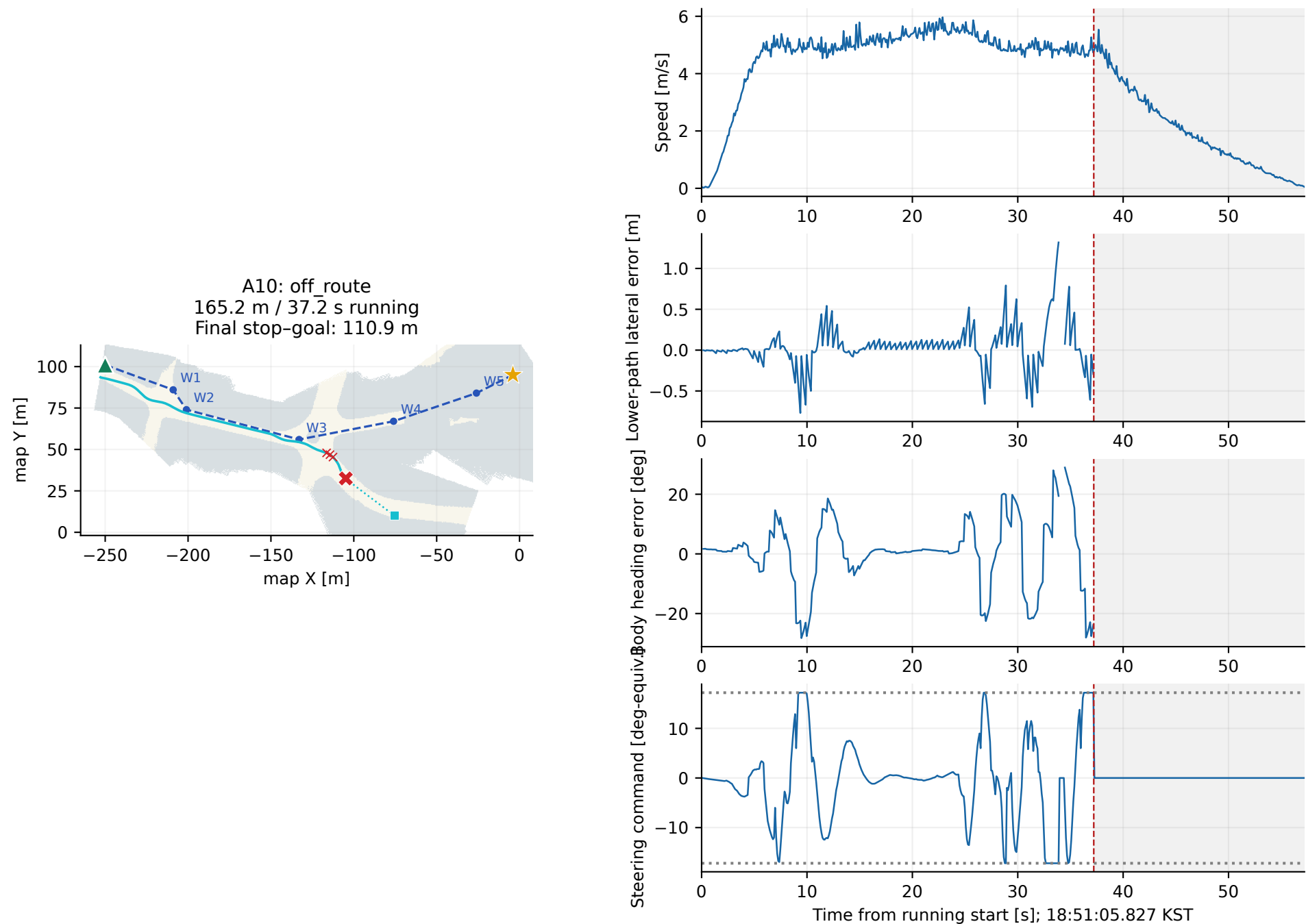
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 09 — 20261008_183952_scenario2_5524df

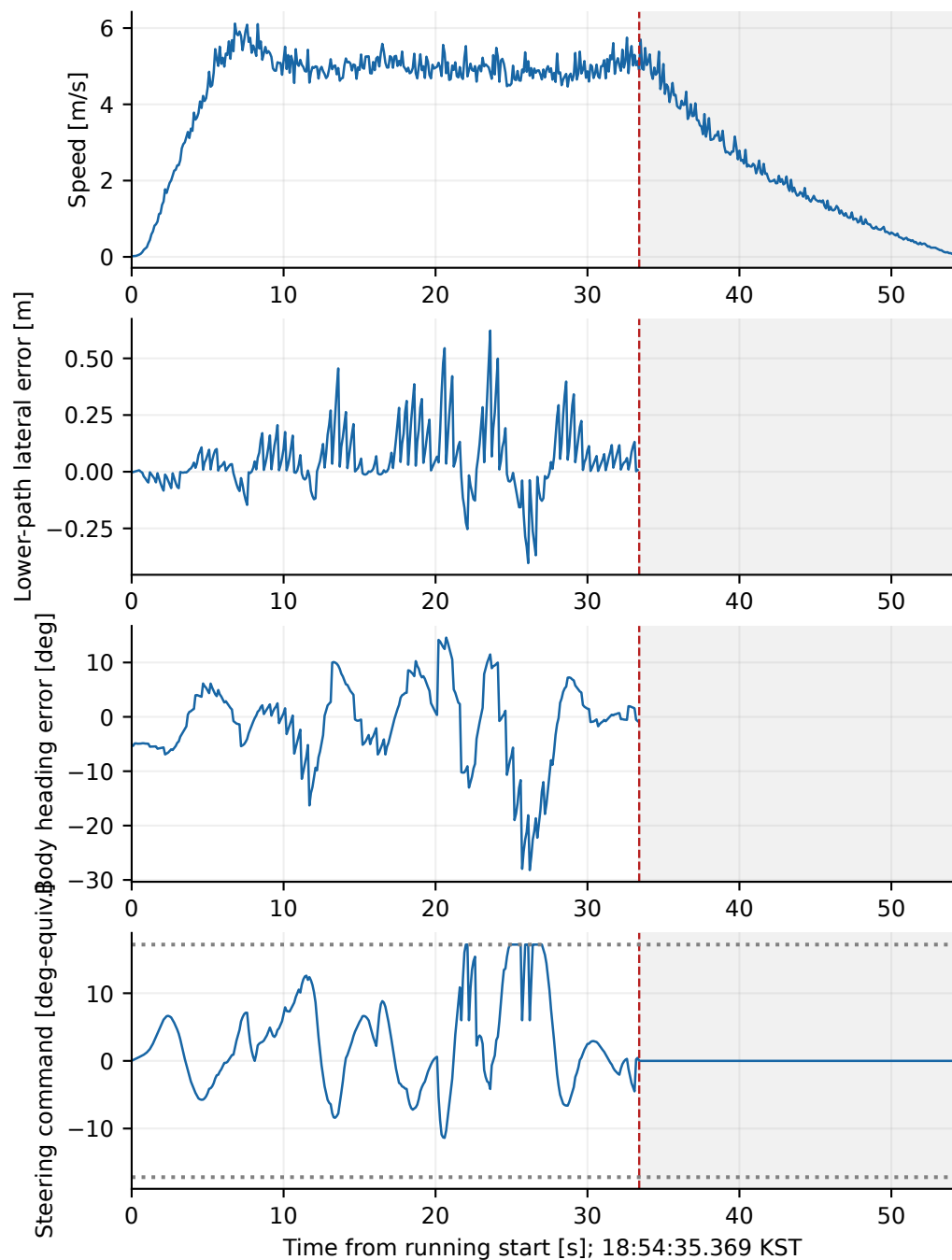
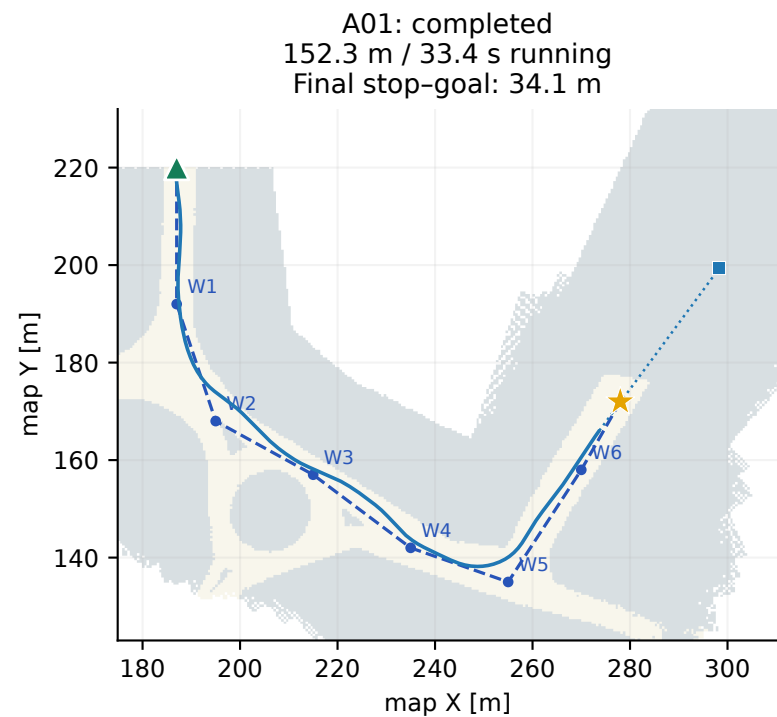


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 2 / Attempt 10 — 20261008_183952_scenario2_5524df

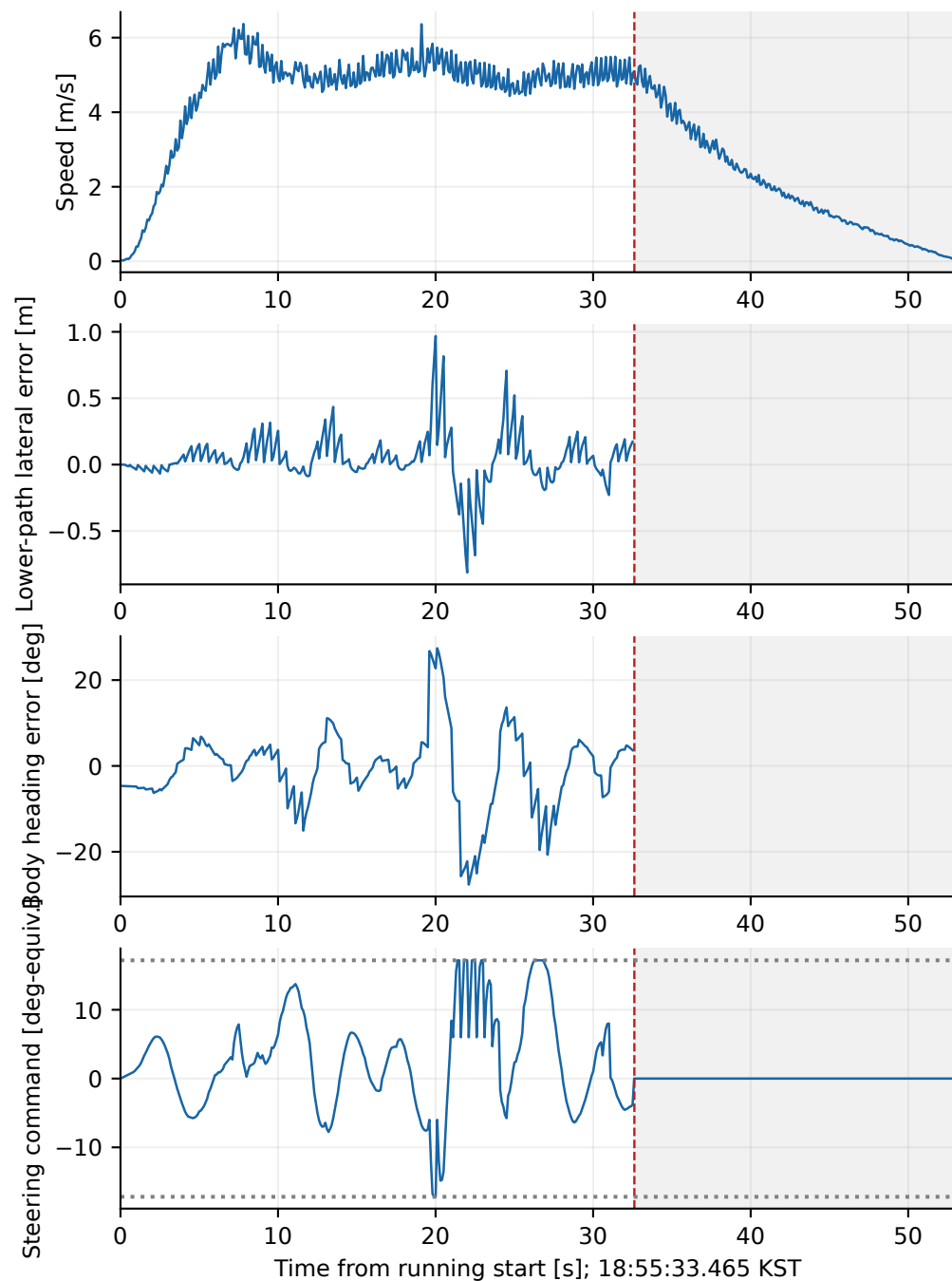
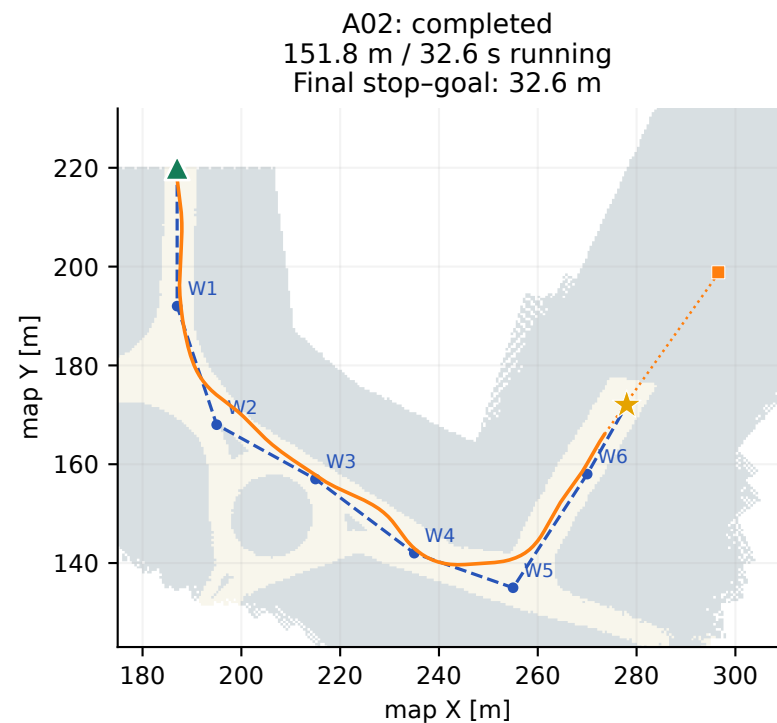


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

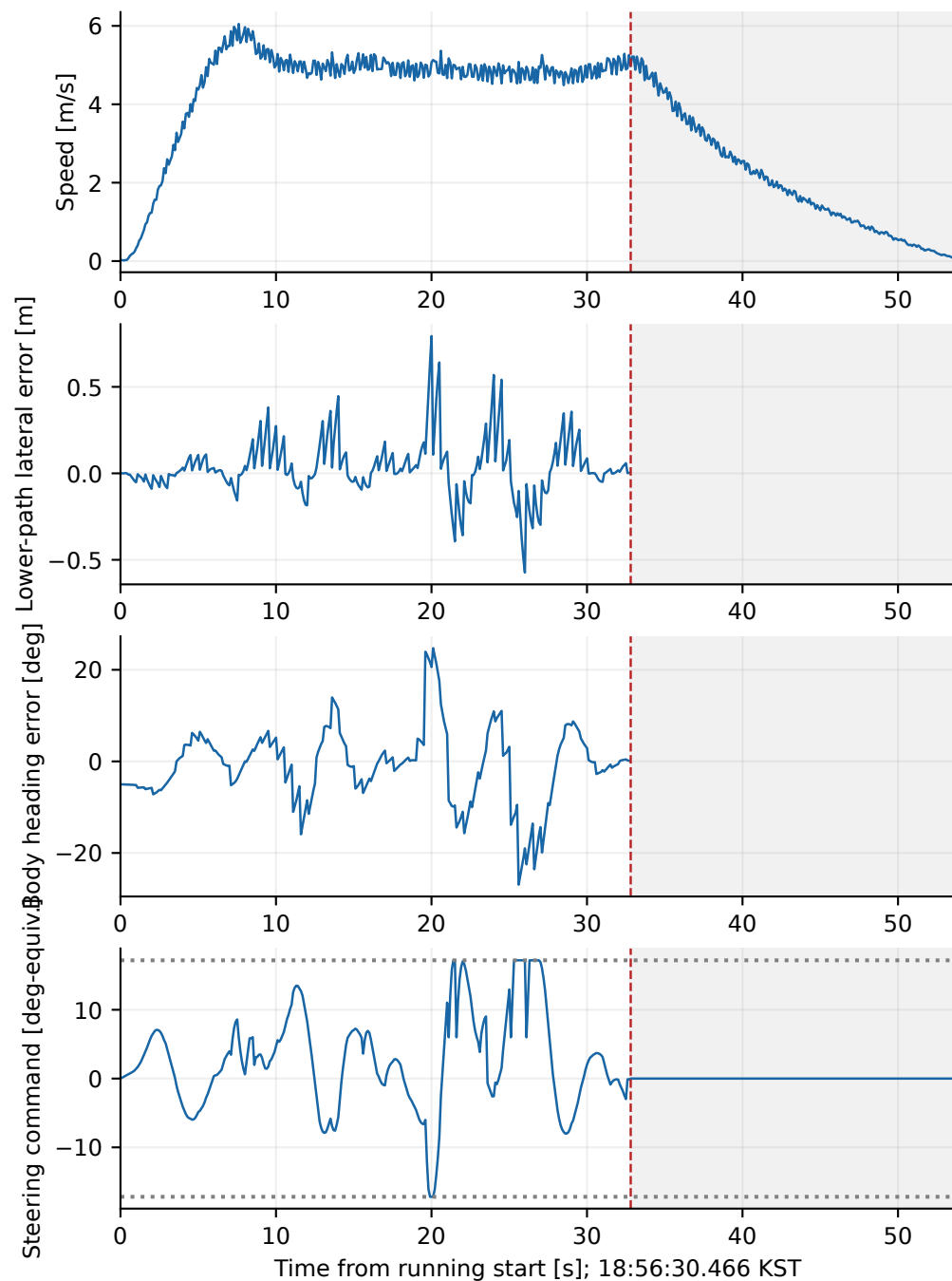
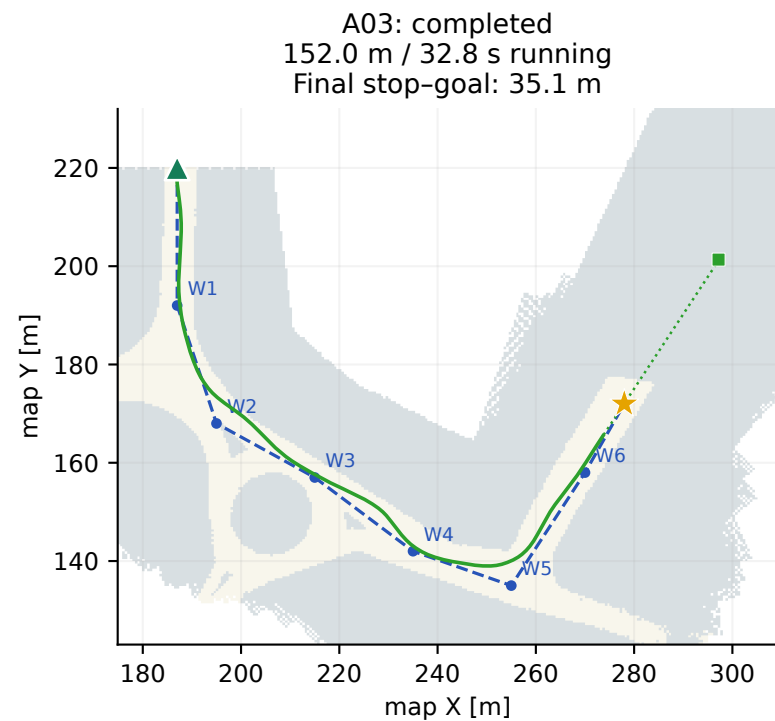


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 3 / Attempt 02 — 20261008_185426_scenario3_72c4df

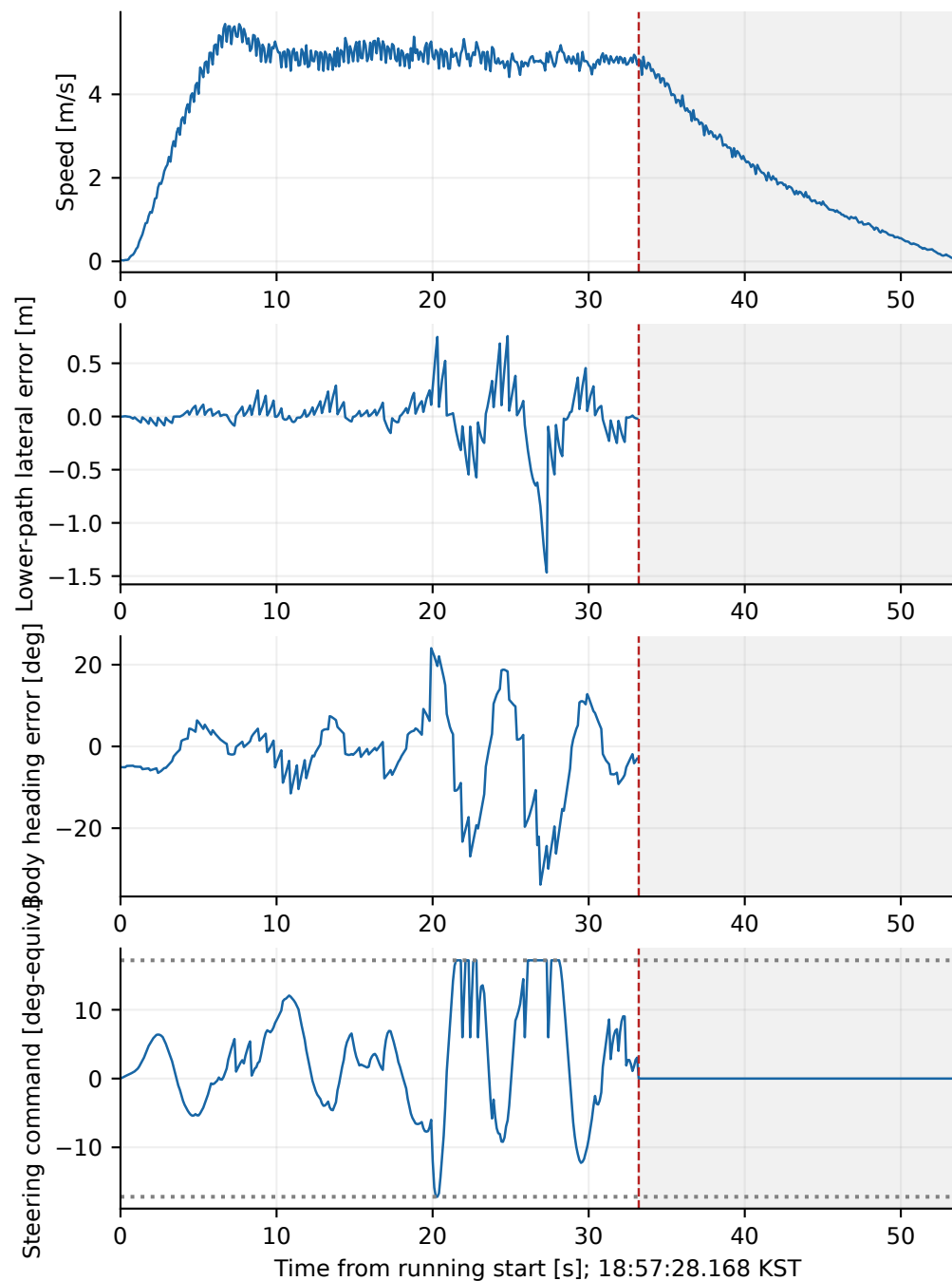
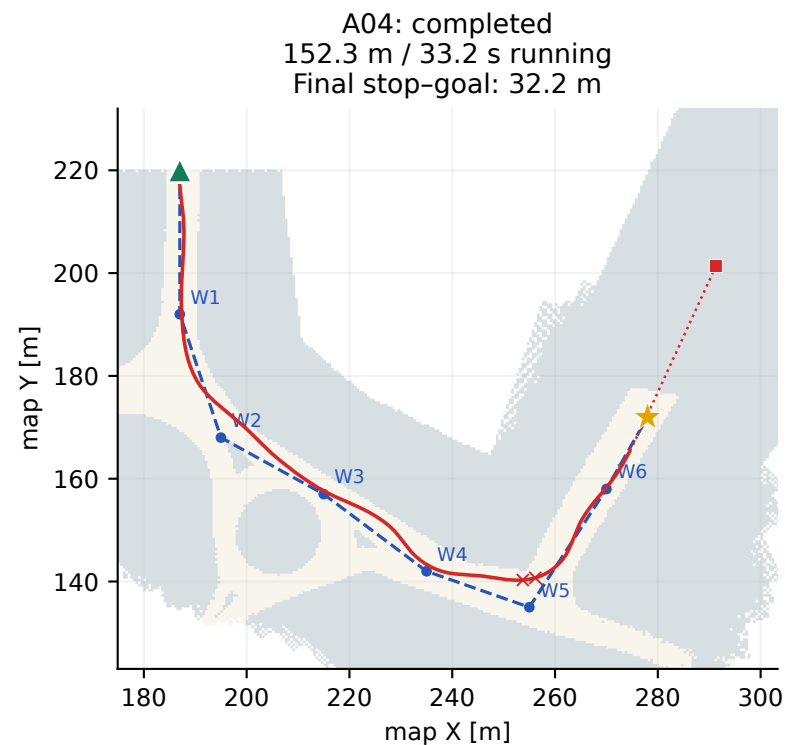


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.



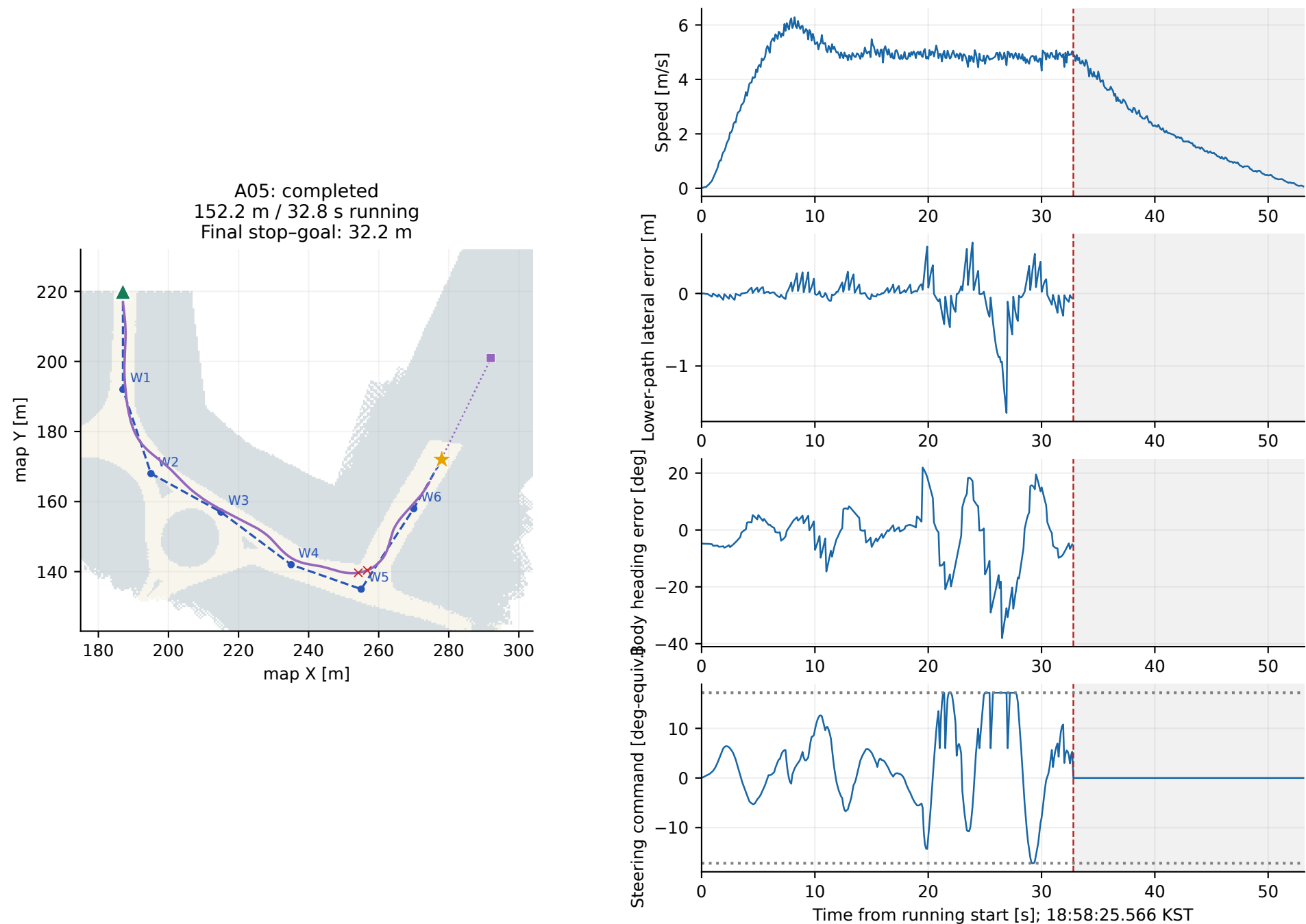
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 3 / Attempt 04 — 20261008_185426_scenario3_72c4df

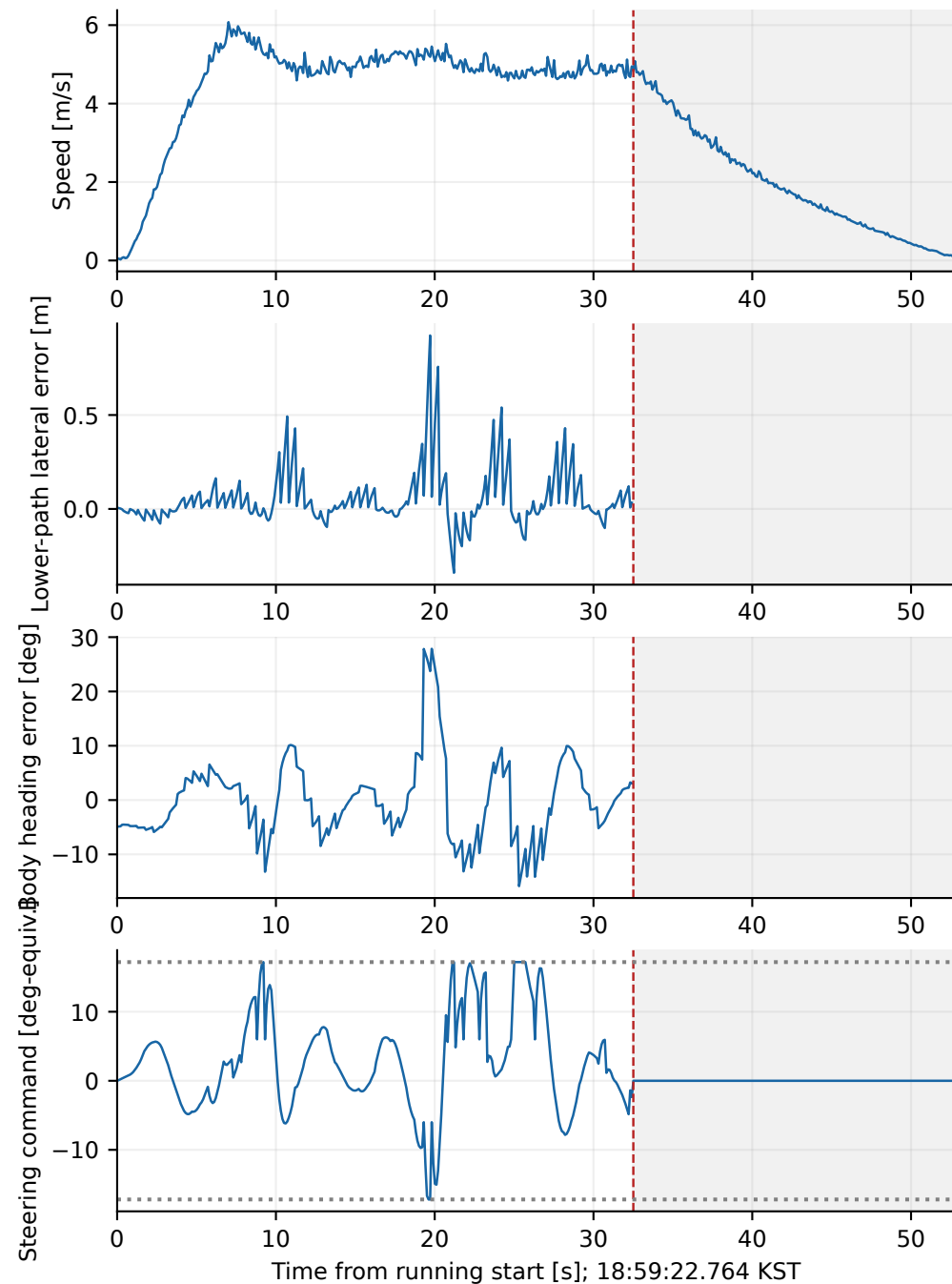
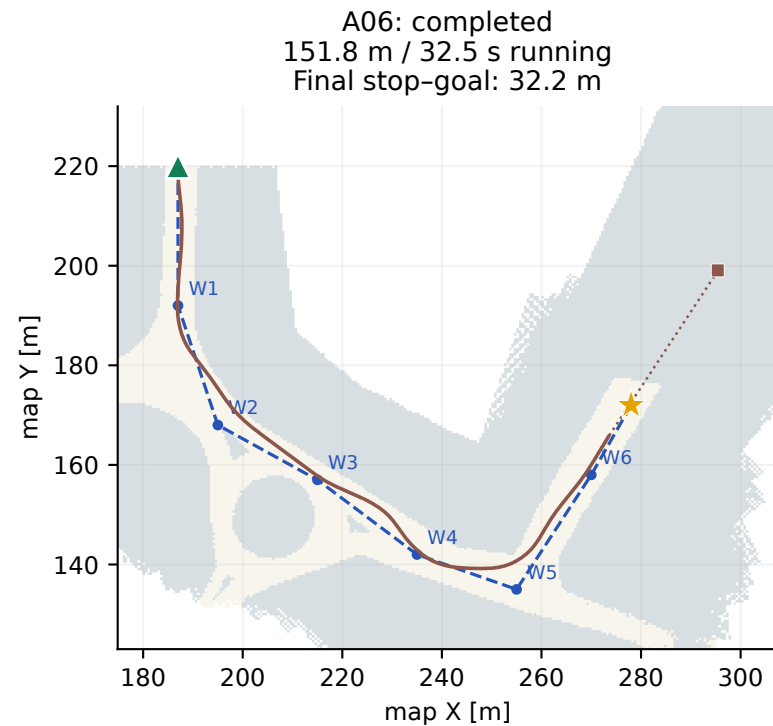


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

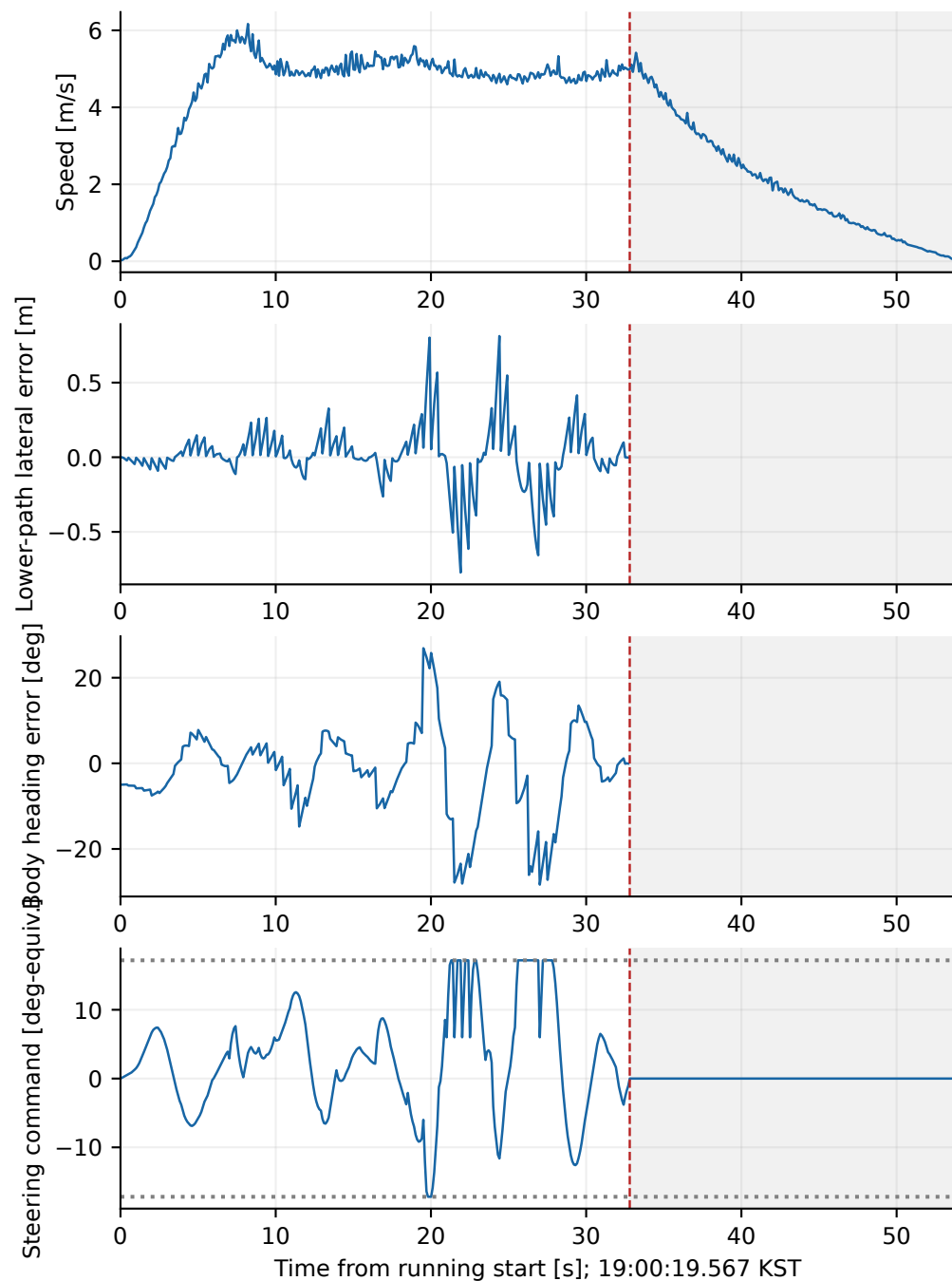
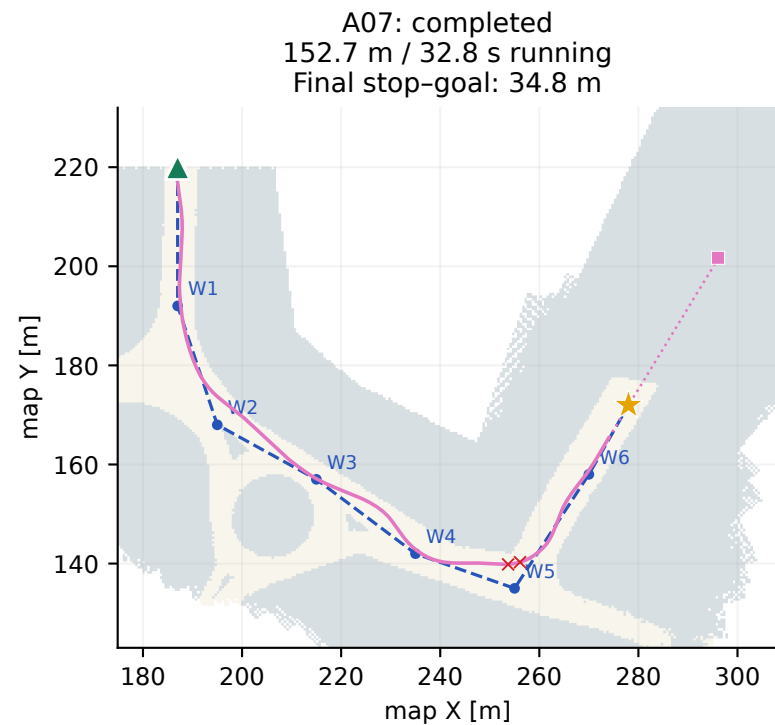
Scenario 3 / Attempt 05 — 20261008_185426_scenario3_72c4df



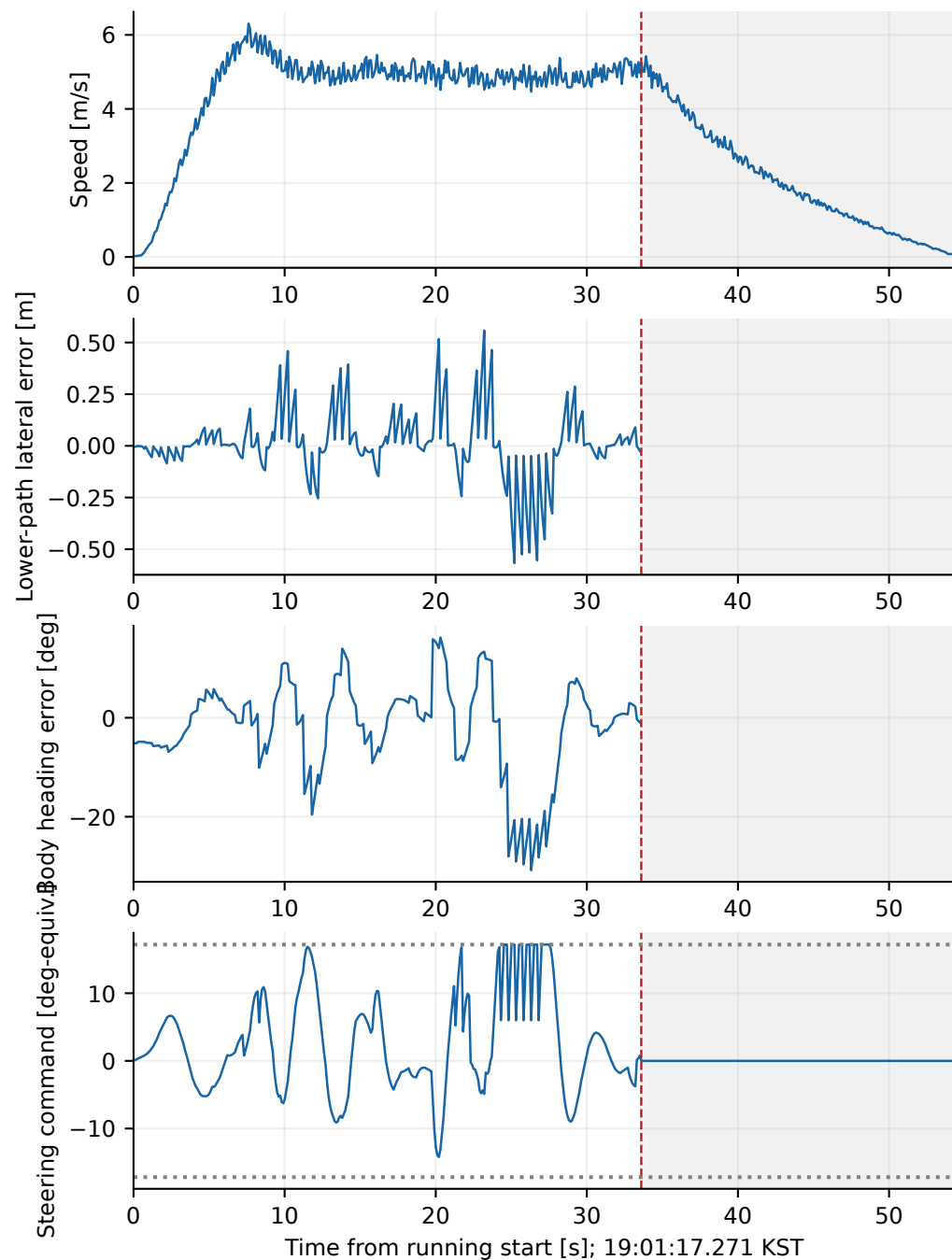
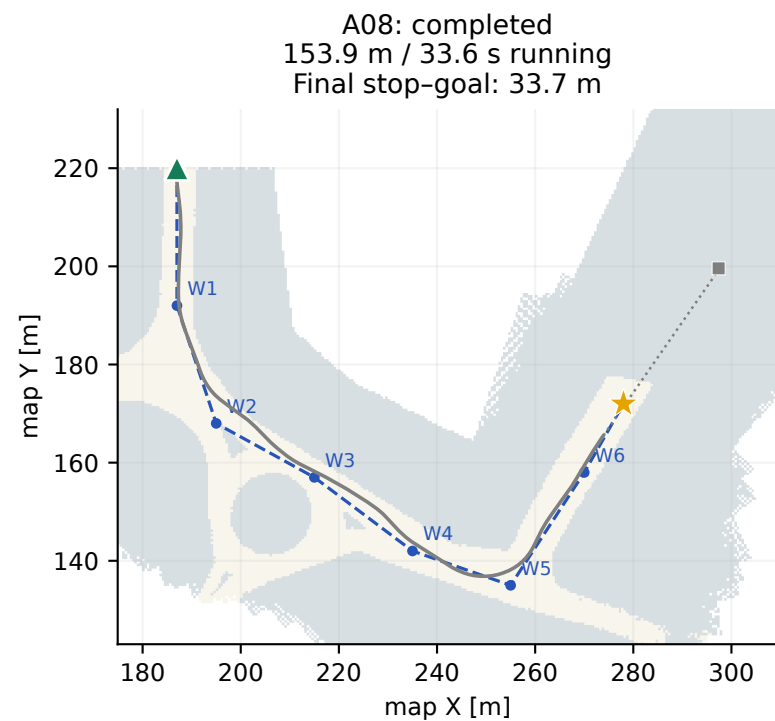
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.



Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

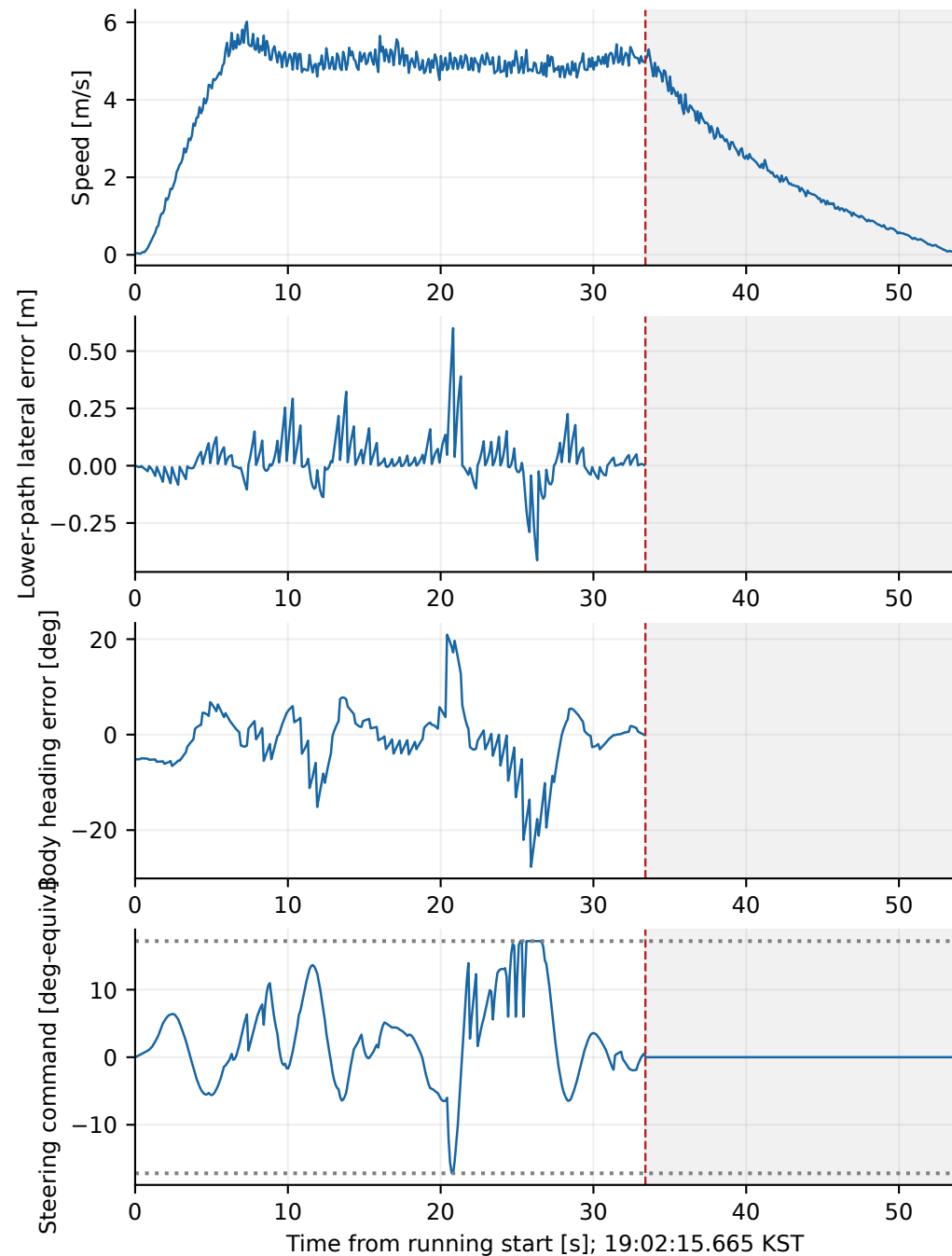
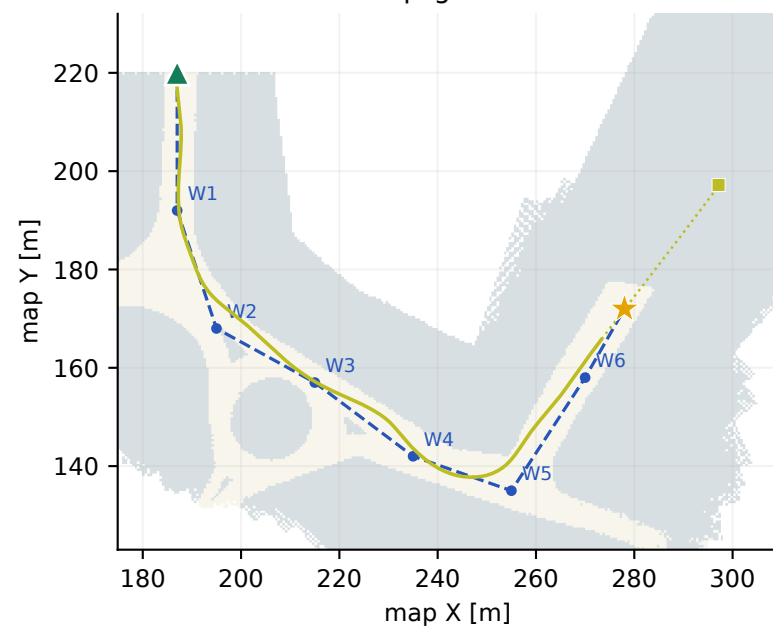


Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.



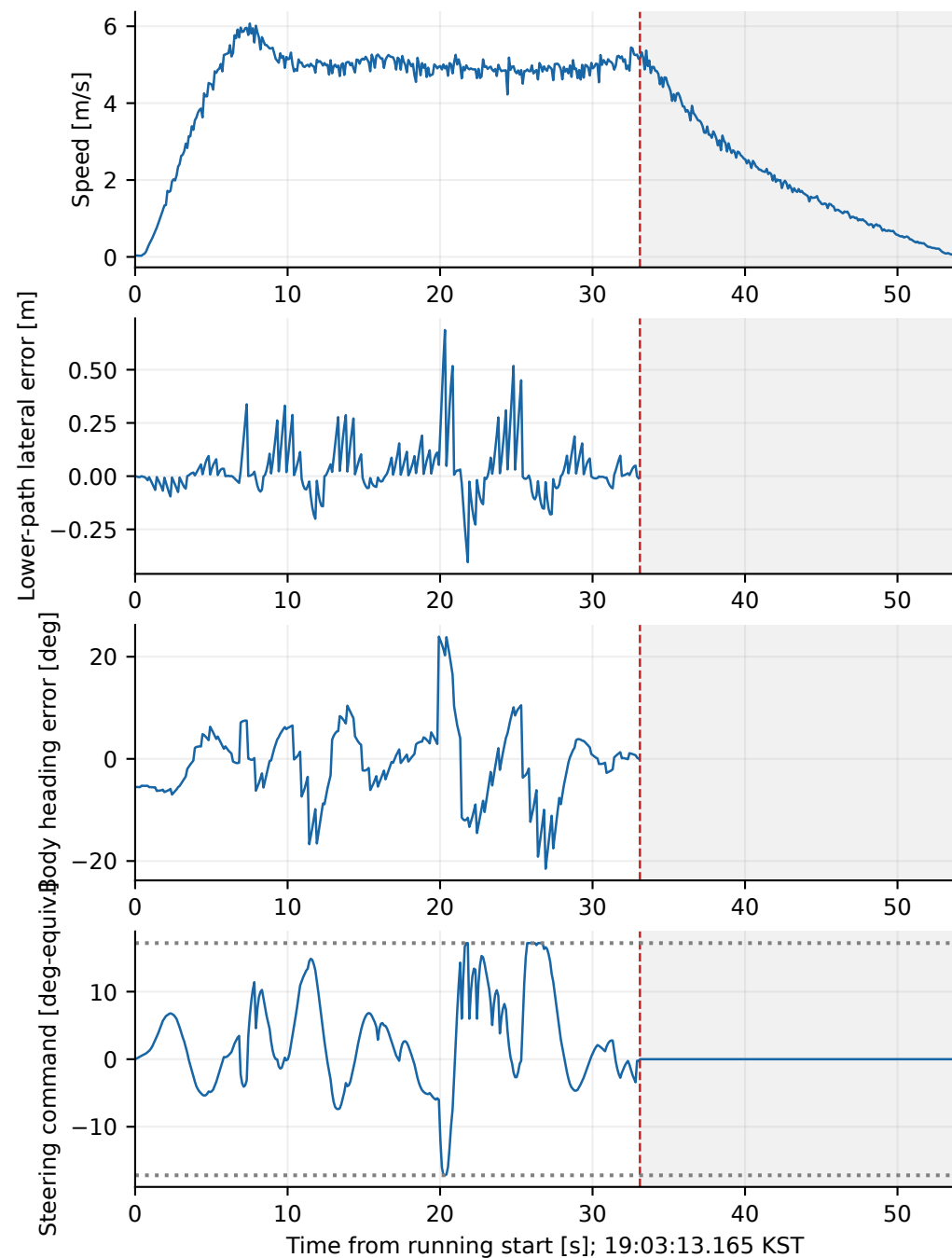
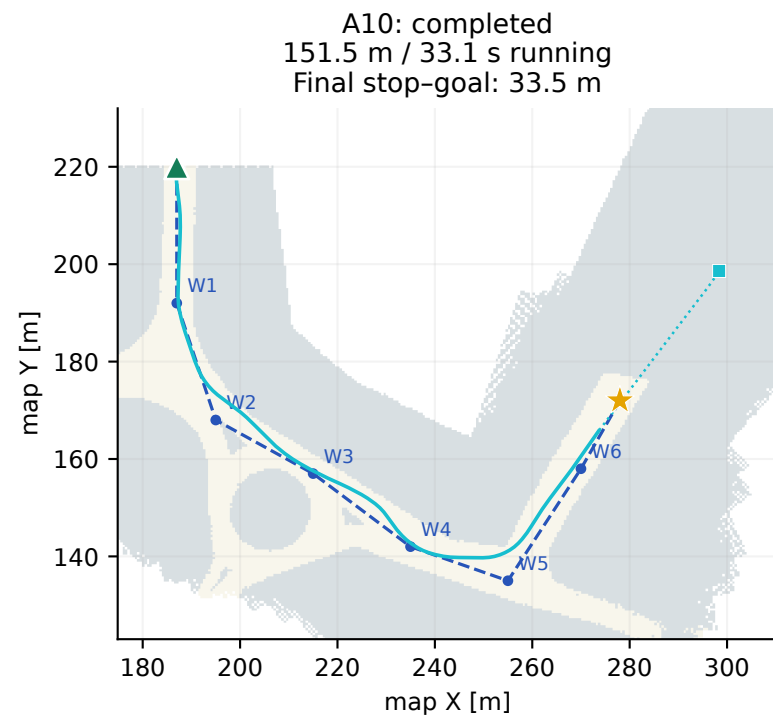
Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

A09: completed
152.0 m / 33.4 s running
Final stop-goal: 31.6 m



Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.

Scenario 3 / Attempt 10 — 20261008_185426_scenario3_72c4df



Recorded EKF-derived map position. Gray time span / dotted route: post-decision coast. Steering is a command; no wheel-angle measurement.